MultiQT notes

November 10, 2026

Notation

Unless stated otherwise, fix one disease, location and forecast horizon; their indices are suppressed.

Forecasts and indices. The indices s, t order forecast reference weeks (`reference_date`, Saturday); j also denotes a historical step inside a re-run. The outcome y_t belongs to the target week of forecast q_t, identified by `target_end_date` in the input forecasts and `target_date` in the prepared cases, rather than necessarily to its reference week. The quantile levels are $\mathcal{A} = \{\alpha_1 < \cdots < \alpha_m\} \subset (0,1)$, with $m = |\mathcal{A}|$. Write $b_t = (b_t^\alpha)_{\alpha\in\mathcal{A}}$ for the base forecasts and $q_t = (q_t^\alpha)_{\alpha\in\mathcal{A}}$ for the recalibrated forecasts. Unless a transformation is shown explicitly, forecasts and outcomes are in outcome units (hospital counts in the code). The methodological recursions first issue q_t, then observe y_t and update for the next step. The coding section specifies when delayed or revised reports become available instead.

Loss and projections. The indicator is $\mathbf{1}\{\cdot\}$ and $x_+ = \max(x, 0)$. Quantile loss takes the prediction as its first argument and the observation as its second:

$$\rho_\alpha(z, r) = \alpha(r - z)_+ + (1 - \alpha)(z - r)_+. \tag{1}$$

The ordered-quantile cone and Euclidean projection are

$$\mathcal{K} = \{q \in \mathbb{R}^m : q^{\alpha_1} \le \cdots \le q^{\alpha_m}\}, \qquad \Pi_C(x) = \arg\min_{z\in C} \|z - x\|^2,$$

where C is nonempty, closed and convex and $\|\cdot\|$ is the Euclidean norm. Assume $b_t \in \mathcal{K}$. For a positive-definite matrix W, write

$$\Pi_{\mathcal{K}}^W(v) = \arg\min_{q\in\mathcal{K}} (q - v)^\top W (q - v).$$

Q_p denotes an empirical p-quantile; I denotes the identity matrix. Write $\mathrm{clip}_{[a,b]}(x) = \min\{b, \max\{a, x\}\}$.

Scores and tracker states. $S_t(y)$ is a score function, $r_t = S_t(y_t)$ its observed value, and c_t its issued threshold in score units. The target coverage is α, with target miscoverage $\delta = 1 - \alpha$; empirical coverage at a step is $\mathrm{cov}_t^\alpha = \mathbf{1}\{y_t \le q_t^\alpha\}$. The superscripts std and stu distinguish standard and studentized methods. A tilde marks hidden, accumulated state; the untilded state is played after projection. In the methodological section, $\tilde\theta_t, \theta_t$ are offsets in outcome units and $\tilde u_t, u_t$ are thresholds in studentized units. In the coding section, the θ offsets are in the units of the selected transform, ψ or χ; the u states remain in studentized units. The finite, positive scales are σ_t^α, collected in

$$A_t = \mathrm{diag}(\sigma_t^{\alpha_1}, \ldots, \sigma_t^{\alpha_m});$$

σ_t denotes a scale common to all levels and σ a constant scale. The tracker step size is η (or η_t when varying), and κ is a rate multiplier (column `multiplier`, labelled m in script 02 plots; that plot label is distinct from $m = |\mathcal{A}|$ here). The selected loss subgradient is written g_α; its form is derived in (8).

Reports, feedback and re-runs. The date ι_t is the information cutoff for forecast q_t: normally the Wednesday of its reference week, with actual generation dates used where known. The notation $y_s^{[v]}$ means the reported count for the target week of q_s in the latest report dated on or before date v (the `version` field of the archive). A valid report is finite and nonnegative. Its first valid publication date is ν_s, and $d \ge 0$ is an additional delay in calendar weeks. The symbols $m_s^{(d)}$, $\tau_d(s)$ and $\mathcal{R}_t^{(d)}$ denote the maturity date, the index of the round that consumes feedback for q_s, and the set of rows consumed before issuing q_t, respectively; their

scheduling rules are given in (28)–(29). The set $I_{20}^{(d)}$ contains the first 20 consumed forecast rows in release order, with their stateful report values

$$\bar{y}_s^{(d)} = y_s^{[\iota_{\tau_d(s)}]}, \qquad \tau_d(s) \text{ exists.} \tag{2}$$

For forecasts and states, a superscript $[t]$ instead identifies the re-run at reference-week index t: $q_s^{[t]}$ is the forecast for row s reconstructed inside that re-run. Thus v in $y_s^{[v]}$ is a report-cutoff date, while t in $q_s^{[t]}$ is a re-run index. Feedback respects the cutoff convention, subject to the retrospective initialization described in Section 2.1.3.

Implementation scales. The tracker transformation is $\psi(x) = x$ or $\psi(x) = \log(1 + x)$. The count divisor $c_{\text{scale}} > 0$ is population divided by 100,000 by default, so y/c_{scale} is a rate per 100,000 residents. The fixed reference constants S_0 (initial error scale) and L_0 (expert loss normalization) are defined in (34) and (63); S_0 is separate from the time-indexed score function S_t. Their numerical floors are ε_S and ε_L. Square-root benchmarks use transform χ, projected tracker forecast ξ_t, fast/slow states $\tilde{\theta}_t^F, \tilde{\theta}_t^S$, and released coverage-error sum ϵ_t^α. $\mathcal{F}_t^{(d)} = \bigcup_{j \le t} \mathcal{R}_j^{(d)}$ contains all rows consumed by issue t; $\mathcal{F}_{t,20}^{(d)}$ contains its last 20 rows in release order, or all rows if fewer are available. c_{pop} always equals population divided by 100,000.

Experts and weighting. Expert k issues $f_{k,t}^\alpha$ at level α in forecast round t. The index n counts processed expert-weight updates, not calendar weeks; $s(n)$ is the forecast row whose feedback is consumed, and Y_n is the report used in that update. For a fixed bank of K experts, $w_{k,n}^\alpha$ is the weight immediately before update n and λ_n^α its learning rate; $\bar{w}_{k,n+1}^\alpha$ is the weight after the exponential step but before optional sharing. Common weights and rates omit α: $w_{k,n}$ and λ_n. Suppress time indices when only one forecast or update is considered; $\hat{q}$ is the mixture before any crossing repair. The unnormalized expert losses are $L_{k,n}^\alpha$; $\ell_{k,n}^\alpha$ are their normalized versions, and $L_{k,n}$ and $\ell_{k,n}$ are the corresponding means over levels. The second moments are M_n^α or M_n, the lookback is H, the starting rate is λ_{start}, and γ is the optional fraction of weight shared uniformly among experts. Superscripts raw and norm distinguish calculations without and with L_0 normalization. For the soft switch, $G_t \in [0.05, 0.95]$ is the gate at issuance; $w_{k,t}^B$ and $w_{k,t}^D$ are the weights in the vectors named `baseline` and `declining`, and $w_{k,t}^{\text{mix}}$ are the combined weights used for that forecast. Here the second index is the issue round t, after all available weight updates, rather than the update index n. Its raw signals, logistic scores and nonnegative weights are $x_{t,j}, p_{t,j}, v_j$, with $j \in \{P, N, R\}$. The letters correspond to `path`, `nowcast` and `revision`. Gate strength, threshold and width are $s_G \in [0, 1]$, $\tau_G \ge 0$ and $w_G > 0$. $Z(t, h; v)$ is $\log(1 +$ base median$)$ at reference date t and horizon h, if that median is finite, nonnegative, available by cutoff v and not filled; otherwise it is missing. Only in these gate formulas, $t - 1$ means exactly one calendar week earlier. Write $Z_{t,h} = Z(t, h; \iota_t)$ and let ι_t^- be the stored cutoff of the horizon--1 nowcast from the preceding week. Its two historical values are $Z_t^- = Z(t - 1, -1; \iota_t^-)$ and $Z_t^{--} = Z(t - 2, -1; \iota_t^-)$. g_t, g_t^- denote current and previous nowcast growth; $r_{t,j}, r_{t,j}^-$ are paired same-target forecast revisions, with available pairs indexed by J_t. a_n^B, a_n^D are the loss multipliers for the two weight vectors.

Implementation and evaluation notation. In coding formulas, $\beta_t^\alpha = \psi(b_t^\alpha)$ is the base in tracker units, and a_t^α is the chosen centre (full base or base median). Φ is the standard normal distribution function. $\mathcal{H}_t^{(d)}$ contains the last five consumed rows by issue t, in release order; $e_s^\alpha = \psi(\bar{y}_s^{(d)}) - \beta_s^\alpha$ is their base residual. Raw scale estimates carry a hat; their bounded versions do not. For paired base widths, $a_\alpha = \min(\alpha, 1 - \alpha)$, except at .5, where it is the nearest lower grid level.

In the diverse bank, $b_{k,t}$ is the starting forecast of expert k before its own offset $\tilde{\theta}_{k,t}$; $\eta_{\text{slow}} = \kappa_{\text{slow}} S_0$ is its step. The growth adjustment is D_t, its damping coefficient is d_k, and the width factor is ω_k. The available report $y_{\text{avail},t}$ is from the most recent forecast row among those consumed under delay 0. The additional common shift on the log scale of the leaky expert is v_t, and Δ_t is elapsed calendar weeks since the previous issue, at least one.

For evaluation, $y_t^\star$ is the latest archived count, T_t the target date, $p(t)$ its nearest retained peak date (earlier at a tie), and $u_t^{\text{peak}} = (T_t - p(t))/7$ its calendar-week offset. E contains evaluation-period rows with finite scoring truth; W is a selected window and $N = |W|$. $\mathcal{L}_{j,c}$ and $\text{CE}_{j,c}$ denote the mean quantile loss and calibration error of method j in case c; $j = 0$ is the original forecast. A case fixes location, horizon and feedback design within one disease. $\mathcal{C}$ is a collection of such cases and J its size. For peak detection, x_i are retained latest counts in date order, z_i their smoothed values, P_i prominence, and l_i, r_i the nearest strictly higher smoothed positions on either side (or the series ends). X is the largest retained count on a forecast target date.

1 Methodological notes

1.1 Studentized score

Writing the score as a signed residual gives the usual additive MultiQT correction. Studentizing that residual gives multiplicatively scaled offsets, or, equivalently, offsets tracked in scaled coordinates. A changing scale rescales the accumulated correction as well as each new increment. If the scale decreases after a peak, it can shrink the outcome-scale contribution of a positive correction accumulated during an upswing. This is the motivation; the hidden accumulated state is not erased.

With quantile-specific scales, the ordering constraint defines a diagonally transformed, shifted isotonic cone. This is generally not a cone itself, but it remains a nonempty closed convex polyhedron with a unique Euclidean projection, computable through weighted isotonic regression in forecast coordinates. The gradient steps use a weighted quantile loss. The coverage criterion and calibration target are unchanged (strictly increasing invertible transformation); calibration is not guaranteed.

1.1.1 Standard MultiQT

The original MultiQT procedure [1] is

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t^{\text{std}} = b_t + \theta_t, \tag{3}$$

$$\text{cov}_t^{\text{std},\alpha} = \mathbf{1}\{y_t \le q_t^{\text{std},\alpha}\}, \qquad \tilde{\theta}_{t+1}^{\alpha} = \tilde{\theta}_t^{\alpha} - \eta(\text{cov}_t^{\text{std},\alpha} - \alpha). \tag{4}$$

1.1.2 Score-threshold tracking

To make the connection explicit, first consider the scalar score-threshold update of Angelopoulos et al. [2]. We then recover the MultiQT update using the signed residual as the score.

The score and threshold determine the prediction set

$$\mathcal{P}_t = \{y : S_t(y) \le c_t\}, \qquad \text{err}_t = \mathbf{1}\{r_t > c_t\}. \tag{5}$$

Update according to

$$\begin{aligned} c_{t+1} &= c_t + \eta(\text{err}_t - \delta) \\ &= c_t - \eta(\mathbf{1}\{r_t \le c_t\} - \alpha). \end{aligned} \tag{6}$$

The second line follows from $\text{err}_t = 1 - \mathbf{1}\{r_t \le c_t\}$ and $\delta = 1 - \alpha$. The prediction set is obtained by score inversion. For the signed affine scores below, the endpoint of that set is the outcome quantile forecast.

Holding the observed score r fixed, the subdifferential of (1) in the threshold is

$$\partial_z \rho_\alpha(z, r) = \begin{cases} \{-\alpha\}, & z < r, \\ [-\alpha, 1-\alpha], & z = r, \\ \{1-\alpha\}, & z > r. \end{cases} \tag{7}$$

Selecting $1 - \alpha$ at equality gives

$$g_\alpha(z; r) = \mathbf{1}\{r \le z\} - \alpha \ \in \ \partial_z \rho_\alpha(z, r). \tag{8}$$

Thus $c_{t+1} = c_t - \eta g_\alpha(c_t; r_t)$ is (6). The distance of the score from the threshold affects the loss; the side of the threshold determines the update direction and size.

1.1.3 MultiQT in residual coordinates

For each level, define the score and its observed value by

$$S_t^{\mathrm{std},\alpha}(y) = y - b_t^\alpha, \qquad r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha. \tag{9}$$

Using θ_t^α as the score threshold gives

$$\{y : S_t^{\mathrm{std},\alpha}(y) \le \theta_t^\alpha\} = (-\infty, b_t^\alpha + \theta_t^\alpha] = (-\infty, q_t^{\mathrm{std},\alpha}]. \tag{10}$$

Consequently,

$$\mathrm{cov}_t^{\mathrm{std},\alpha} = \mathbf{1}\{y_t \le b_t^\alpha + \theta_t^\alpha\} = \mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\}. \tag{11}$$

Substitution into (4) gives

$$\boxed{\tilde{\theta}_{t+1}^\alpha = \tilde{\theta}_t^\alpha - \eta(\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha).} \tag{12}$$

The original MultiQT loss has the exact residual-coordinate representation

$$\begin{aligned} \ell_t^{\mathrm{std}}(\theta) &= \sum_{\alpha\in\mathcal{A}} \rho_\alpha(b_t^\alpha + \theta^\alpha, y_t) \\ &= \sum_{\alpha\in\mathcal{A}} \rho_\alpha(\theta^\alpha, y_t - b_t^\alpha) = \sum_{\alpha\in\mathcal{A}} \rho_\alpha(\theta^\alpha, r_t^{\mathrm{std},\alpha}). \end{aligned} \tag{13}$$

The threshold c_t of the scalar score tracker corresponds to the offset θ_t^α when the score is $y - b_t^\alpha$. For a single level the constraint is all of $\mathbb{R}$, so the hidden and played states coincide and (12) is exactly scalar quantile tracking. For multiple levels, MultiQT uses a joint projection: evaluate g_α at the played θ_t^α, but apply the increment to the hidden $\tilde{\theta}_t^\alpha$.

1.1.4 Studentized residuals and their update

Now replace the residual score by the studentized residual.

For this construction, choose the scales from information available at issuance and hold them fixed when optimizing the threshold. Define

$$S_t^{\mathrm{stu},\alpha}(y) = \frac{y - b_t^\alpha}{\sigma_t^\alpha}, \qquad r_t^{\mathrm{stu},\alpha} = \frac{y_t - b_t^\alpha}{\sigma_t^\alpha}. \tag{14}$$

Inverting the score gives

$$\begin{aligned} S_t^{\mathrm{stu},\alpha}(y) \le u_t^\alpha \quad &\Longleftrightarrow \quad y \le b_t^\alpha + \sigma_t^\alpha u_t^\alpha, \\ q_t^{\mathrm{stu},\alpha} = b_t^\alpha + \sigma_t^\alpha u_t^\alpha, &\qquad \mathrm{cov}_t^{\mathrm{stu},\alpha} = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\}. \end{aligned} \tag{15}$$

Standard MultiQT tracks the outcome-unit residual threshold θ_t^α and issues $b_t^\alpha + \theta_t^\alpha$. Studentization tracks u_t^α and issues $b_t^\alpha + \sigma_t^\alpha u_t^\alpha$. The factor σ_t^α enters through the inverse score map.

Apply the same quantile-loss construction to the studentized scores:

$$\begin{aligned} \ell_t^{\mathrm{stu}}(u) &= \sum_{\alpha\in\mathcal{A}} \rho_\alpha(u^\alpha, r_t^{\mathrm{stu},\alpha}) \\ &= \sum_{\alpha\in\mathcal{A}} \frac{1}{\sigma_t^\alpha} \rho_\alpha(b_t^\alpha + \sigma_t^\alpha u^\alpha, y_t). \end{aligned} \tag{16}$$

Using (8), its selected subgradient is

$$g_t^{\mathrm{stu},\alpha}(u_t) = \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha = \mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha. \tag{17}$$

The chain rule gives the same result. Away from ties, for one level,

$$\frac{\partial}{\partial u^\alpha}\left[\frac{\rho_\alpha(b_t^\alpha+\sigma_t^\alpha u^\alpha, y_t)}{\sigma_t^\alpha}\right] = \frac{1}{\sigma_t^\alpha}(\mathbf{1}\{y_t \le b_t^\alpha + \sigma_t^\alpha u^\alpha\} - \alpha)\sigma_t^\alpha$$
$$= \mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u^\alpha\} - \alpha. \tag{18}$$

At a tie use the selected subgradient in (8). The inverse scale in the loss cancels the scale in the inverse score map.

Retaining the lazy gradient update of MultiQT therefore gives

$$\boxed{\tilde{u}_{t+1}^\alpha = \tilde{u}_t^\alpha - \eta g_t^{\mathrm{stu},\alpha}(u_t) = \tilde{u}_t^\alpha - \eta\left(\mathbf{1}\left\{\frac{y_t - b_t^\alpha}{\sigma_t^\alpha} \le u_t^\alpha\right\} - \alpha\right).} \tag{19}$$

Initialize $\tilde{u}_1 = 0$. The score is compared with the *played* threshold; the increment is applied to the *hidden* threshold. The projection that produces u_t is specified next.

	Standard MultiQT	Studentized extension
Observed score	$r_t^{\mathrm{std},\alpha} = y_t - b_t^\alpha$	$r_t^{\mathrm{stu},\alpha} = (y_t - b_t^\alpha)/\sigma_t^\alpha$
Gradient	$\mathbf{1}\{r_t^{\mathrm{std},\alpha} \le \theta_t^\alpha\} - \alpha$	$\mathbf{1}\{r_t^{\mathrm{stu},\alpha} \le u_t^\alpha\} - \alpha$
Update origin	$\tilde{\theta}_t^\alpha$	$\tilde{u}_t^\alpha$
Issued forecast	$b_t^\alpha + \theta_t^\alpha$	$b_t^\alpha + \sigma_t^\alpha u_t^\alpha$

The gradient has the same form because both losses apply quantile loss to a score threshold. The observed scores, state units, and issued forecasts change. An unnormalized outcome-unit loss would instead have gradient $\sigma_t^\alpha(\mathrm{cov}_t^{\mathrm{stu},\alpha} - \alpha)$ with respect to u^α, and would define a different update.

1.1.5 Ordering constraints and projection

The constraint always concerns the outcome forecasts: $q_t^{\alpha_1} \le \cdots \le q_t^{\alpha_m}$. For the ordinary residual score, this requires

$$b_t + \theta \in \mathcal{K} \iff \theta \in C_t^{\mathrm{std}} := \mathcal{K} - b_t, \tag{20}$$
$$\theta^{\alpha_i} - \theta^{\alpha_{i+1}} \le b_t^{\alpha_{i+1}} - b_t^{\alpha_i}, \qquad i = 1, \ldots, m-1. \tag{21}$$

For studentization, $q_t^{\mathrm{stu}} = b_t + A_t u_t$, so

$$\boxed{C_t^{\mathrm{stu}} = \{u \in \mathbb{R}^m : b_t + A_t u \in \mathcal{K}\} = A_t^{-1}(\mathcal{K} - b_t).} \tag{22}$$

This is a closed convex polyhedron given by linear inequalities. It contains zero when $b_t \in \mathcal{K}$. The shift means that, like $\mathcal{K} - b_t$, it is generally not a cone. For unequal scales, $A_t^{-1}\mathcal{K} \neq \mathcal{K}$ in general, so the set cannot be simplified to $\mathcal{K} - A_t^{-1}b_t$. The standard constraint $b_t + \theta \in \mathcal{K}$ is replaced by $b_t + A_t u \in \mathcal{K}$, and the Euclidean projection is well defined:

$$u_t = \Pi_{C_t^{\mathrm{stu}}}(\tilde{u}_t), \qquad q_t^{\mathrm{stu}} = b_t + A_t u_t. \tag{23}$$

Substituting $u = A_t^{-1}(q - b_t)$ into the projection objective yields

$$q_t^{\mathrm{stu}} = \arg\min_{q \in \mathcal{K}} \|A_t^{-1}(q - b_t) - \tilde{u}_t\|^2$$
$$= \arg\min_{q \in \mathcal{K}} \sum_{\alpha \in \mathcal{A}} \frac{(q^\alpha - b_t^\alpha - \sigma_t^\alpha \tilde{u}_t^\alpha)^2}{(\sigma_t^\alpha)^2}. \tag{24}$$

In weighted-projection notation, the exact identity is

$$\boxed{b_t + A_t \Pi_{A_t^{-1}(\mathcal{K}-b_t)}(\tilde{u}_t) = \Pi_{\mathcal{K}}^{A_t^{-2}}(b_t + A_t \tilde{u}_t).} \tag{25}$$

Thus the implementation is weighted isotonic regression of $b_t^\alpha + \sigma_t^\alpha \tilde{u}_t^\alpha$ with weights $1/(\sigma_t^\alpha)^2$.

The ordinary residual case has forecast-space metric I; the studentized case has metric A_t^{-2}. These weights follow from the chosen Euclidean geometry of the tracked score coordinates.

1.1.6 Common and constant scales

For $S_t^\alpha(y) = (y - b_t^\alpha)/\sigma_t$, we have $A_t = \sigma_t I$. Positive scalar multiplication preserves $\mathcal{K}$:

$$C_t^{\text{stu}} = \mathcal{K} - b_t/\sigma_t, \qquad q_t^{\text{stu}} = \Pi_{\mathcal{K}}(b_t + \sigma_t \tilde{u}_t). \tag{26}$$

All projection weights are equal and cancel. Equivalently, run standard MultiQT on base b_t/σ_t and outcome y_t/σ_t, then multiply the issued forecast by σ_t.

If $\sigma_t = \sigma$ is constant, standard MultiQT on the original inputs is also equivalent after converting the state and learning rate:

$$\tilde{\theta}_t = \sigma \tilde{u}_t, \qquad \eta_{\text{std}} = \sigma \eta_{\text{stu}}. \tag{27}$$

1.2 Questions and ideas

- What other scores would be reasonable to consider, i.e. what property of the base forecast do I need to recalibrate? Instead of asking for 1d PIT calibration, could I ask for 1d hdr-calibration? With studentized scores, or in general strictly increasing and invertible scores, the coverage indicator and calibration target do not change

2 Coding Updates

2.1 Data, feedback, tuning and evaluation

2.1.1 Data preparation and the longer-history design

`R/00_base_forecasts.R` now separates downloading and generating base forecasts from case preparation in `R/01_data_overview.R`. Inputs are the CMU–TimeSeries hub quantiles and the NHSN archive of dated report versions. Prepared cases retain the full chronology, actual generation cutoffs where known, report revisions, feedback schedules for delays 0–4, populations, and flags for filled forecasts. Newly generated data, tuned settings and scores carry the protocol tag `latest-at-use-v2`; readers reject mismatched or missing tags.

Active and extended designs. The active `hub_first_months` design uses hub forecasts for both diseases from 23 November 2024. Periods follow the *target date*: tuning runs through 31 May 2025, and evaluation uses later weeks in the October–September seasons 2024/25 and 2025/26. Later plotted weeks outside those seasons are not part of the reported evaluation. The latest report date in the prepared archive is 30 October 2026; "latest" means latest in this archive, not necessarily never revised again.

The implemented `seasons` alternative starts on 14 October 2023, tunes on 2023/24, and evaluates on 2024/25 and 2025/26. Our COVID base uses the unchanged, pinned `windowed_seasonal` code and pinned packages from Delphi, with production solver `fn`. These forecasts have been generated. Flu still uses hub forecasts: generating its own earlier forecasts requires non-public auxiliary data from Delphi. The extended design has not yet supplied the results reported below.

Report versions begin on 19 November 2024 with a retrospective bulk load. For the earlier 2023/24 experiment, base forecasts therefore use latest counts restricted to the dates then observable. Feedback for target weeks through 27 April 2024 is also the latest count, assigned a synthetic Wednesday release after the target week and marked `synthetic_final`. This does not reconstruct historical reporting revisions. The own horizon -1 forecasts have zero interval width because their target is already a predictor. There are no forecasts across the May–November 2024 reporting gap; stateful methods carry their state across it.

Cleaning and filling. Preparation removes the identified broken submissions (COVID 23 November 2024; COVID horizon -1 on 25 April 2026; US flu on 14 and 21 December 2024 and 4 January 2025), and all-zero submissions across every location and quantile of a disease–round–horizon. Forecasts of the incomplete-reporting target weeks 4 May–2 November 2024 are excluded from learning and scoring.

Gaps of at most six missing rounds can be filled using the most recent earlier forecast of the *same target week*: a missing horizon-h forecast at reference date r uses horizon $h + k$ from $r - 7k$ days, with the smallest available $k > 0$. This uses no future forecast or interpolated observation. Horizon 3 has no longer horizon to fill from. Filled rows keep their source date and are scored by default; unsuitable or longer gaps remain missing. Plot lines break across gaps longer than 28 days.

Locations and aggregation. Every disease–location–horizon case learns independently. Scripts 02 and 04 show detailed pages for US, California, Texas, New York, Pennsylvania and Illinois, plus all-location overviews. This fixed list is not the population top five. Script 06 ranks the stored 2019 populations and shows US, California, Texas, Florida, New York and Pennsylvania. The results section focuses on the US and the first four of those states. All-location summaries include 53 locations, including the US; this geographically overlapping set is not a partition of the country and does not define a jointly fitted forecasting method.

2.1.2 Feedback handling: Stateful, re-run and delay

The feedback design is independent of the recalibration method and uses the report and cutoff conventions in Notation.

Stateful (learn once). Wait d calendar weeks from the first valid report. At the next forecast cutoff ι_t with $t > s$, look up the latest report. If it is invalid, wait for a later forecast; never substitute an older report.

$$m_s^{(d)} = \nu_s + 7d \text{ days}, \tag{28}$$

$$\tau_d(s) = \min\{t > s : \iota_t \geq m_s^{(d)},\ y_s^{[\iota_t]} \text{ is valid}\}, \qquad \mathcal{R}_t^{(d)} = \{s : \tau_d(s) = t\}. \tag{29}$$

An empty $\mathcal{R}_t^{(d)}$ means no update. For each released row, use $y_s^{[\iota_t]}$. If a missing forecast postpones the update so that $\iota_t > m_s^{(d)}$, select the report at ι_t, rather than fixing it at $m_s^{(d)}$. Feedback for each q_s is used at most once; revisions after that update are ignored.

For plain MultiQT on counts, start at $\tilde{\theta}_0 = 0$; when learning is active, update using previously issued q_s^α, then issue q_t:

$$\tilde{\theta}_t^\alpha = \tilde{\theta}_{t-1}^\alpha - \eta_t \sum_{s \in \mathcal{R}_t^{(d)}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^\alpha\} - \alpha), \tag{30}$$

$$\theta_t = \Pi_{\mathcal{K}-b_t}(\tilde{\theta}_t), \qquad q_t = b_t + \theta_t. \tag{31}$$

Sum with η_t; keep the hidden state unprojected. For the `log1p` tracker, apply the same update in ψ coordinates. The default retrospective initialization of S_0 is described in Section 2.1.3.

Weekly re-run. At each reference-week index t, re-run MultiQT from the start using the latest reports $y_s^{[\iota_t]}$. Only the new forecast $q_t^{[t]}$ is issued; previously issued forecasts remain unchanged. Starting from $\tilde{\theta}_0^{[t]} = 0$, replay $j = 1, \ldots, t$ with feedback only at $j = \tau_d(s)$, retaining delay d; omit invalid reports:

$$\tilde{\theta}_j^{\alpha,[t]} = \tilde{\theta}_{j-1}^{\alpha,[t]} - \eta_j^{[t]} \sum_{\substack{s \in \mathcal{R}_j^{(d)} \\ y_s^{[\iota_t]} \text{ valid}}} (\mathbf{1}\{y_s^{[\iota_t]} \leq q_s^{\alpha,[t]}\} - \alpha), \tag{32}$$

$$\theta_j^{[t]} = \Pi_{\mathcal{K}-b_j}(\tilde{\theta}_j^{[t]}), \qquad q_j^{[t]} = b_j + \theta_j^{[t]}. \tag{33}$$

For each case and feedback delay, estimate S_0 and the reference expert loss L_0 once from the first 20 feedback rows consumed by the stateful run. Every weekly re-run reuses these constants. With warmup replay enabled, the constants are applied before the 20th feedback row is released, making initialization retrospective. See Section 2.1.3 for S_0 and initialization timing, and Section 2.2.4 for L_0.

Studentized methods use $\tilde{u}$, q^{stu} and projection (23), issuing the base until their scale is available; re-runs reconstruct the scale history. All methods are evaluated against the same latest reported counts, separately from the reports used for learning.

2.1.3 Tuning and initialization

The reference scales are estimated from $I_{20}^{(d)}$ and its stateful feedback values $\bar{y}_s^{(d)}$; these 20 rows need not span 20 calendar weeks. Method parameters, such as a rate multiplier or an expert-weighting lookback, are selected on a tuning period. These are distinct steps in the implementation. The expert reference loss L_0 and its effect on weight updates are described in Section 2.2.4.

Initial error scale S_0. Pool absolute base errors across the reference rows and all quantile levels, and take their empirical 90th percentile:

$$S_0 = \max\Big\{\varepsilon_S, Q_{0.9}\Big(\Big\{|\psi(b_s^\alpha) - \psi(\bar{y}_s^{(d)})| : s \in I_{20}^{(d)},\ \alpha \in \mathcal{A}\Big\}\Big)\Big\}. \tag{34}$$

Here $\varepsilon_S = 10^{-6}$ is a numerical floor. The fixed-rate tracker uses $\eta = \kappa S_0$, measured in tracker units. The studentized rate is instead chosen directly in standardized units. In the code, S_0 also sets the studentized-scale bounds $[0.05S_0, S_0]$.

Retrospective initialization. The default `replay_warmup = TRUE` computes both constants from the eventual first 20 releases, then applies them from the start. Forecasts before the 20th release can therefore depend on information unavailable at their issue time through these constants. This initialization is retrospective even though feedback updates follow the schedule above. With replay disabled, learning waits until the warmup is complete; if fewer than 20 rows are released, learning remains inactive. Once estimated, S_0 and L_0 remain fixed across weekly re-runs. Corrections, expert weights and time-varying studentized scales are still reconstructed in each re-run. The reference scales are computed by `error_scale()` and `weighting_loss_scale()` in `R/03_functions.R`.

Tuning procedure. `R/05_tuning.R` runs COVID–`log1p`, COVID–`identity`, flu–`log1p`, then flu–`identity`. For each pair, it completes stateful delay 0, stateful delay 1, then weekly re-run with delay 0. Parameters are chosen independently for these 12 configurations, with eight stages each (96 stages in the full schedule).

The `log1p` tracker updates on $\log(1 + \text{counts})$ and minimizes quantile loss on that scale. The `identity` tracker updates on raw counts and minimizes quantile loss per 100,000 residents. For each candidate setting, forecasts run online under the chosen feedback design; only tuning weeks are scored, against the latest reports. Mean losses are pooled equally over location–horizon cases to select one value per method and configuration. Choose the smallest loss, with the smaller parameter value at ties. The criterion is not divided by L_0; calibration error is reported separately.

Stage	What is tuned, and what happens next
1	Tune the ordinary trackers and expert mixtures: tracker rates, weighting lookback H, fixed weighting rate for restart experts, and the slow rate of the diverse experts. Search one setting at a time for each method, then retain its best value.
2–7	Tune the soft switch in order: strength, threshold, width, then the forecast-path, nowcast and revision signal weights. Keep each selected value when moving to the next setting, using the expert bank tuned in stage 1.
8	Tune the outer MultiQT correction rate on each already tuned inner mixture. Then reset to defaults for the next disease–tracker–feedback configuration.

All prepared locations contribute. The hub design tunes on horizons $-1, \ldots, 3$ through 31 May 2025; the seasons design uses horizons $0, \ldots, 3$, with -1 reusing those choices. Configurations without usable tuning data skip the remaining stages.

Candidate grids. The following grids are used in `R/05_tuning.R` for each disease–tracker–feedback configuration. Setting names match the code.

Stage	Setting	Candidate values
1	`fixed_multiplier`	0.001, 0.0025, 0.005, 0.01, 0.02, 0.05, 0.1, 0.2, 0.4
1	`slope_min_multiplier`	0.002, 0.004, 0.01, 0.02, 0.05, 0.1
1	`median_rate`	0.003, 0.01, 0.03, 0.1, 0.3, 1
1	`H`	2, 4, 8, 16, 32
1	`restart_lambda`	0.25, 0.5, 1, 2.72, 7, 20
1	`diverse_slow_multiplier`	0.001, 0.002, 0.004, 0.008, 0.02, 0.05, 0.1, 0.2
2	`switch_strength`	0, 0.25, 0.5, 0.75, 1
3	`switch_threshold`	0.02, 0.05, 0.1, 0.2
4	`switch_width`	0.02, 0.05, 0.1, 0.2
5	`switch_weight_path`	0, 0.5, 1
6	`switch_weight_nowcast`	0, 0.5, 1
7	`switch_weight_revision`	0, 0.5, 1
8	`outer_multiplier`	0.001, 0.002, 0.004, 0.01, 0.02, 0.05, 0.1, 0.2

The four `multiplier` grids scale S_0; the slope multiplier controls only the floor term of its capped, slope-dependent rate. `median_rate` is a step in standardized units, H counts weight updates, and `restart_lambda` is a dimensionless rate for losses normalized by L_0. The switch settings control the gate.

Single-pass sequential grid search. At each search, vary one setting and select its lowest-loss candidate, holding all other settings fixed. Retain choices from earlier stages; settings not yet tuned keep their defaults. Stage 1 searches the different methods separately; stages 2–7 successively tune the six settings of the same soft-switch method; stage 8 tunes each outer correction on its already tuned inner mixture.

For the six soft-switch settings, a full Cartesian search would evaluate

$$5 \times 4 \times 4 \times 3 \times 3 \times 3 = 2160$$

joint combinations. The sequential search instead makes

$$5 + 4 + 4 + 3 + 3 + 3 = 22$$

candidate evaluations per location–horizon case, with some configurations possibly evaluated again. These counts cover only stages 2–7; all 12 disease–tracker–feedback configurations are still tuned separately. This is one pass: earlier choices are not revisited after later settings change. It is cheaper, but parameter interactions can make the result depend on the search order, and the best joint combination is not guaranteed.

Tuning diagnostics and boundary selections. `multiqt_tuning.pdf` plots the tuning criterion across each grid, with one page per disease and feedback design. Each panel shows $100(\overline{\mathcal{L}}_j(\vartheta)/\overline{\mathcal{L}}_0 - 1)$: the percentage change in the equally pooled location–horizon mean loss at candidate ϑ, relative to the original forecasts. Negative is better. Solid curves use the criterion for the `log1p` tracker; dashed curves use the criterion for the `identity` tracker, so the two line types do not represent the same loss scale. Circles mark selected values, but the current soft-switch panels can also circle values chosen for another gate parameter; use the tuning CSV to identify the selection for the setting in that panel. These are tuning-period diagnostics, not

evaluation results; later-stage curves are conditional on the settings already selected. The saved `on_grid_edge` flag identifies choices at either end of a grid. Such a choice is the best tested value and can motivate extending the grid; it does not establish an interior optimum. All six pages are loaded at notes page 41.

2.1.4 Evaluation changes: peaks, windows and scores

How peaks are determined. `find_peaks()` in script 01 uses local maxima of the latest counts, rather than one selected winter or summer maximum. After removing nonfinite values and the incomplete-reporting interval, it smooths the retained series with

$$z_i = (x_{i-1} + x_i + x_{i+1})/3, \qquad P_i = z_i - \max\left\{ \min_{l_i \le j \le i} z_j, \min_{i \le j \le r_i} z_j \right\}.$$

At an endpoint, the unsmoothed value replaces the unavailable mean. A candidate must be the largest smoothed value within eight retained observations on each side, with the earliest equal maximum preferred, and must satisfy

$$P_i > 0, \qquad P_i \ge .30 z_i, \qquad P_i \ge .10 X.$$

The peak is then placed at the largest raw count within one retained observation of that candidate, earliest at a tie. Only dates between the first and last forecast target dates are kept. The window widths used by `find_peaks()` count retained observations; they equal calendar weeks only where the retained series is consecutive. Peaks can depend on reports and observations after the evaluated forecast. They are hindsight labels, used by the methods only in the explicitly marked target-oracle benchmark, not by the deployable trackers or soft switch.

Before, after and overall. Each target row belongs to its nearest retained peak, so it is counted once even where windows around different peaks overlap. The windows are

$$\begin{aligned} W_{\text{overall}} &= E, \\ W_{\text{before}} &= \{t \in E : -7 \le u_t^{\text{peak}} \le 0\}, \\ W_{\text{after}} &= \{t \in E : 1 \le u_t^{\text{peak}} \le 8\}. \end{aligned} \tag{35}$$

Thus "up to peak" includes the peak week; "after peak" excludes it. The overall score also includes evaluation weeks outside both peak windows. Missing forecasts, missing truth and period boundaries shorten windows; the code reports missing weeks without discarding the peak. It tests period membership on the target row, while its nearest peak can lie in another period. Season-specific scores intersect these sets with that season; the main plots combine the two evaluation seasons.

How the displayed values are obtained. For any one method and case, average over the selected rows and all 23 quantile levels:

$$\mathcal{L}^{\text{nat}}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(q_t^\alpha / c_{\text{scale}}, y_t^\star / c_{\text{scale}}), \tag{36}$$

$$\mathcal{L}^{\log}(W) = \frac{1}{mN} \sum_{t \in W} \sum_{\alpha \in \mathcal{A}} \rho_\alpha(\log(1 + \max\{q_t^\alpha, 0\}), \log(1 + y_t^\star)), \tag{37}$$

$$\text{CE}(W) = \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left| \frac{1}{N} \sum_{t \in W} \mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha \right|. \tag{38}$$

The evaluation columns are `quantile_loss`, `log_quantile_loss` and `coverage_error`, respectively. The tuning tables use `loss` and `log_loss` for the two loss scales, and retain `coverage_error` for CE. Weeks are averaged *before* taking the absolute coverage residual. All feedback designs are scored against the same latest reports, although they learned from different reports. Empty windows have missing scores, not zero. Both losses are saved; the trade-off plots from script 04 and the trees from script 06 display quantile loss per 100,000 for `identity` and quantile loss on the log scale for `log1p`. The fixed-rate exploration in script 02 displays quantile loss per 100,000. Evaluation losses are not normalized by L_0.

Diagnostics at individual quantile levels. `multiqt_by_level.pdf` repeats the fixed-rate trade-off from script 02 for each α separately. Its axes are

$$\mathcal{L}_\alpha^{\text{nat}}(W) = \frac{1}{N}\sum_{t\in W}\rho_\alpha(q_t^\alpha/c_{\text{scale}}, y_t^\star/c_{\text{scale}}),$$
$$\text{CE}_\alpha(W) = \left|\frac{1}{N}\sum_{t\in W}\mathbf{1}\{y_t^\star \le q_t^\alpha\} - \alpha\right|. \tag{39}$$

Each page fixes disease and quantile level, with location rows and horizon columns; phase symbols distinguish overall, up to peak and scores after peak. This reveals levels whose errors are hidden by averaging across quantiles. The source has 23 COVID pages followed by 23 flu pages, in increasing α order. Pages for $\alpha = .1, .5, .9$ from each disease are loaded at notes page 35.

All-location trade-off plots. In `multiqt_peak_reasoning_tradeoff.pdf`, columns are horizons $-1, 0, 1, 2, 3$; row blocks are delay 0, delay 1 and weekly re-run, each with rows for overall, up to peak and after peak. The point for one location is $(\mathcal{L}_{j,c}, \text{CE}_{j,c})$; lower left is better. The first page shows small points for each location and a large point for each method at

$$\left(J^{-1}\sum_c \mathcal{L}_{j,c},\ J^{-1}\sum_c \text{CE}_{j,c}\right).$$

These are equal-weight means of location scores, not scores of averaged forecasts or CE obtained by pooling all coverage indicators first. The first row of `multiqt_error_tradeoff.pdf` from script 02 uses the same distinction for each fixed rate.

Each panel has its own axes. The all-location axes use the 1st–99th percentile spread and the large mean points; extremes can be outside the visible region. The loss range is capped near four times the base loss, with overflowing displayed method points marked `>`. Read the axes when comparing panels. The original is an open circle; mixtures with common weights use triangles; outer corrections use squares or diamonds. Exact forecast duplicates are hidden. Each feedback design uses its own tuned settings, so comparisons include retuning as well as the changed feedback schedule.

Evaluation tree and its aggregation. Script 06 first pools the cases for one disease, then splits by feedback design, then horizon. Each node has one bar chart per window. Its loss reduction and fraction of cases improved are

$$R_j(\mathcal{C}) = 100\left[1 - \frac{\sum_{c\in\mathcal{C}} N_c\mathcal{L}_{j,c}}{\sum_{c\in\mathcal{C}} N_c\mathcal{L}_{0,c}}\right], \tag{40}$$
$$P_j(\mathcal{C}) = J^{-1}\sum_{c\in\mathcal{C}}\mathbf{1}\{\mathcal{L}_{j,c} < \mathcal{L}_{0,c}\}. \tag{41}$$

These quantities are saved as `reduction` and `share_improved` in the evaluation tree table. Positive reduction means improvement. The first expression is a ratio of week-weighted losses, not a mean of percentage gains; the second weights cases equally and counts ties as not improved. Shares are printed only for nodes with at least five cases. The tree does not show CE, and its overlapping horizons and feedback cases are not independent replicates. Panels on one tree page share axes; clipped bars end in `<` or `>`. By contrast, the phase-rate plots from script 02 show percentage *loss change*, $100(\mathcal{L}_j/\mathcal{L}_0 - 1)$: negative is better.

Forecast curves alongside average scores. `multiqt_method_forecasts.pdf` shows each issued quantile ($\alpha = .1, \ldots, .9$) and its weekly quantile loss per 100,000 at a fixed horizon. Gray regions mark tuning, dotted lines mark peaks, and gaps break lines. `multiqt_season_`

`forecasts.pdf` shows selected fixed rates with coverage and losses over time. These sequences differ from `multiqt_base_forecast_paths.pdf`: each curve there is the median trajectory from one submission across horizons -1 to 3, with darker curves issued later. Its flu winter 2024/25 peak is in the tuning period, so that panel is a descriptive diagnostic, not held-out flu evidence. Weight and soft-switch timelines explain changes in the mixtures; they are not additional performance scores.

2.1.5 Single trackers and reference benchmarks

The slope tracker and target-oracle control were already present in the October workflow; the scaling of the slope-dependent rate by S_0 was updated on 5 November. The two Delphi production benchmarks were added on 10 November. They complement the fixed-rate and studentized methods.

Slope-dependent rate: `absolute_slope`. The tracker uses the usual delayed MultiQT update, with rate

$$\eta_t = \min\left\{\kappa_{\max} S_0,\ \kappa_{\min} S_0 + \left|\frac{\beta_t^5 - \beta_{t-1}^5}{\Delta_t}\right|\right\}. \tag{42}$$

The first slope is zero. The default floor multiplier is $\kappa_{\min} = .004$, selected separately by tuning; the cap multiplier stays at $\kappa_{\max} = .08$. The signal compares base medians from successive issue dates at the same horizon, per elapsed calendar week. It uses no future outcomes and differs from the path across horizons within one submission. A larger rate changes new increments; it does not rescale the accumulated offset as studentization does.

Target oracle: `oracle_target`. This hindsight control returns

$$q_t^{\text{oracle}} = \begin{cases} q_t^{\text{fixed}}, & T_t \le p(t), \\ b_t, & T_t > p(t). \end{cases} \tag{43}$$

The rule bypasses correction for every postpeak target, including a target forecast before its peak. The underlying fixed tracker keeps learning from its own forecasts; the substitution changes only the forecast returned by the control. It asks whether bypassing postpeak correction could help with perfect phase information. It is not a deployable peak detector. Setting the rate to zero would instead freeze the accumulated offset and give a different control.

Delphi benchmarks: separate scales and initialization. Both benchmarks below run for both diseases and all feedback designs, independently of the main identity/log tracker setting. Their states start at zero, without an S_0 estimate or a 20-row warmup; their imported constants are excluded from our tuning search. For each benchmark, form the released coverage-error sum in its own units:

$$\epsilon_t^\alpha = \sum_{s \in \mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\chi(\bar{y}_s^{(d)}) \le \xi_s^\alpha\}\right). \tag{44}$$

In a weekly re-run, use the reports and reconstructed historical forecasts of that re-run in this expression, as in the feedback section.

Adaptive fast and slow offsets: `ref_op`. This version of the COVID production configuration of Delphi starts with zero offsets and uses $\chi(x) = \sqrt{x}$ on raw counts. Its rates are

$$\eta_t^F = \max\left\{10^{-3},\ .03Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_{t,20}^{(d)},\ \alpha \in \mathcal{A}\right\}\right)\right\},$$
$$\eta_t^S = .003Q_{.9}\left(\left\{|\sqrt{\bar{y}_s^{(d)}} - \sqrt{b_s^\alpha}| : s \in \mathcal{F}_t^{(d)},\ \alpha \in \mathcal{A}\right\}\right). \tag{45}$$

Both are zero before the first release. The fast rate uses up to 20 recent consumed rows; the slow rate uses all consumed rows. Both pool absolute base errors over levels and are recomputed as feedback arrives. The hidden states and forecast are

$$\tilde{\theta}_t^{F,\alpha} = .9\tilde{\theta}_{t-1}^{F,\alpha} + \eta_t^F \epsilon_t^\alpha, \qquad \tilde{\theta}_t^{S,\alpha} = \tilde{\theta}_{t-1}^{S,\alpha} + \eta_t^S \epsilon_t^\alpha,$$
$$\xi_t = \Pi_{\mathcal{K}}(\sqrt{b_t} + \tilde{\theta}_t^F + \tilde{\theta}_t^S), \qquad q_t^\alpha = (\max\{\xi_t^\alpha, 0\})^2. \tag{46}$$

The fast correction loses 10% each issue round, including rounds without new feedback; the slow correction persists. Decay is per issue round, not raised to the number of weeks in a calendar gap. This combines a temporary response with a persistent correction. Re-runs reconstruct both rates and states using the then-current reports. The rolling scale is separate from our frozen S_0.

Constant square-root rate: `sqrt_constant`. The flu production benchmark of Delphi uses $\chi(x) = \sqrt{x/c_{\text{pop}}}$, with

$$\tilde{\theta}_t^\alpha = \tilde{\theta}_{t-1}^\alpha + .018\epsilon_t^\alpha,$$
$$\xi_t = \Pi_{\mathcal{K}}\left(\sqrt{b_t/c_{\text{pop}}} + \tilde{\theta}_t\right), \qquad q_t^\alpha = c_{\text{pop}}(\max\{\xi_t^\alpha, 0\})^2. \tag{47}$$

The rate .018 is in square-root counts-per-100,000 units. There is no adaptive rate, decay or second state. Delphi selected this constant using flu seasons 2024/25 and 2025/26, which overlap our evaluation seasons. Although we do not retune it, this is not an independently selected held-out benchmark for those seasons.

Zero boundary and plots. Both square-root methods return zero when $\xi_t^\alpha < 0$, but their feedback updates use the untruncated ξ_s^α. At an outcome of zero, learning coverage can therefore differ from coverage of the returned zero-count forecast. This convention is specific to the square-root benchmarks. They appear in the "Single trackers" group of script 04 and in the evaluation tree of script 06 after those outputs are regenerated; their results are absent from the saved method comparisons described in Section 3.

2.1.6 Studentized scores and non-additive updates in the code

The six implemented studentized variants. The implementation applies the studentized construction in tracker units. The centre is either the full base or its median at every level. After processing the feedback available before issue t, it returns

$$q_t = \psi^{-1}\left[\Pi_{\mathcal{K}}^{A_t^{-2}}(a_t + A_t\tilde{u}_t)\right]. \tag{48}$$

For a common scale, the projection is the ordinary isotonic projection. After warmup, the delayed update is applied to the hidden state,

$$\tilde{u}_t^{\alpha} = \tilde{u}_{t-1}^{\alpha} + \eta \sum_{s\in\mathcal{R}_t^{(d)}} \left(\alpha - \mathbf{1}\{\bar{y}_s^{(d)} \le q_s^{\alpha}\}\right). \tag{49}$$

Each report is compared with its own historical issued forecast. Only the forecasts are projected; the hidden state is neither projected nor reset at a peak or season boundary. In a re-run, the reports and internal forecasts in this formula are the ones reconstructed for that re-run. The step η is in standardized units, with default .1 and a separately tuned value for each studentized method.

Scales from recent errors or the current forecast. The residual window contains five consumed reports, not necessarily five consecutive calendar weeks. The default scales are

$$\widehat{\sigma}_{t,\text{med}} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}} (e_s^{.5})^2\right]^{1/2}, \tag{50}$$

$$\widehat{\sigma}_{t,\text{res}}^{\alpha} = \left[\frac{1}{|\mathcal{H}_t^{(d)}|}\sum_{s\in\mathcal{H}_t^{(d)}} (e_s^{\alpha})^2\right]^{1/2}. \tag{51}$$

Both require at least five consumed rows. These are root mean squared errors about zero, not sample standard deviations after subtracting the residual mean. Thus persistent bias also increases the scale. An optional robust estimator replaces RMS by median $|e_s^{\alpha}|/\Phi^{-1}(.75)$; RMS is the current default.

The common scale inferred from the current base is

$$\widehat{\sigma}_{t,\text{base}} = \frac{\beta_t^{.9} - \beta_t^{.1}}{\Phi^{-1}(.9) - \Phi^{-1}(.1)}. \tag{52}$$

The denominator is approximately 2.563. The level-specific base scales are

$$\widehat{\sigma}_{t,\text{pair}}^{\alpha} = \frac{\beta_t^{1-a_\alpha} - \beta_t^{a_\alpha}}{\Phi^{-1}(1-a_\alpha) - \Phi^{-1}(a_\alpha)}. \tag{53}$$

This construction requires a quantile grid symmetric about .5. Base-width scales respond immediately to changes in the issued forecast; residual scales respond only after the corresponding reports arrive.

Every raw scale is first floored at 10^{-6} and, once S_0 exists, clipped to $[.05S_0, S_0]$. Denote the resulting scale by σ with the corresponding subscripts. The lower bound prevents accumulated standardized offsets from having almost no effect for a long stretch; the upper bound limits their subsequent amplification by a large scale. Both bounds are current defaults, rather than consequences of the studentized construction.

Table 1: Studentized methods. Scales refer to the bounded versions above. Initial states are specified before the first update.

Method ID	Centre and scale	Initialization and purpose
`studentized`	$a_t^\alpha = \beta_t^{.5}$; common $\sigma_{t,\text{med}}$	$\tilde{u}_{\text{init}}^\alpha = \Phi^{-1}(\alpha)$. Reconstruct the distribution around the base median using recent errors.
`studentized_base`	$a_t^\alpha = \beta_t^{.5}$; common $\sigma_{t,\text{base}}$	Normal initial state. Reconstruct the distribution from the median and the central width of the current base forecast.
`median_constant`	$a_t^\alpha = \beta_t^{.5}$; common constant σ	Normal initial state; freeze $\sigma_{t,\text{med}}$ at the first issue after both required warmups. Separates replacing the base by its median from subsequently changing the scale.
`studentized_full`	$a_t^\alpha = \beta_t^\alpha$; common $\sigma_{t,\text{base}}$	$\tilde{u}_{\text{init}} = 0$. Keep the full base and rescale its accumulated corrections together.
`studentized_levels`	$a_t^\alpha = \beta_t^\alpha$; $\sigma_{t,\text{pair}}^\alpha$	Zero initial state. Allow the centre and tails of the base forecast to determine different correction scales; weighted isotonic projection.
`studentized_level_rms`	$a_t^\alpha = \beta_t^\alpha$; $\sigma_{t,\text{res}}^\alpha$	Zero initial state. Set the scale at each level from recent errors at that level; weighted isotonic projection.

The normal initial state in the two residual-median methods can instead be $t_5^{-1}(\alpha)$; the base-width median method always uses normal quantiles. A base-width normal reconstruction does not generally reproduce the original forecast: it keeps its median and central width, then imposes a symmetric normal shape in tracker units, subject to the scale bounds. Until a required residual scale is available, the method issues the original full base. Reports consumed during those waiting rounds are not later replayed into the hidden state; reports on those forecasts that arrive after waiting ends are compared with the original forecasts that were issued.

Studentization changes the size of the accumulated correction whenever σ_t changes. It does not merely choose a larger or smaller step for the newest report. For the full-base methods with the `log1p` tracker, in the absence of a crossing,

$$q_t^\alpha = (1 + b_t^\alpha)\exp(\sigma_t^\alpha \tilde{u}_t^\alpha) - 1, \tag{54}$$

so the outcome-scale change is non-additive. With the `identity` tracker, the correction is additive in counts but its multiplier is dynamic. Quantile-specific scales retain the same coverage target; the code does not establish a calibration guarantee for these variants.

2.2 Weighting and adaptive weighting of experts

An expert is a complete forecasting procedure, such as MultiQT with a particular rate or start date. An expert bank is the set of procedures whose forecasts are combined. The mixture updates their weights using quantile losses on released reports. Even a fixed learning rate gives changing weights; adaptive weighting additionally changes the learning rate. Each location–horizon case is learned separately. The formulas below first consider a fixed bank whose experts are all eligible for every update.

2.2.1 Mixtures and feedback losses

With weights specific to each quantile level, issue

$$\hat{q}^{\alpha} = \sum_{k=1}^{K} w_k^{\alpha} f_k^{\alpha}, \qquad w_k^{\alpha} \geq 0, \quad \sum_{k=1}^{K} w_k^{\alpha} = 1, \qquad q = \Pi_{\mathcal{K}}(\hat{q}). \tag{55}$$

Different levels can favor different experts. This can introduce crossings, so the code repairs the mixture by isotonic projection. With common weights, replace w_k^{α} by w_k at every level. If the quantiles of every expert are ordered, the mixture with common weights is already ordered and needs no repair. The forecasts are combined in count units, including when the experts use the `log1p` tracker; the combination is not an average of log-transformed forecasts.

On receiving feedback, score the expert forecasts for the released row:

$$L_{k,n}^{\alpha} = \rho_{\alpha}\left(f_{k,s(n)}^{\alpha}/c_{\text{scale}}, Y_n/c_{\text{scale}}\right), \qquad L_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} L_{k,n}^{\alpha}, \tag{56}$$

$$\ell_{k,n}^{\alpha} = L_{k,n}^{\alpha}/L_0, \qquad \ell_{k,n} = \frac{1}{m}\sum_{\alpha\in\mathcal{A}} \ell_{k,n}^{\alpha}. \tag{57}$$

The reference L_0 is estimated as in Section 2.2.4 and is common to all experts and levels within the case and feedback design. In stateful learning, score forecasts actually issued for that row; a re-run instead scores the forecasts reconstructed inside that re-run. Feedback timing follows Section 2.1.2.

2.2.2 Exponential updates and optional sharing

Before optional sharing, update separately at each level:

$$\bar{w}_{k,n+1}^{\alpha} = \frac{w_{k,n}^{\alpha}\exp(-\lambda_n^{\alpha}\ell_{k,n}^{\alpha})}{\sum_{j=1}^{K} w_{j,n}^{\alpha}\exp(-\lambda_n^{\alpha}\ell_{j,n}^{\alpha})}. \tag{58}$$

With common weights, there is one update using the mean loss:

$$\bar{w}_{k,n+1} = \frac{w_{k,n}\exp(-\lambda_n\ell_{k,n})}{\sum_{j=1}^{K} w_{j,n}\exp(-\lambda_n\ell_{j,n})}. \tag{59}$$

An expert with lower loss gains weight relative to one with higher loss. A larger learning rate gives a stronger response to the same loss gap. The initial weights are uniform. The option $\lambda = 0$ keeps a uniform average of the active experts.

When sharing is enabled, follow the exponential step by

$$w_{k,n+1}^{\alpha} = (1-\gamma)\bar{w}_{k,n+1}^{\alpha} + \gamma/K. \tag{60}$$

The version with common weights again omits α. This gives previously downweighted experts a chance to recover when conditions change. The default is $\gamma = 1/(2H)$; without sharing, $\gamma = 0$. The *Sharing* column of Table 2 refers to this redistribution of weight among experts. The *Level weights* column separately specifies per-quantile or common weights across levels.

2.2.3 Adaptive learning rates

The first H processed updates use λ_{start}. Record the second moment of each update using its weights before the exponential step and sharing:

$$M_n^\alpha = \sum_{k=1}^{K} w_{k,n}^\alpha (\ell_{k,n}^\alpha)^2, \qquad M_n = \sum_{k=1}^{K} w_{k,n} \ell_{k,n}^2. \tag{61}$$

For per-quantile weighting, the rate is

$$\lambda_n^\alpha = \begin{cases} \lambda_{\text{start}}, & 1 \le n \le H, \\ \sqrt{\dfrac{\log(2KH)+1}{\sum_{r=n-H}^{n-1} M_r^\alpha}}, & n > H. \end{cases} \tag{62}$$

Thus update $H+1$ is the first to use an adaptive rate. Only the previous H moments enter, excluding the loss being processed now. If their sum is zero, retain the previous finite rate. Larger recent second moments give a smaller rate, moderating the response to large losses.

With common weights, use (62) without α and with M_r. The loss of each expert is averaged over levels *before* it is squared in (61); the result is not the mean of the per-quantile second moments. Per-quantile weighting maintains a separate weight vector, moment history and adaptive rate for each α. Common weighting maintains one of each and learns a compromise across levels; its weights are not an average of separately learned per-quantile weights.

Later starts and implementation. For restart banks, only experts that issued the released forecast are eligible for its update. Apply the formulas to their weights normalized to sum to one, then restore their original total mass; other experts keep their mass. Use the number of eligible experts in the rate formula. Each new expert receives $1/K$ of the total weight of the enlarged set of active experts, preserving the relative weights of the existing experts. Updates with fewer than two eligible experts do not count toward H. These rules are implemented in `exp_weights()` in `R/03_functions.R`. The formulas describe weighting without the soft switch. The soft switch multiplies each loss by a value determined by the gate when the forecast was issued. Its second moments use those multiplied losses, feedback with a zero multiplier is skipped, and the sharing fraction depends on the same multiplier, as specified below.

2.2.4 Expert loss normalization

Using the quantile loss in (1), define one reference loss per case and delay:

$$L_0 = \max\left\{ \varepsilon_L, \frac{1}{20|\mathcal{A}|} \sum_{s \in I_{20}^{(d)}} \sum_{\alpha \in \mathcal{A}} \rho_\alpha\left(\frac{b_s^\alpha}{c_{\text{scale}}}, \frac{\bar{y}_s^{(d)}}{c_{\text{scale}}} \right) \right\}, \qquad \varepsilon_L = 10^{-6}. \tag{63}$$

Previously, expert weighting used unnormalized quantile losses. Now (57) expresses the loss of every expert relative to this initial base loss. A value of 0.8 means 80% of the initial mean loss of the base forecast, not necessarily a 20% improvement over the base in the current week. Normalized losses can exceed one. These weighting losses use counts divided by c_{scale} even for a `log1p` tracker.

Initial and fixed-rate weight updates. The exponent in an update is $-\lambda L_{k,n}^\alpha / L_0$ for per-quantile weights, or $-\lambda L_{k,n} / L_0$ for common weights. The common divisor preserves expert rankings but makes the response depend on loss gaps relative to initial base performance. Quantile loss scales linearly with a positive change of units; the current loss and L_0 therefore

scale together. The normalized losses and sequence of weights are unchanged, provided the numerical floor is inactive.

This matters throughout a fixed-rate run and during the first H updates of a adaptive-rate run, which use λ_{start}. Those H updates are distinct from the 20 feedback rows used to estimate L_0. For example, two equally weighted experts with losses 80 and 100, $L_0 = 100$ and $\lambda_{\text{start}} = 1$ receive weights about 0.55 and 0.45 before sharing. Using those raw losses with the same numerical starting rate would put almost all weight on the first expert after one update.

Effect on adaptive learning rates. After H updates, the rule in (62) already adapts to loss magnitude. Conditional on identical previous weights and a positive moment sum, normalization gives, separately at each level,

$$\begin{aligned} M_r^{\alpha,\text{norm}} &= M_r^{\alpha,\text{raw}}/L_0^2, & \lambda_n^{\alpha,\text{norm}} &= L_0\lambda_n^{\alpha,\text{raw}}, \\ \lambda_n^{\alpha,\text{norm}}\ell_{k,n}^{\alpha} &= \lambda_n^{\alpha,\text{raw}}L_{k,n}^{\alpha}. \end{aligned} \tag{64}$$

The same identity holds for common weights after omitting α. The reciprocal changes in the adaptive rate and loss cancel, so the exponential update is the same under that comparison. If $L_0 < 1$, normalization increases losses and decreases the adaptive rate; if $L_0 > 1$, it does the reverse. This does not make two runs with the same numerical starting rate identical: normalization can change their first H weight updates, and those weights affect both later second moments and subsequent updates. Equivalent raw and normalized runs require $\lambda_{\text{start}}^{\text{raw}} = \lambda_{\text{start}}^{\text{norm}}/L_0$ and otherwise matching settings. If a moment sum is zero, the code retains the previous rate; equivalence then also requires those retained rates to be scaled alike.

We keep L_0 normalization throughout adaptive-rate runs. It standardizes initialization and keeps stored moments and incoming losses in the same units. Switching it off after H updates without converting the stored moments and rates would make those units inconsistent. For adaptive rates, it is not an additional reduction in update strength once the adaptive rule compensates for it.

Effect across quantile levels and re-runs. The single L_0 is pooled across levels; it is not a separate baseline for each α and does not equalize their typical losses. With per-quantile weights, $\ell_{k,n}^{\alpha}$ drives the update at that level, and its own second moments determine λ_n^{α} after initialization. With common weights, the mean $\ell_{k,n}$ drives one update and one adaptive rate, as described in Section 2.2.3. During a re-run, losses use the current reports and reconstructed expert forecasts, while L_0 stays at its stateful value. Normalization sets the scale of learning; it does not guarantee better accuracy or calibration, and evaluation losses are not divided by L_0.

2.2.5 Expert combinations implemented in the code

Table 2 collects the 12 mixtures and five outer-MultiQT variants selected by `apply_methods()` in `R/03_functions.R` for scripts 04 and 05. Adaptive rates follow (62); fixed rates are tuned before evaluation and then held constant within a run. Both update the expert weights. Equal averaging uses $\lambda = 0$, so weight learning and sharing are inactive. "Per quantile" means a separate weight vector at each level; "common" means the same learned vector at every level.

Table 2: Current expert combinations. Bank names and outer corrections are described below. "Sharing" redistributes mass among experts.

Method ID	Expert bank	Weighting rate	Sharing	Level weights
Rate-grid experts				
`uniform`	Rate grid	0 (equal)	–	Identical
`weighted`	Rate grid	Adaptive	No	Per quantile
`weighted_common`	Rate grid	Adaptive	No	Common
`shared`	Rate grid	Adaptive	Yes	Per quantile
`shared_common`	Rate grid	Adaptive	Yes	Common
Weekly restart banks				
`restart_uniform`	Restart	0 (equal)	–	Identical
`restart_weighted`	Restart	Fixed, tuned	Yes	Per quantile
`restart_grid`	Restart grid	Fixed, tuned	Yes	Per quantile
Diverse experts and soft switch				
`diverse_uniform`	Diverse	0 (equal)	–	Identical
`diverse_shared`	Diverse	Adaptive	Yes	Per quantile
`diverse_shared_common`	Diverse	Adaptive	Yes	Common
`diverse_soft_switch`	Diverse	Adaptive for each vector	Yes	Common for each vector
Mixtures followed by an outer MultiQT				
`shared_outer`	Rate grid	Adaptive	Yes	Per quantile
`shared_common_outer`	Rate grid	Adaptive	Yes	Common
`restart_uniform_outer`	Restart	0 (equal)	–	Identical
`restart_weighted_outer`	Restart	Fixed, tuned	Yes	Per quantile
`restart_grid_outer`	Restart grid	Fixed, tuned	Yes	Per quantile

Expert banks. The following bank sizes and numerical values are the current defaults in `multiqt_settings()` and `method_parameters()`. Tuned values can differ by method, disease, tracker and feedback design.

- **Rate grid:** 11 persistent MultiQT experts, all starting in the first round, with $\eta = \kappa S_0$ and $\kappa \in \{0, .01, .015, .02, .03, .05, .075, .1, .15, .2, .3\}$. The zero-rate expert reproduces the base forecast.
- **Restart and restart grid:** fresh MultiQT copies start at every issue week after warmup; with successful initialization and warmup replay, starts begin in the first round. Restart uses $\kappa = .05$; the grid uses $\kappa \in \{.01, .05, .2\}$ at every start. Each copy starts from the base and learns only from feedback on its own forecasts. These banks grow over time and have no permanent zero-rate base expert.
- **Diverse:** eight members, specified in Table 3 below.

Eight diverse experts and their intended roles. The bank contains the eight procedures in Table 3. All start in the first forecast round. The *Starting forecast* column gives $b_{k,t}$, computed from the current base forecast at each issue; it is not a one-time initialization. Each of the seven experts other than `base` maintains a separate hidden MultiQT offset, initially zero, and learns from its own issued forecasts. For experts other than `base` and `leaky`, the issued forecast is

$$f_{k,t} = \psi^{-1}\Big[\Pi_{\mathcal{K}}(\psi(b_{k,t}) + \tilde{\theta}_{k,t})\Big]. \tag{65}$$

The `leaky` expert applies the additional shift described below before issuance. Every coverage update uses the final issued forecast.

After warmup, each of these seven accumulated offsets follows the delayed coverage update,

$$\tilde{\theta}^{\alpha}_{k,t} = \tilde{\theta}^{\alpha}_{k,t-1} + \eta_{\text{slow}} \sum_{s \in \mathcal{R}^{(d)}_t} (\alpha - \mathbf{1}\{\bar{y}^{(d)}_s \le f^{\alpha}_{k,s}\}).$$

The `base` expert is the exception: it has no learned correction.

Here the workflow default is $\kappa_{\text{slow}} = .008$; tuning can select a different value for each ordinary diverse mixture. The label `slow` identifies this component; its tuned rate need not be smaller than every other tuned rate. Experts are not added or reset at peaks in this bank. During any S_0 warmup, all eight issue the original base.

Growth adjustment. The growth signal uses the available delay-0 report even when the MultiQT update waits an additional week. In the stateful design the value is the report current at the delay-0 update of that row; a re-run replaces it by the current report. Define the implied growth, clipped to $[-1, 1]$,

$$D_t = \min\Big(1, \max\Big[-1, \log(1 + b^{.5}_t) - \log(1 + y_{\text{avail},t})\Big]\Big), \tag{66}$$

with $D_t = 0$ if no such report exists. For damping coefficient d_k,

$$b^{\alpha}_{k,t} = \exp(\max\{0, \log(1 + b^{\alpha}_t) - d_k D_t\}) - 1. \tag{67}$$

Thus positive damping moves the starting median toward the available report, while negative damping extends the implied growth or decline. Every quantile receives the same log adjustment before flooring. When $d_k = 1$ and clipping is inactive, the starting median equals that report. With clipping, this equality need not hold. These adjustments are always defined in $\log(1 + x)$, including when the subsequent tracker operates in counts.

Width adjustment. For factor ω_k, the starting forecast is

$$b^{\alpha}_{k,t} = \psi^{-1}\Big(\max\{0, \beta^{.5}_t + \omega_k(\beta^{\alpha}_t - \beta^{.5}_t)\}\Big). \tag{68}$$

This contracts or expands distances from the base median in tracker units. The starting forecasts are floored at zero counts before their separate slow MultiQT corrections are applied. The factors .75 and 1.5 describe these starting distances; the separate accumulated corrections can change the final interval widths.

Table 3: The eight members of the diverse bank. Intended roles motivate the alternatives; they do not prescribe the learned weights.

Expert	Starting forecast	Intended role
`base`	Original b_t	Return $f_{\text{base},t} = b_t$ without correction.
`slow`	Original b_t	Accumulate a persistent coverage correction.
`leaky`	Original b_t	Add a temporary common log shift that can react to recent coverage errors and then decay.
`damp=-0.5`	(67), $d_k = -.5$	Extend the growth or decline implied by the base relative to the available report.
`damp=0.5`	(67), $d_k = .5$	Partly damp the implied change.
`damp=1`	(67), $d_k = 1$	Remove the implied change up to the clipping limit.
`width=0.75`	(68), $\omega_k = .75$	Offer narrower intervals.
`width=1.5`	(68), $\omega_k = 1.5$	Offer wider intervals.

The leaky expert. Besides its persistent slow offsets, this expert has one common shift on the log scale, initially zero. After warmup, its update is

$$v_t = \text{clip}_{[-.2,.2]}\left[2^{-\Delta_t} v_{t-1} + .2 \sum_{s \in \mathcal{R}_t^{(d)}} \frac{1}{m} \sum_{\alpha \in \mathcal{A}} \left(\alpha - \mathbf{1}\{\bar{y}_s^{(d)} \le f^{\alpha}_{\text{leaky},s}\}\right)\right], \tag{69}$$

The shift halves per calendar week, including across gaps. Each report contributes an average over levels, but multiple reports in a release batch are summed. The cap is applied after the batch. Only this additional shift decays; the ordinary MultiQT offsets of this expert persist.

With the `log1p` tracker, v_t is added to the projected tracker forecast before inversion. With the `identity` tracker, if $q^{\alpha}_{\text{plain},t}$ denotes the ordinary projected count forecast before this shift, the exact implemented map is

$$f^{\alpha}_{\text{leaky},t} = (1 + (q^{\alpha}_{\text{plain},t})_+)e^{v_t} - 1 + \min(q^{\alpha}_{\text{plain},t}, 0). \tag{70}$$

For nonnegative $q^{\alpha}_{\text{plain},t}$, this is the same log-shift as above. The additional term gives the extension to negative count forecasts used in the code; final MultiQT forecasts are not clipped for these trackers.

How phase information enters. `diverse_soft_switch` learns two weight vectors over the same eight diverse experts, named `baseline` and `declining` in the code. Each vector is common across quantile levels. At issuance, it combines them as

$$w^{\text{mix}}_{k,t} = (1 - G_t) w^{B}_{k,t} + G_t w^{D}_{k,t}. \tag{71}$$

The gate uses information available at that issue cutoff: the shape of the base forecast path, changes in nowcasts, and changes between forecasts for the same target. It describes declining context rather than a known peak date or a probability of having passed the peak. It uses $\log(1 + \text{base median})$ in counts for either tracker, never observed outcomes or hindsight peak dates. Exact reference and target dates must match; filled forecasts and forecasts unavailable at the relevant cutoff are excluded.

The three soft-switch signals. The forecast-path signal is the largest decrease between adjacent horizons in the current submission, rather than its overall slope:

$$x_{t,P} = \max_{\substack{h=-1,\ldots,2 \\ Z_{t,h}, Z_{t,h+1} \text{ available}}} (Z_{t,h} - Z_{t,h+1}). \tag{72}$$

All four horizons $0, \ldots, 3$ are required; the decrease from horizon -1 to 0 is included only when available. The nowcast signal is

$$\begin{aligned} g_t &= Z_{t,-1} - Z_t^-, \qquad g_t^- = Z_t^- - Z_t^{--}, \\ x_{t,N} &= \max\{-g_t,\ \min(g_t^-, g_t^- - g_t)\}. \end{aligned} \tag{73}$$

It detects falling nowcasts or a slowdown from positive growth. Both growth values must be available. The current horizon--1 nowcast is issued at t, although its target is the preceding week. Historical values use the cutoff of the preceding nowcast ι_t^-; if that cutoff is missing or later than ι_t, this signal and the revision signal below are missing.

For $j = 0, \ldots, 3$, form the paired revisions

$$\begin{aligned} r_{t,j} &= Z_{t,-1} - Z(t-j-1, j; \iota_t), \\ r_{t,j}^- &= Z_t^- - Z(t-j-2, j; \iota_t^-), \\ x_{t,R} &= \min\Big\{-\operatorname{median}_{j\in J_t} r_{t,j},\ -\operatorname{median}_{j\in J_t}(r_{t,j} - r_{t,j}^-)\Big\}. \end{aligned} \tag{74}$$

Each difference compares forecasts for the same target week; the current and previous revision pairs concern consecutive targets. At least two paired values are required. The signal is positive when revisions are downward and have become more downward. The second term is a median of paired changes, not a difference of medians. These are revisions of forecasts, not revisions of hospital reports.

From signals to a gate. For $j \in \{P, N, R\}$, convert each signal to a bounded score:

$$\begin{aligned} p_{t,j} &= \left[1 + \exp\left\{-\frac{x_{t,j} - \log(1+\tau_G)}{\log(1+w_G)}\right\}\right]^{-1}, \\ \bar{p}_t &= \frac{\sum_j \upsilon_j p_{t,j}}{\sum_j \upsilon_j}, \qquad G_t = \operatorname{clip}_{[.05,.95]}\left\{\tfrac{1}{2} + s_G(\bar{p}_t - \tfrac{1}{2})\right\}. \end{aligned} \tag{75}$$

A missing signal gets score .5 and retains its weight in the denominator, pulling towards neutrality. If all signal weights are zero, set $\bar{p}_t = .5$. Defaults before tuning are $s_G = 1$, $\tau_G = w_G = .05$ and three weights equal to one. The threshold locates the transition, the width controls its smoothness, and strength shrinks the result towards .5 before clipping. The tuning section gives their sequential search grids.

Context-specific loss updates and sharing. When feedback for row $s(n)$ arrives, use its saved issuance-time gate:

$$a_n^B = 2(1 - G_{s(n)}), \qquad a_n^D = 2G_{s(n)}.$$

For either weight vector $C \in \{B, D\}$, replace $\ell_{k,n}$ in its exponential update by $a_n^C \ell_{k,n}$ and use the moment

$$M_n^C = \sum_k w_{k,n}^C (a_n^C \ell_{k,n})^2. \tag{76}$$

Thus each vector has a separate adaptive rate based on the multiplied losses. Its sharing step uses

$$\gamma_n^C = 1 - (1-\gamma)^{a_n^C}, \qquad w_{k,n+1}^C = (1-\gamma_n^C)\bar{w}_{k,n+1}^C + \gamma_n^C / K. \tag{77}$$

The gate affects both the loss multiplier and the amount shared uniformly. With strength zero, $G_t = .5$, both multipliers are one, and the method equals `diverse_shared_common` when their settings match. By default, the soft switch uses the expert parameters tuned for `diverse_shared_common`.

Outer corrections and the restart control. Each `_outer` method takes the mixture with the same name without that suffix and adds a further MultiQT correction. The table describes its inner mixture. The outer tracker learns from its own final forecasts, with continuous offsets even when new restart experts join. Its default rate is $.004S_0$, tuned after the inner mixture. The related `restart_persistent` control keeps only the first expert with $\kappa = .05$ from the restart construction; it is a single tracker, not an expert combination.

Which settings are tuned. The four adaptive mixtures of rate-grid experts tune H. The two learned restart mixtures tune their fixed `restart_lambda`; the default is 2.72. The ordinary diverse mixtures tune their slow tracker rate. Soft-switch tuning selects gate strength, threshold, width and three signal weights in sequence. Outer methods tune their added correction rate. Adaptive weighting starts at $\lambda_{\text{start}} = 2.72$ with default $H = 8$ unless overridden; sharing uses $\gamma = 1/(2H)$. Equal weights remain identical across levels even though the code stores them in per-quantile columns. All combinations use the same loss and feedback conventions described above.

3 Results and where to see them

All numerical comparisons here use the hub design, identity tracking, latest reported counts and quantile loss per 100,000 residents. Except for the delay comparison, they use stateful delay 0. "25 cases" means US, California, Texas, Florida and New York, each at horizons -1 to 3. A quoted median percentage change is the median of the 25 paired relative loss changes: negative is better. It is not the pooled reduction shown by the tree.

3.1 Effect of increasing feedback delay

Provisional finding: an extra week does not have a uniform effect, and loss and CE can move in opposite directions. For COVID, delay 1 versus delay 0 gives the following overall comparison:

Method	Median loss change	Lower loss	Lower CE
`fixed`	−0.15%	20/25	24/25
`shared`	+5.08%	4/25	22/25
`diverse_shared`	+2.97%	8/25	18/25

For flu, `shared` also has higher median loss (+3.72%), despite lower CE in 16/25 cases. These comparisons include separate tuning for each feedback design. Waiting changes both feedback freshness and the report learned from; it does not simply remove one update. The available script 02 output with `latest-at-use-v2` uses delay 0 only, so a refreshed delay comparison remains pending.

Plots: compare the delay-0 and delay-1 row blocks in `multiqt_peak_reasoning_tradeoff.pdf`, especially pages 1 (all locations) and 2 (US), loaded from notes page 47. The tree separates delay 0, delay 1 and re-run; re-running is a separate design, not a larger delay.

3.2 US and the four largest states

The requested states are California, Texas, Florida and New York. **Provisional results vary by location and method.** At horizon 2, delay 0, the overall COVID loss changes against the original are:

Location	studentized_base	diverse_soft_switch
US	−4.24%	+3.52%
California	−1.51%	+2.34%
Texas	−3.04%	+3.76%
Florida	−2.94%	+7.22%
New York	−5.62%	−1.46%

Thus an all-location summary can hide a method helping New York while hurting other large locations. Each row here contains 66 evaluated target weeks; the matching flu cases have 32.

Plots: tree source pages 2–6 show US, California, Texas, Florida and New York, loaded after the pooled page at notes page 49. The detailed trade-off pages are US 2, California 3, Texas 4 and New York 5. Florida has all-location scores and a tree page, but no saved individual forecast/trade-off page. The tree displays selected methods and omits `studentized_base`; use the trade-off and studentized forecast pages for that method.

3.3 Extending the data history: results pending

The longer-history design is specified in Section 2.1.1, but the current figures still use the hub design.

3.4 Adapted trade-offs and forecast curves

Evidence from `latest-at-use-v2`: a rate that helps up to the peak can hurt after it. For COVID with $\kappa = .03$, loss up to peak improves in 20/25 cases (median change −23.25%), but loss after peak worsens in 23/25 (median change +17.41%). Overall, only 9/25 improve and the median change is +3.78%. For the US at horizon 2, the respective loss changes are −10.55%, +27.87% and +17.62%. This is the observed motivation for adapting corrections to changing conditions; it does not show that the proposed soft switch solves it.

Better empirical calibration need not mean lower loss. At $\kappa = .3$, overall loss worsens in all 25 cases for both diseases, while CE improves in 18/25 COVID and 25/25 flu cases. A smaller flu rate $\kappa = .01$ improves overall loss in 17/25 cases (median −1.51%) and CE in 24/25. The all-location clouds reveal variation around these summaries; their large points alone do not identify which places improved.

Plots: `multiqt_phase_rate_tradeoffs.pdf`, source page 1 (notes page 33), separates the three windows for each horizon and both diseases. Read it with the US `multiqt_season_forecasts.pdf` page 1 (notes page 34) to see the actual forecast levels and weekly losses. The base forecast paths page (notes page 32) shows trajectories across horizons from one issue date. Those trajectories and the fixed-horizon forecast sequences answer different questions.

3.5 Studentized scores and non-additive updates

Provisional finding: the choice of centre and scale matters more than the label "studentized". Overall loss changes against the original, with the number of cases with lower loss in parentheses, are:

Method	COVID	Flu
`studentized`	+6.61% (2/25)	+11.18% (6/25)
`studentized_base`	−2.96% (24/25)	+2.21% (12/25)
`median_constant`	+13.68% (1/25)	+27.31% (1/25)
`studentized_full`	+0.15% (9/25)	−0.19% (13/25)
`studentized_levels`	+0.10% (9/25)	−0.59% (15/25)
`studentized_level_rms`	−0.04% (15/25)	−1.58% (18/25)

For COVID, `studentized_base` improves loss after peak in all 25 cases (median −7.57%), but loss up to peak in only 9/25 (median +2.39%). Its overall CE improves in just 5/25, despite its 24 cases with lower loss. Flu reverses the phase pattern: median change up to peak −9.64%, after peak +7.46%, and CE improves overall in all 25 despite mixed loss. These results do not establish a general advantage of studentization or of non-additive updates; the saved comparisons here use the identity tracker, not the `log1p` tracker.

Plots: `multiqt_method_forecasts.pdf`, US pages 5 (median variants) and 6 (full-base variants), loaded from notes page 55. The weekly losses below each quantile show when a method has higher loss despite a smoother forecast curve, and when `studentized` has large changes at lower quantile levels. Compare the rows labelled "Up to peak and "After peak on trade-off source page 2 for the corresponding average scores. These saved detailed plots show COVID only.

3.6 The diverse expert bank and soft switch

Provisional finding: the diverse experts can help, but the soft switch gives small, mixed gains. Compared with `shared`, `diverse_shared` has lower overall loss in 13/25 COVID cases (median change −3.85%) and 20/25 flu cases (−4.96%). This compares two tuned banks, not a controlled change in bank size alone.

Compared with `diverse_shared_common`, `diverse_soft_switch` has lower loss in 15/25 COVID cases (median −0.17%) and 11/25 flu cases (+0.03%); CE improves in only 7/25 and 5/25. The two weight vectors therefore do not consistently improve on a single vector of common weights. They use the same eight experts, not distinct banks selected using the true pre-/post-peak labels. Intended roles in Table 3 should not be read as demonstrated phase-specific superiority.

Plots: US method-forecast source page 4 (loaded from notes page 55) compares the diverse combinations. The soft-switch timeline, source page 1 (notes page 61), shows the gate, the two learned weight vectors and the issued mixture. It diagnoses how switching works; scoring evidence comes from the trade-off points and score tables.

3.7 Weights shared across quantile levels

Provisional finding: common weights are competitive, but the effect depends on the bank. Paired overall loss changes when replacing per-quantile weights with common weights are:

Bank and weighting	COVID	Flu
Rate grid, no sharing	−1.18% (18/25)	−0.26% (13/25)
Rate grid, sharing	−1.18% (14/25)	−1.20% (18/25)
Diverse bank, sharing	+0.36% (10/25)	−0.50% (14/25)

Parentheses again count lower-loss cases. Each variant uses its own tuned settings; for the diverse bank this includes different slow rates, so the comparison does not isolate the choice between common and per-quantile weights. For rate-grid mixtures with sharing, common weights also reduce COVID CE in 21/25 cases, whereas diverse common weights improve CE in only 10/25. Common weights give a coherent mixture across all quantiles and avoid crossings between ordered experts, but that structural property does not guarantee a lower quantile loss.

Plots: `multiqt_weight_timelines.pdf`, US source pages 1–2 (notes page 59), contrasts the individual quantile rows with the bottom row with common weights. Method-forecast source page 2 compares rate-grid forecast curves; the trade-off pages use triangles for common weights and circles for per-quantile weights.

3.8 Loaded figures and updating them

Source PDF in `output/pdf/`	Source pages	Notes start
`multiqt_base_forecast_paths.pdf`	1	32
`multiqt_phase_rate_tradeoffs.pdf`	1 (US)	33
`multiqt_season_forecasts.pdf`	1 (US)	34
`multiqt_by_level.pdf`	4, 12, 20, 27, 35, 43	35
`multiqt_tuning.pdf`	1–6	41
`multiqt_peak_reasoning_tradeoff.pdf`	1–2	47
`multiqt_evaluation_tree.pdf`	1–6	49
`multiqt_method_forecasts.pdf`	2, 4, 5, 6 (US)	55
`multiqt_weight_timelines.pdf`	1–2 (US)	59
`multiqt_soft_switch_timelines.pdf`	1 (US)	61

The loaded per-quantile pages show $\alpha = .1, .5, .9$ for COVID, then flu. Tuning pages are COVID delay 0, delay 1 and re-run, followed by the same three flu configurations. The tuning PDF was regenerated on 10 November with protocol `latest-at-use-v2`; its curves concern the tuning sample and do not replace the provisional method-evaluation results above. Rebuilding the notes reloads these source PDFs; numerical result statements require a separate check when the evaluation outputs change.

References

[1] Tiffany Ding, Isaac Gibbs, and Ryan J. Tibshirani. *Calibrated Multi-Level Quantile Forecasting.* arXiv:2512.23671v1, 2025. MultiQT paper.

[2] Anastasios N. Angelopoulos, Emmanuel J. Candès, and Ryan J. Tibshirani. *Conformal PID Control for Time Series Prediction.* Advances in Neural Information Processing Systems 36, 2023. NeurIPS paper.

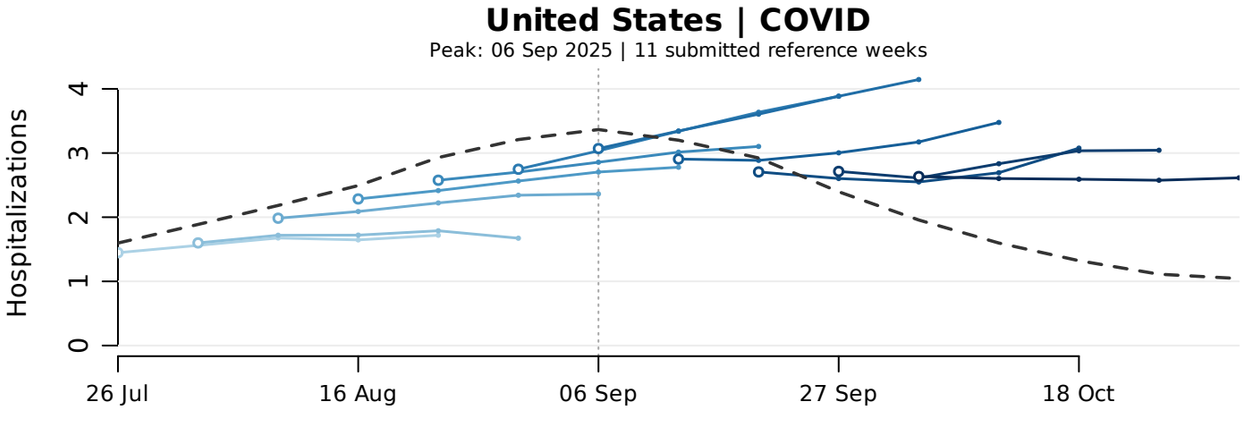
[3] CMU Delphi. *Delphi hub forecasting: calibration experiments and tracker code.* Pinned source revision `9abebcfb`, inspected 9 November 2026. E09: constant learning rate and revision delay; tracker implementation.

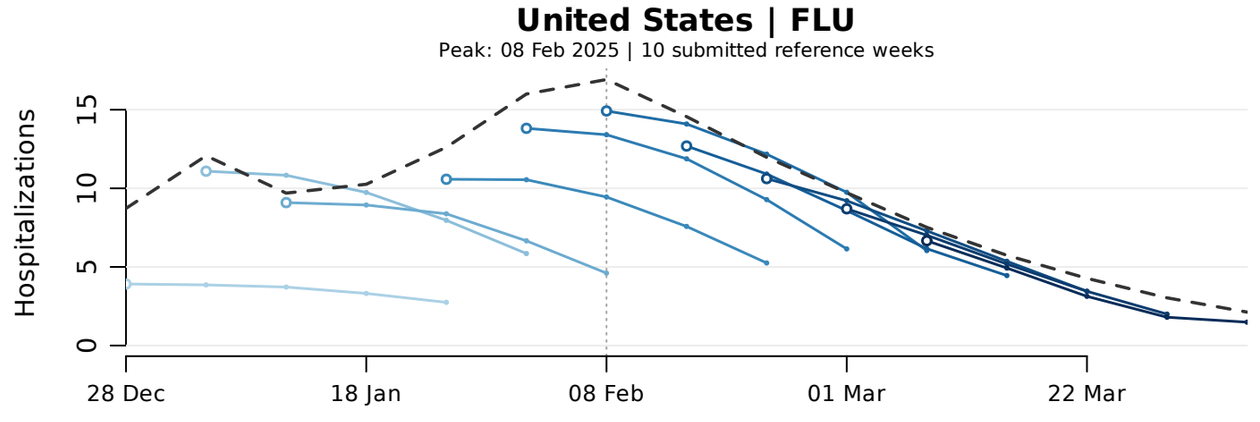
A Loaded plot pages

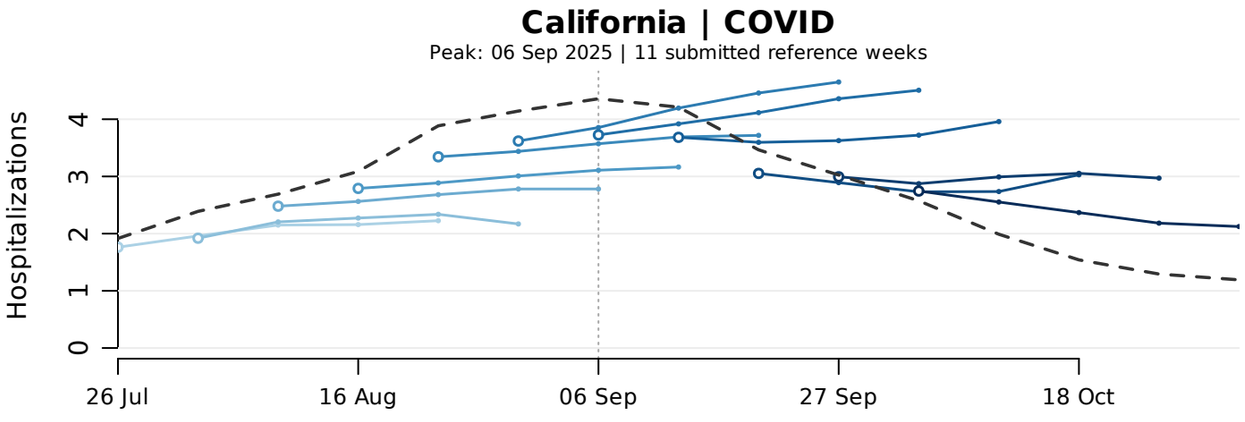
CMU base forecasts: five-horizon median curves around the selected peaks

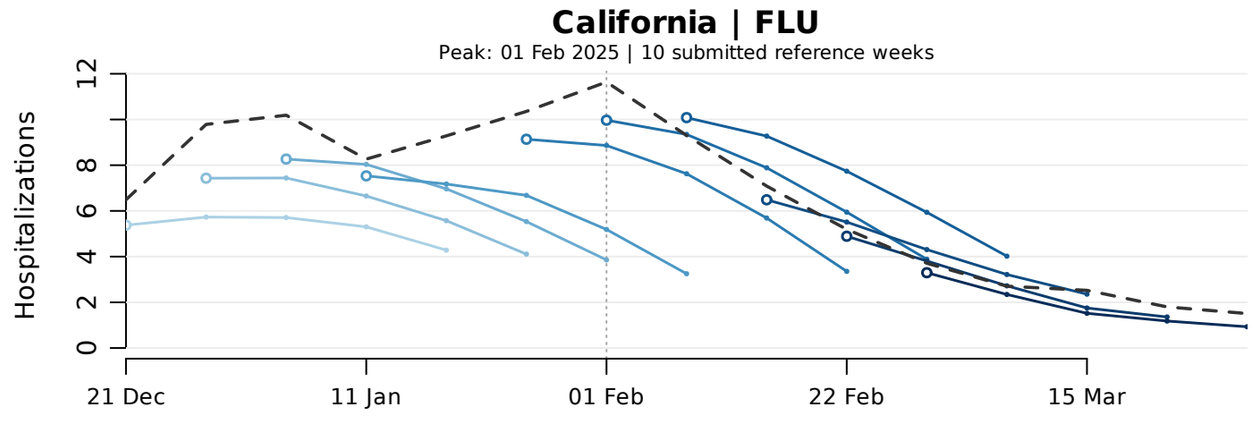
COVID 2024/25 summer | FLU 2024/25 winter | counts per 100k residents

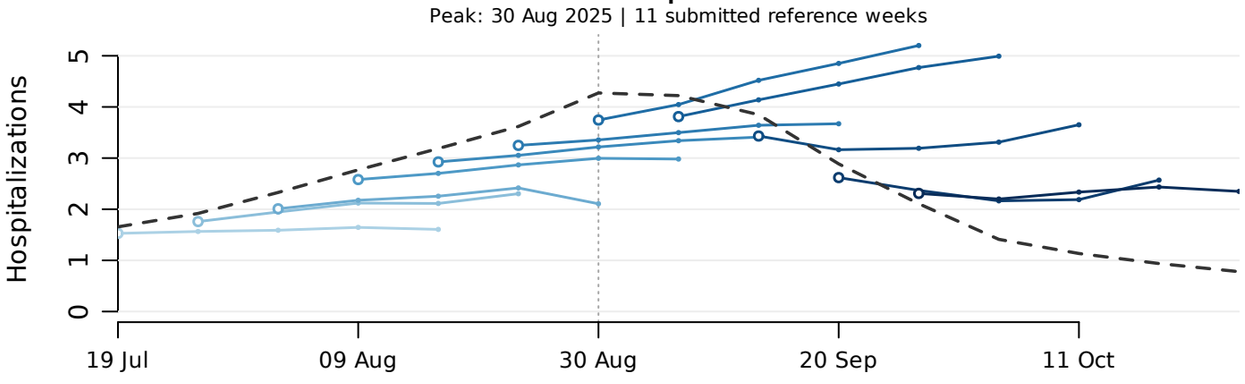
Each curve: one reference week, horizons -1 to 3. Darker = later reference week relative to the local peak.

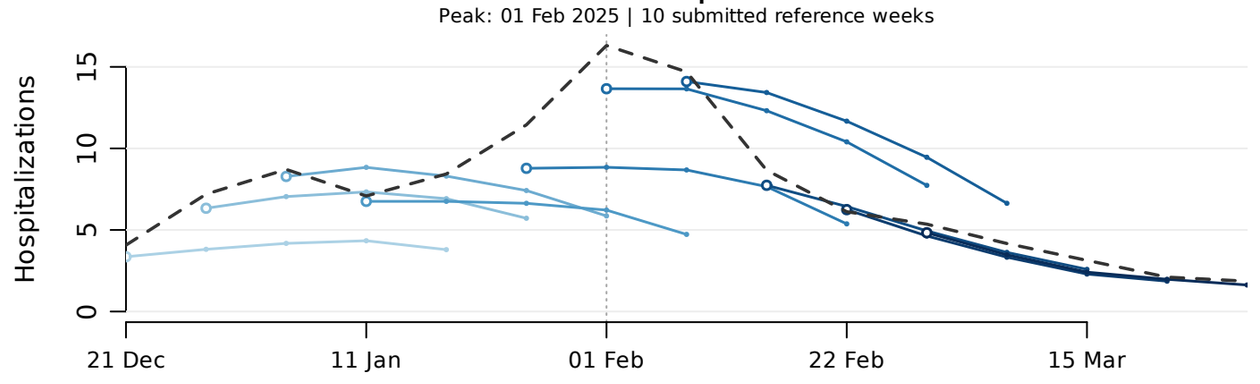


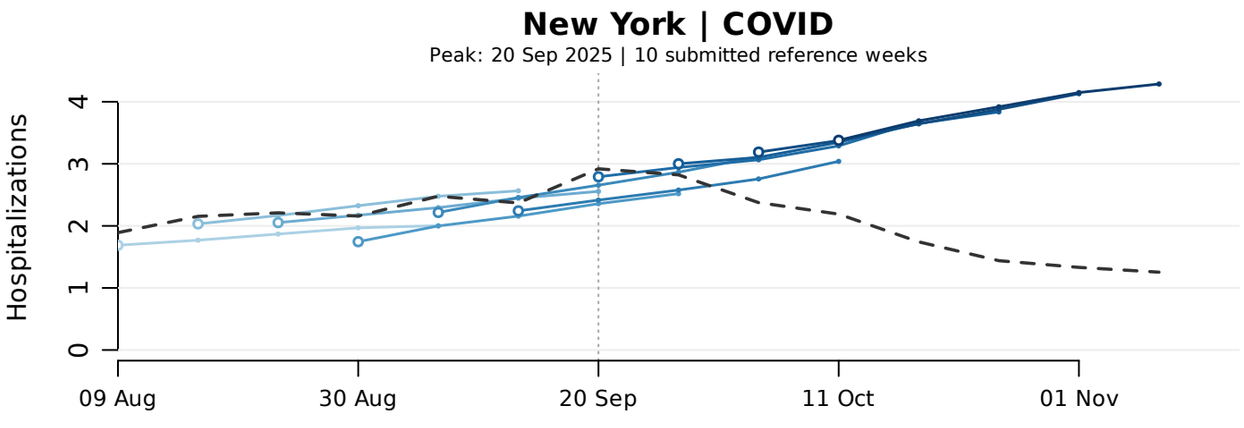


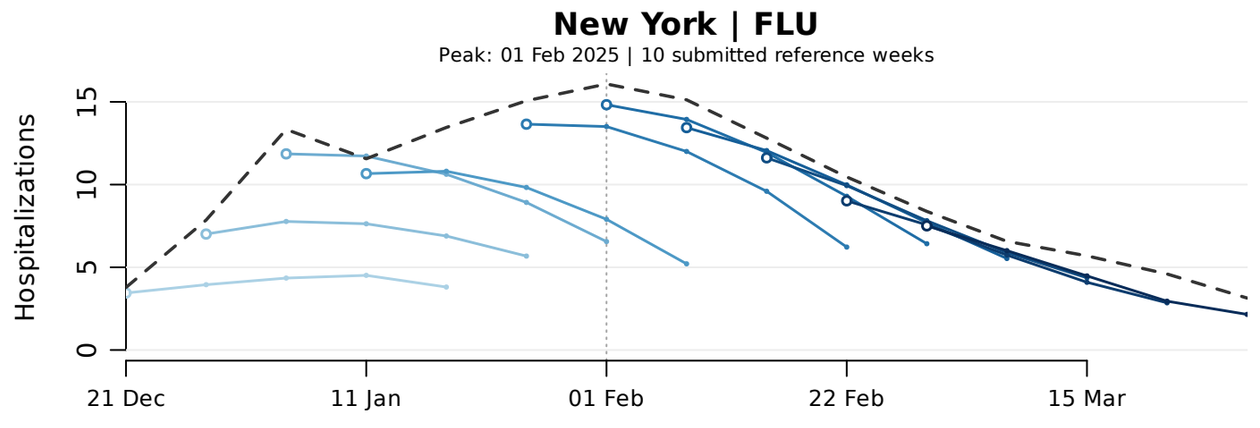


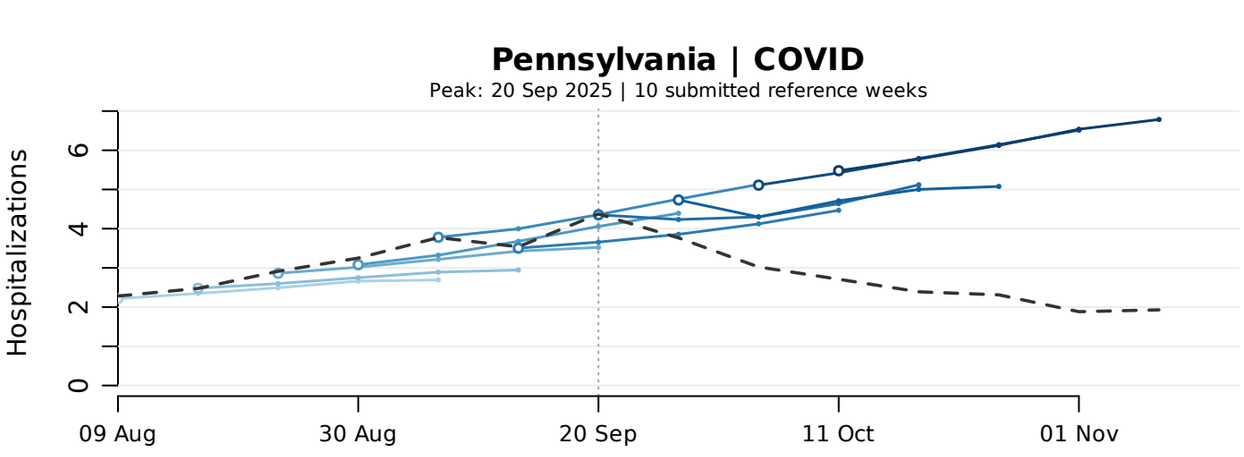


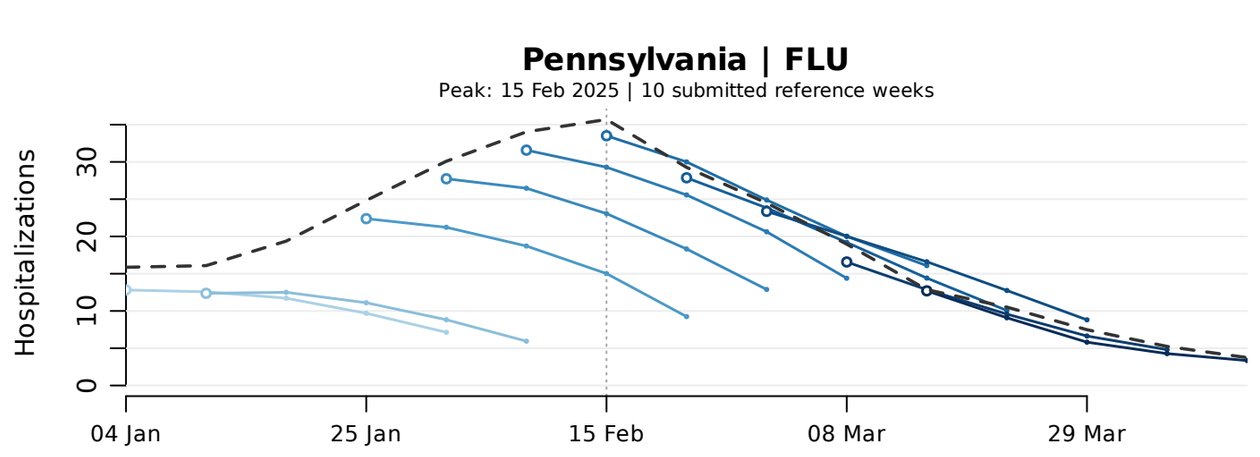


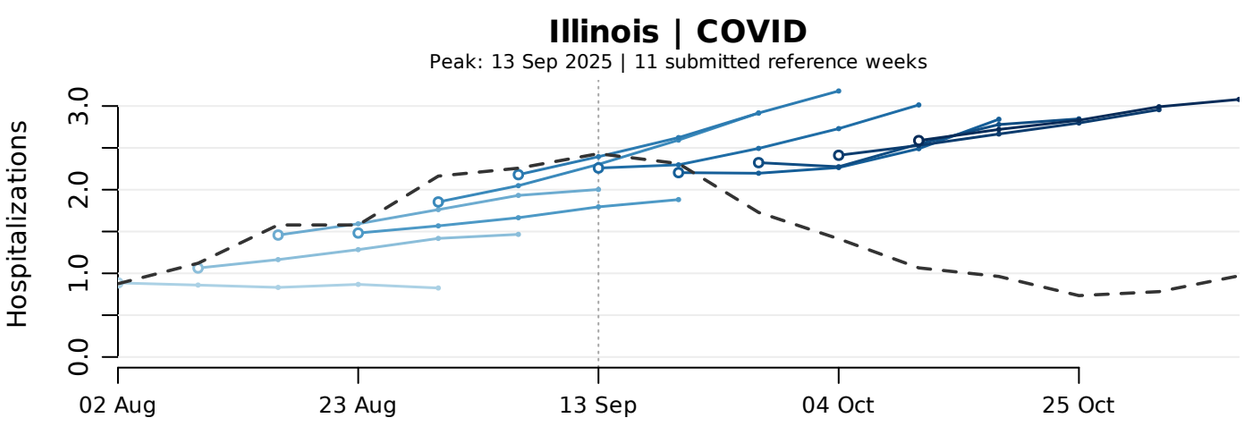


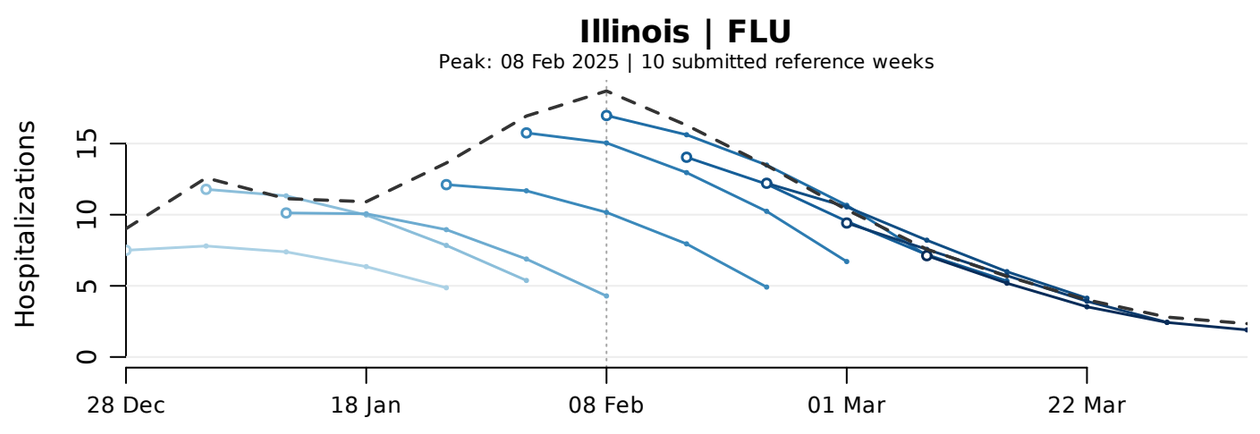


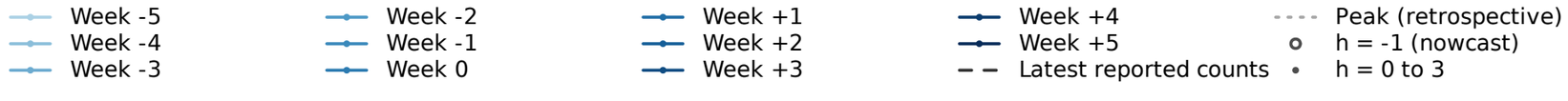










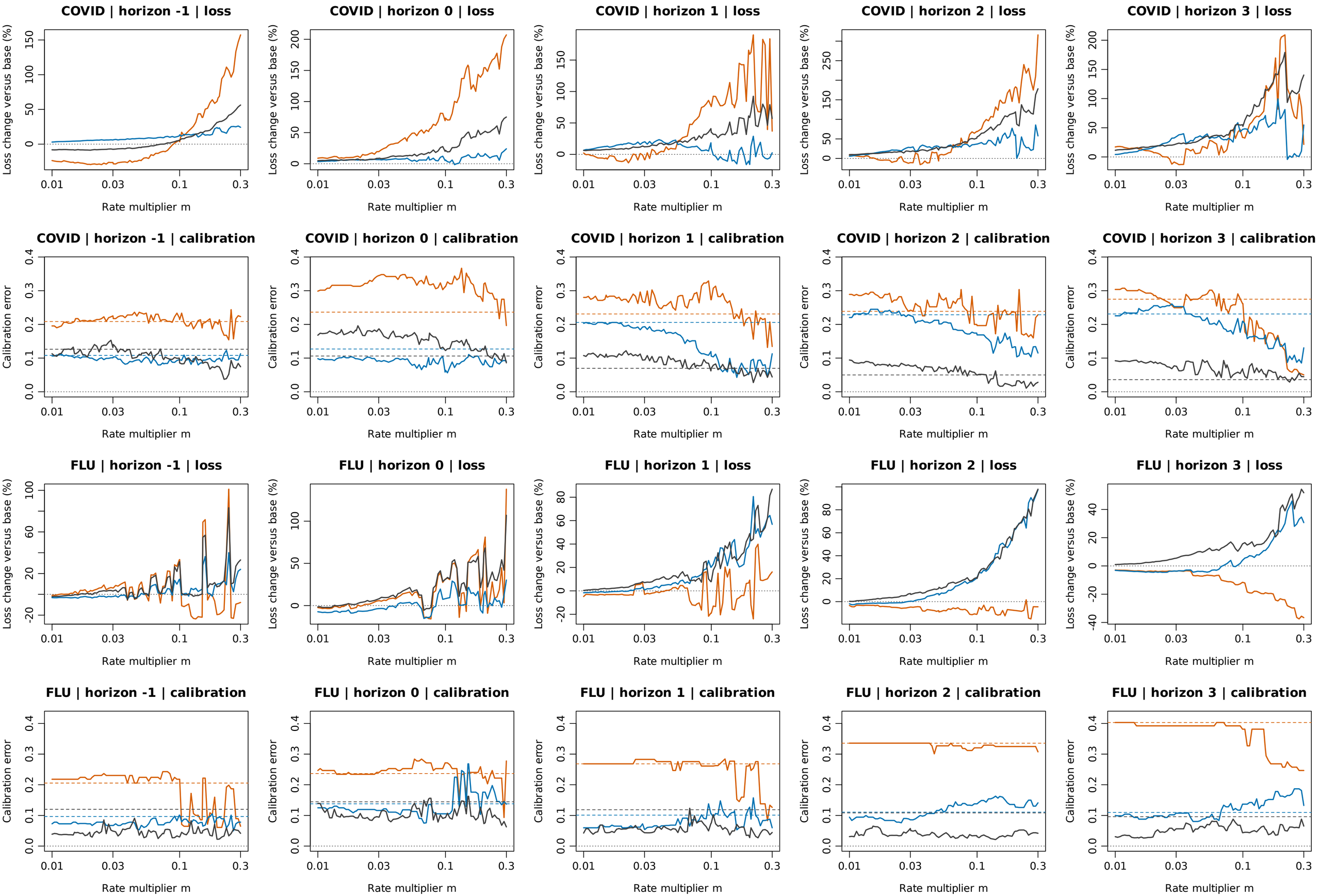


Horizontal axis: target week ending. Target date = reference date + 7h days. Raw CMU submissions; missing rounds stay missing.

Observed counts and peak dates use the latest stored reports and are retrospective. Reference dates label weeks, not exact submission times.

United States | Fixed-rate MultiQT: loss and calibration before and after the peak

Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents

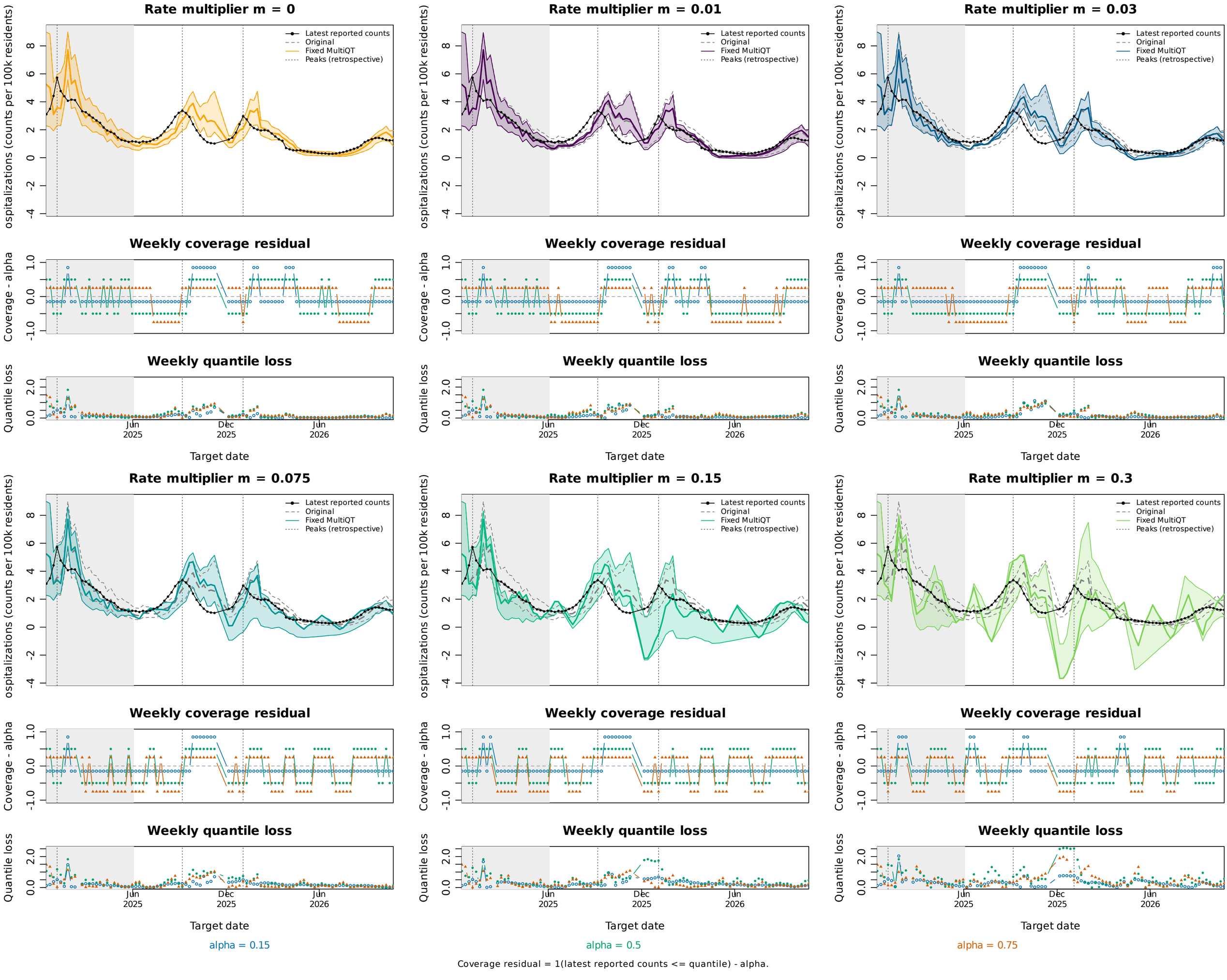


Before peak (weeks -7 to 0) — After peak (weeks 1 to 8) — Overall (Jun 2025 to the end of season 2025/26)

Loss: % change from m = 0 in the same phase (negative is better). Calibration: mean |coverage - alpha| over levels (lower is better); dashed: m = 0.

covid | United States (US) | horizon 2 | 2024-12-14 to 2026-10-24

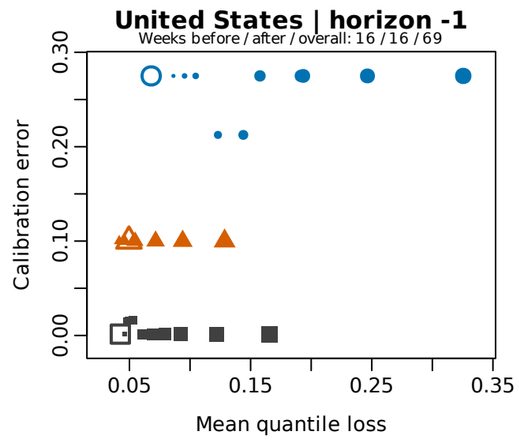
Count tracking | eta = m * S0, S0 = 17100 | Scores/plots: counts per 100k residents | Dotted lines: peaks (local maxima of the latest counts) | Gray: tuning period (not scored)

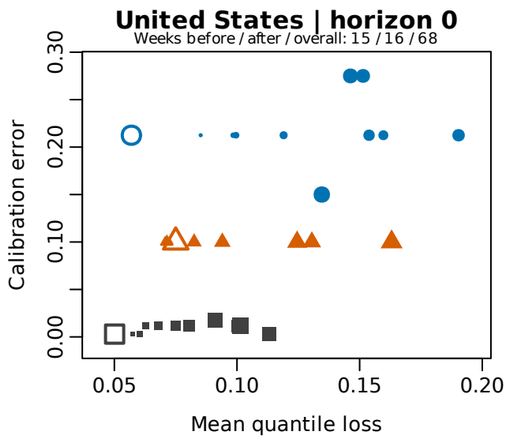


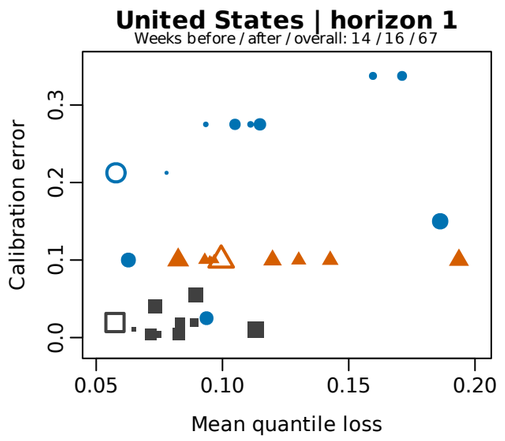
Coverage residual = 1(latest reported counts <= quantile) - alpha.

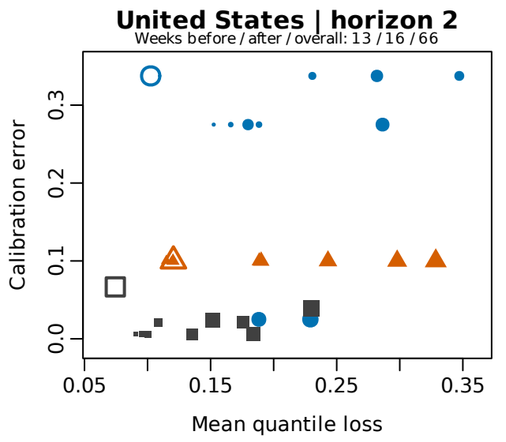
COVID | Fixed-rate MultiQT at quantile level alpha = 0.1: |coverage - alpha| vs quantile loss

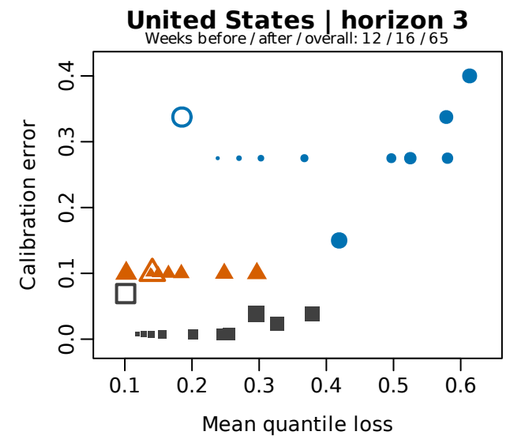
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

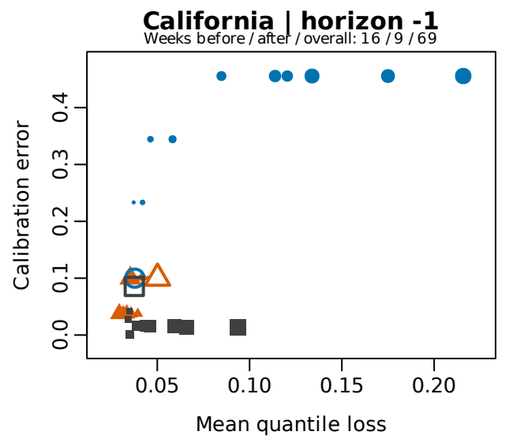


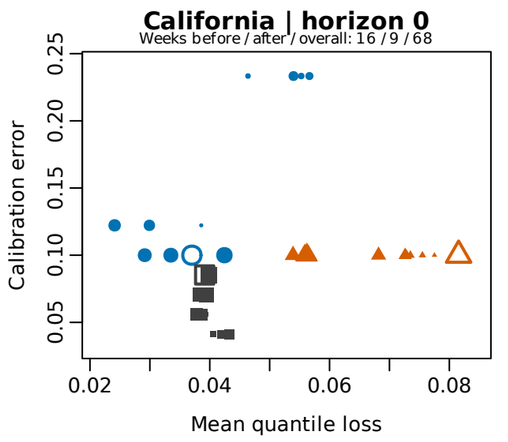


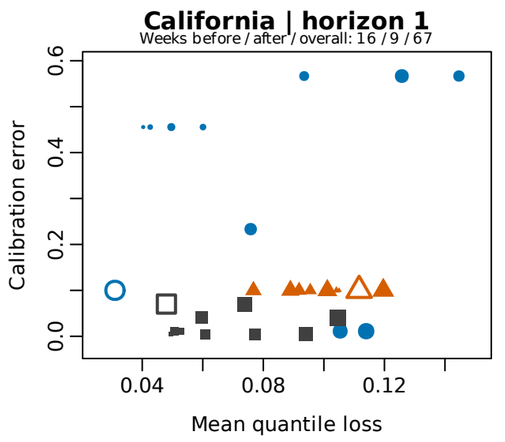


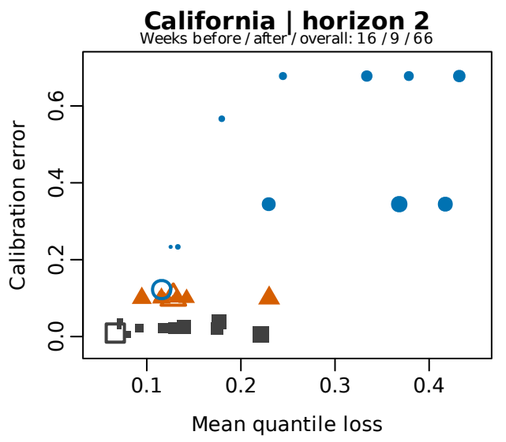


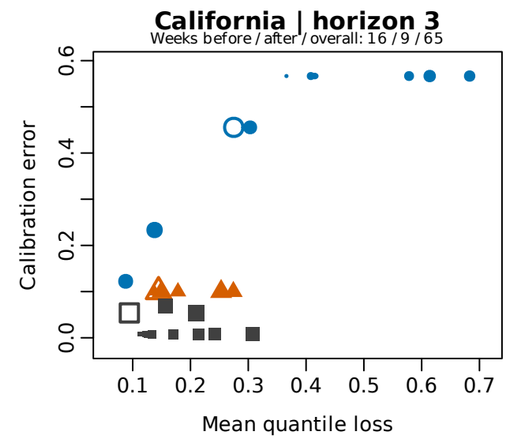


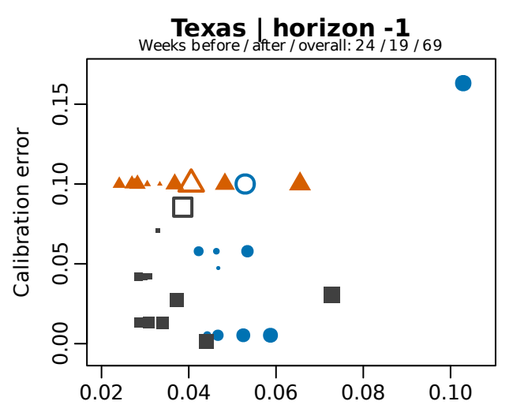


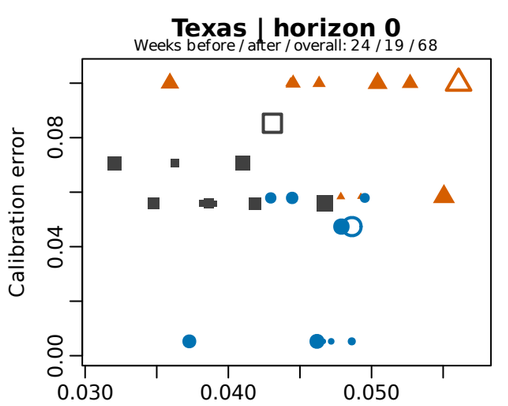


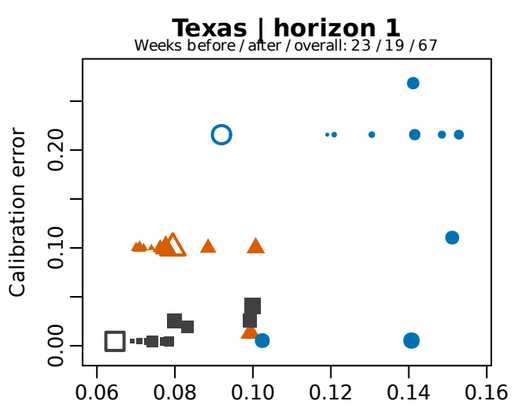


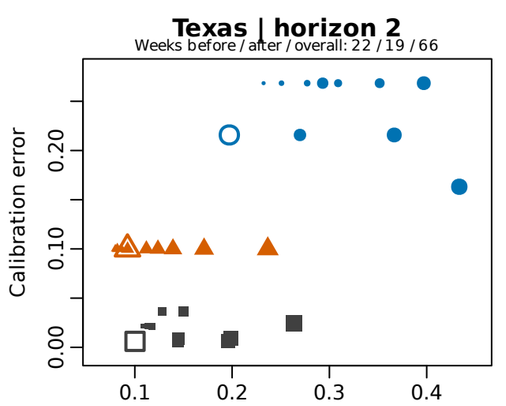


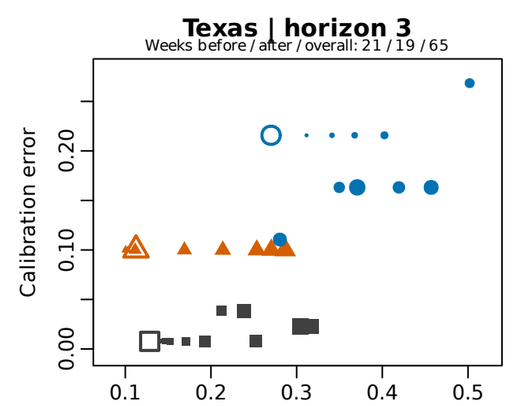


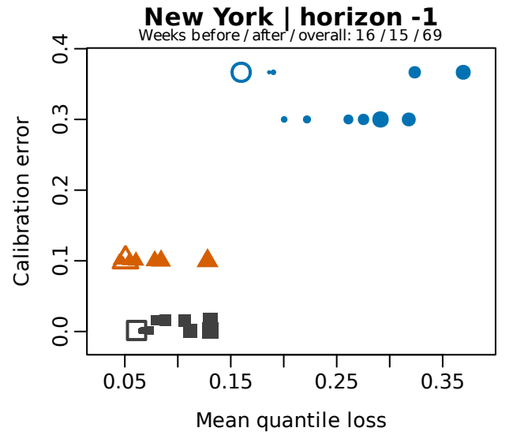


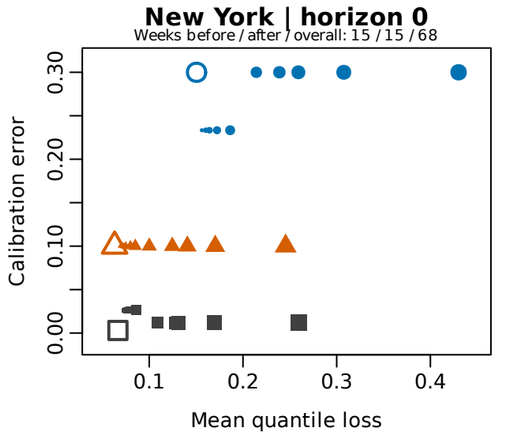


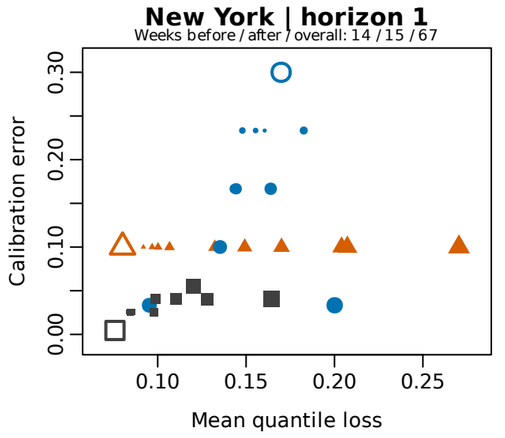


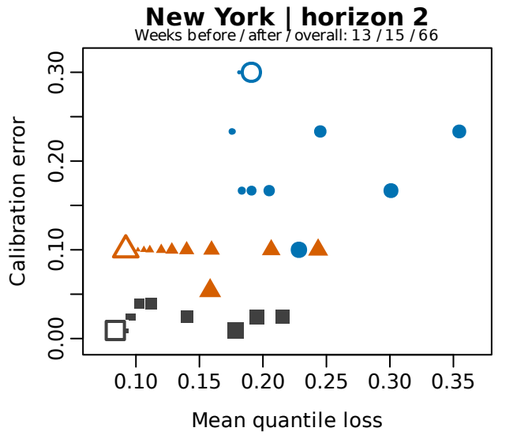


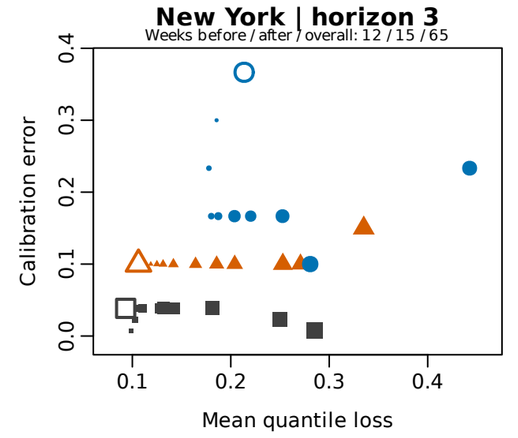


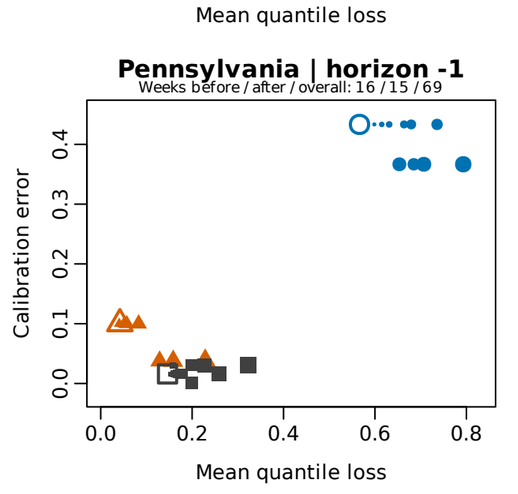


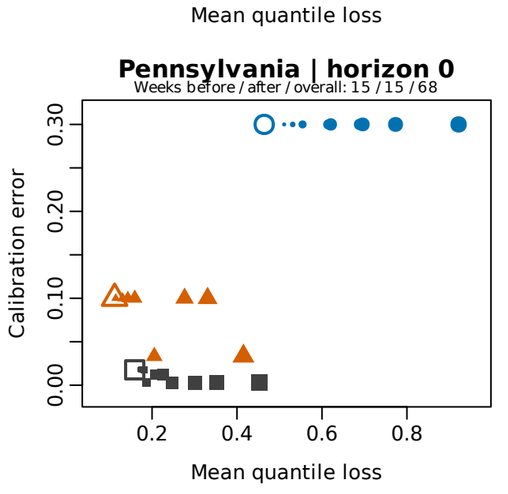


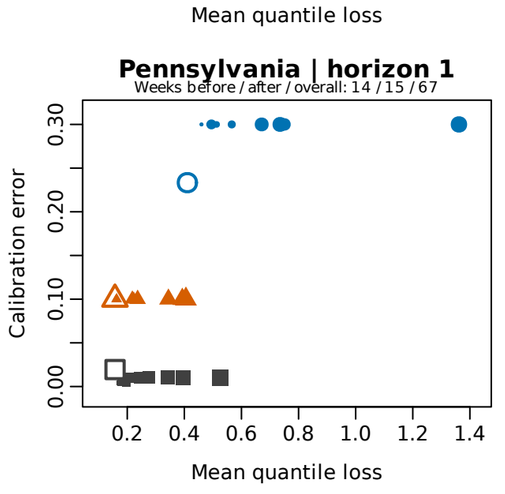


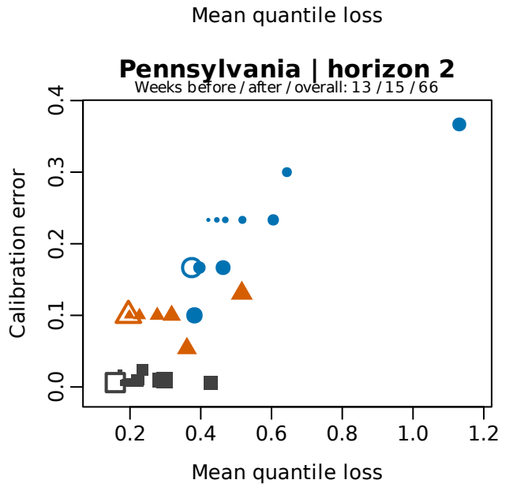


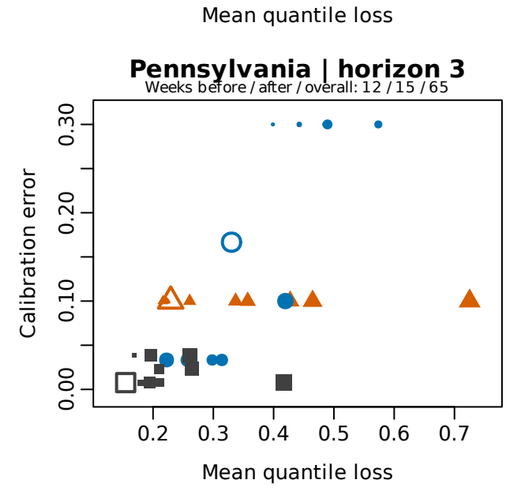


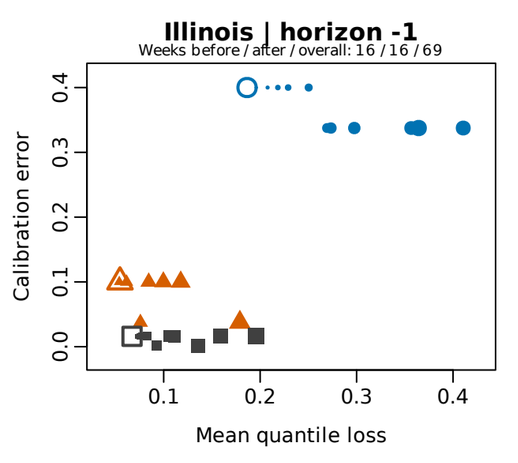


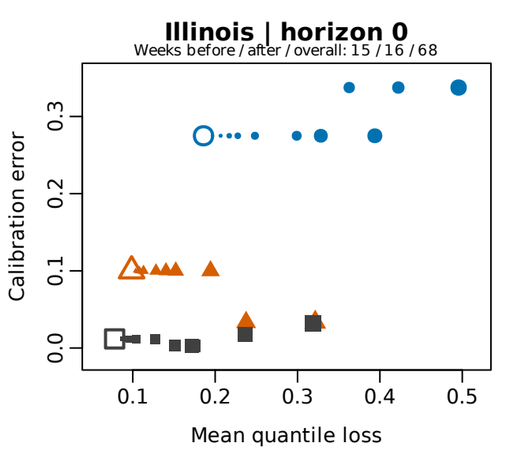


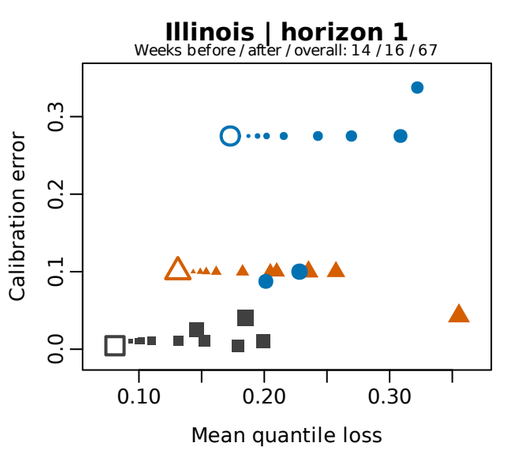


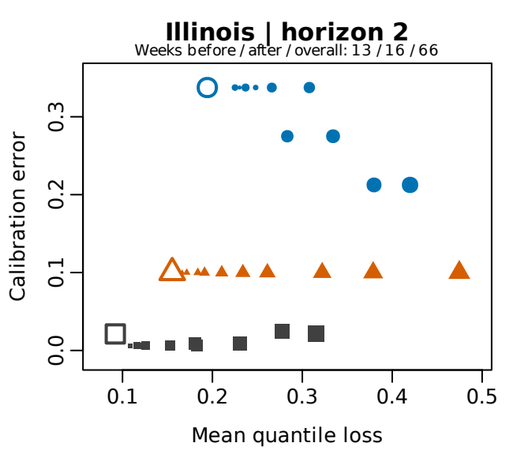


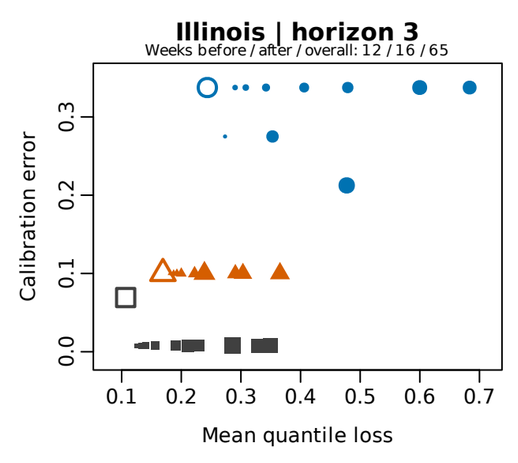












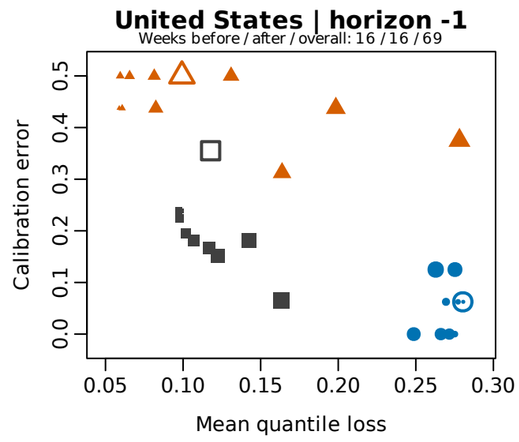
▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)

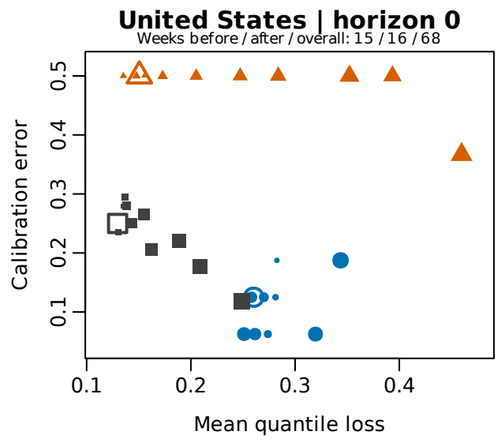
■ Overall (Jun 2025 to the end of season 2025/26)
○ Open symbol: m = 0 (original forecasts)

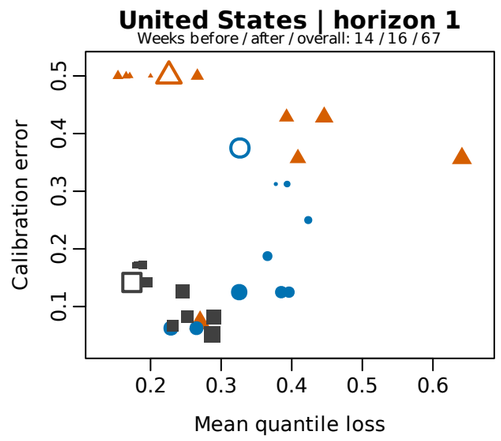
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

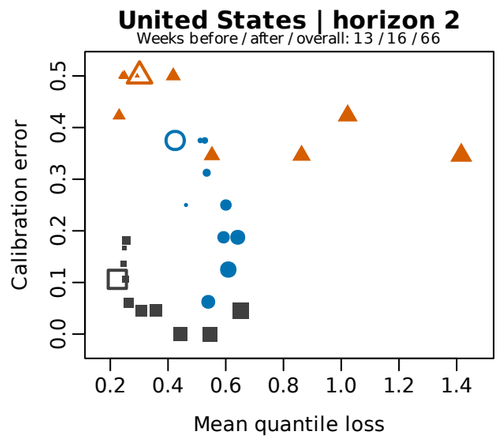
COVID | Fixed-rate MultiQT at quantile level alpha = 0.5: |coverage - alpha| vs quantile loss

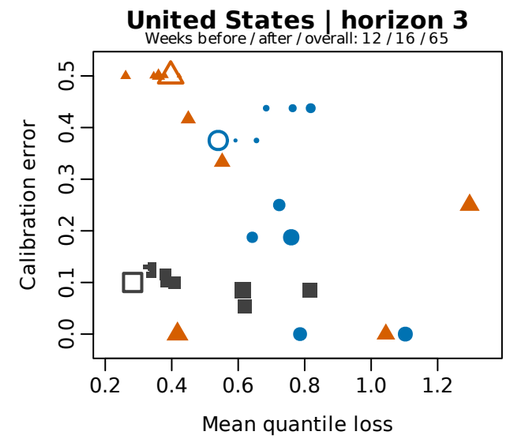
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

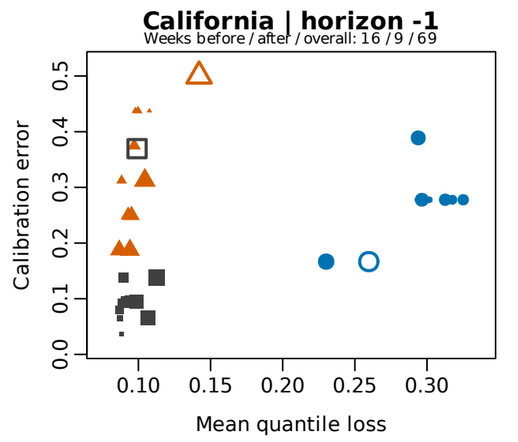


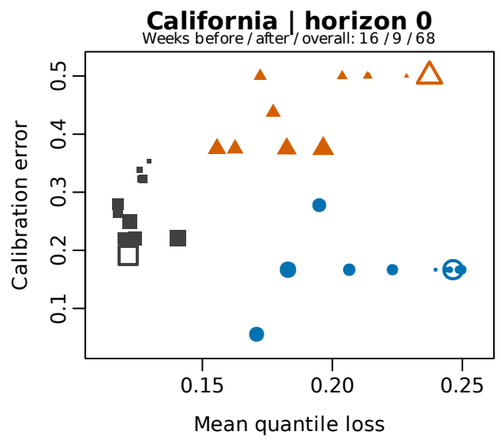


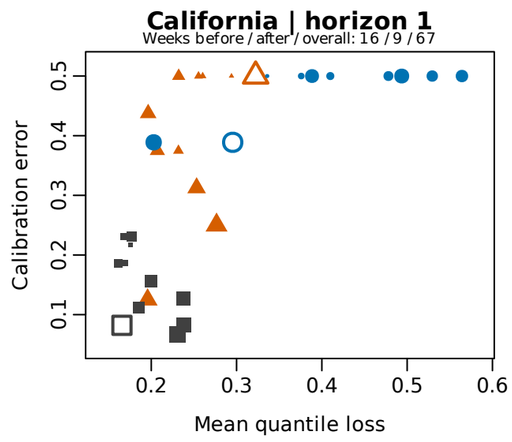


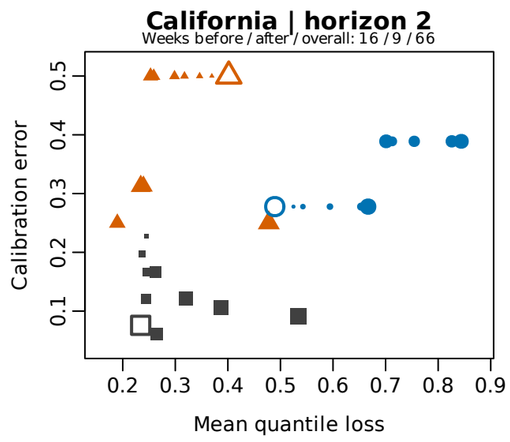


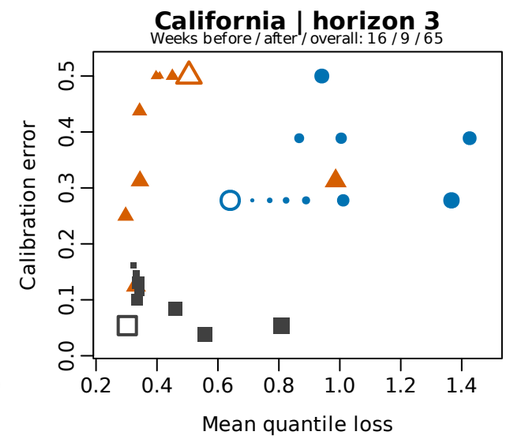


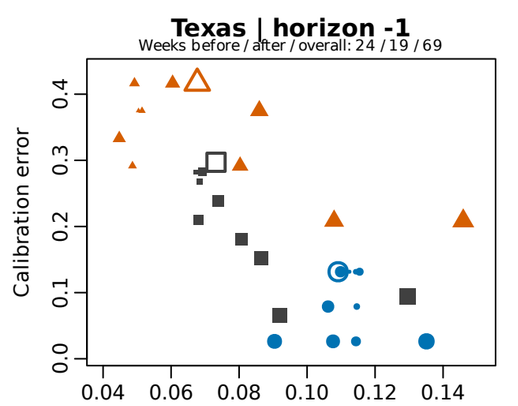


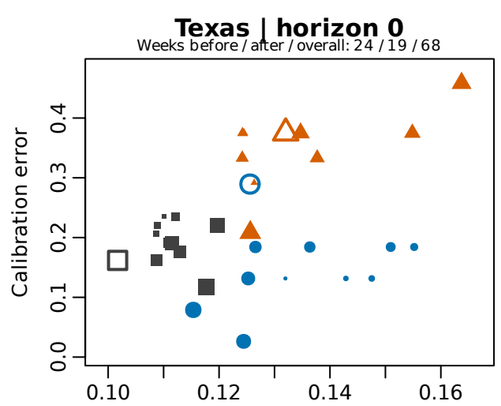


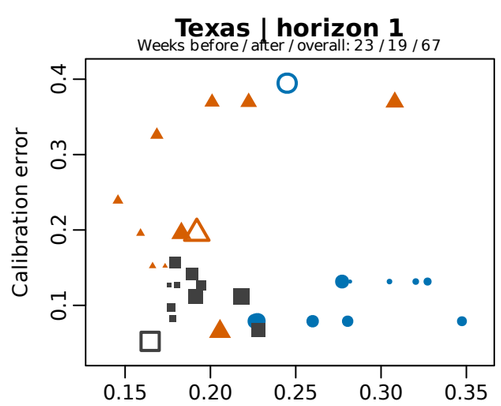


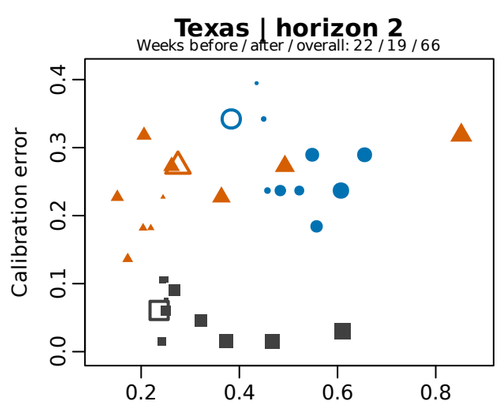


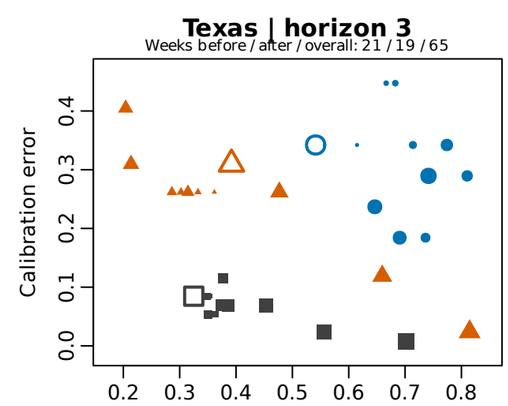


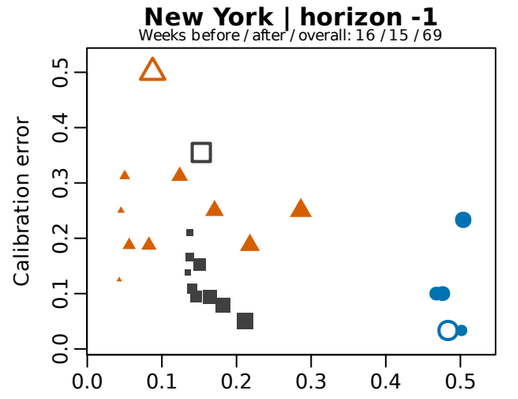


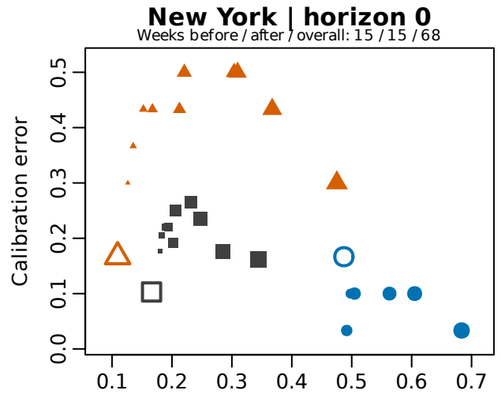


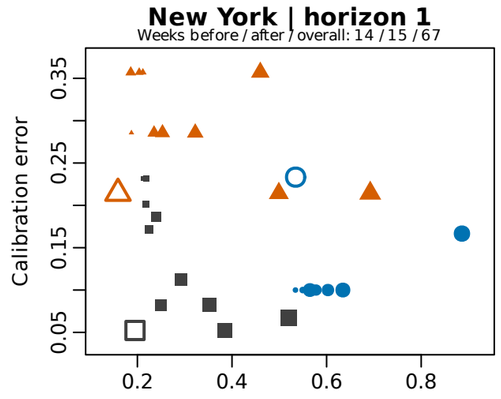


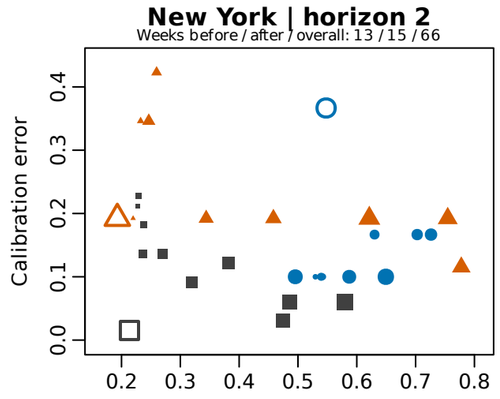


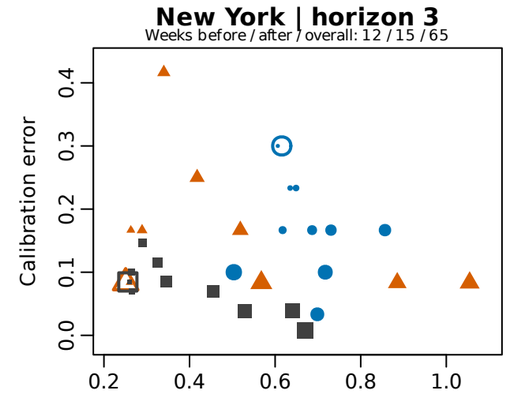


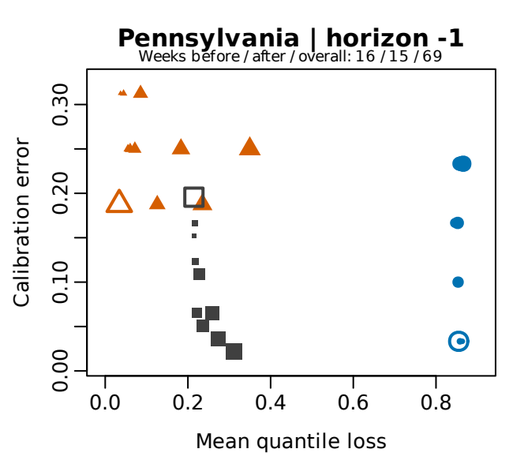


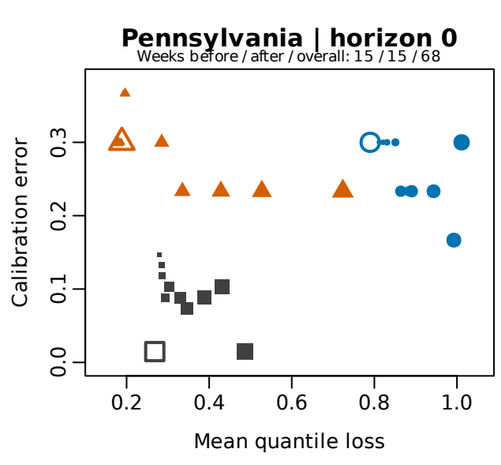


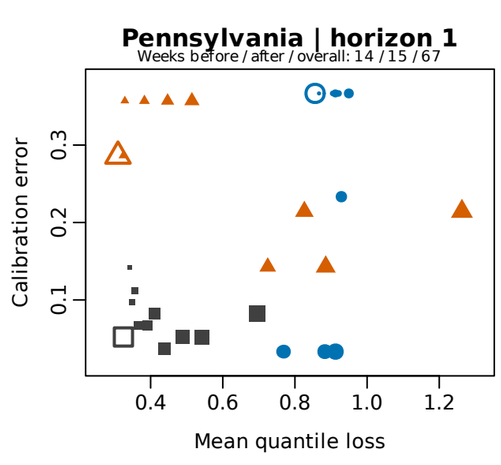


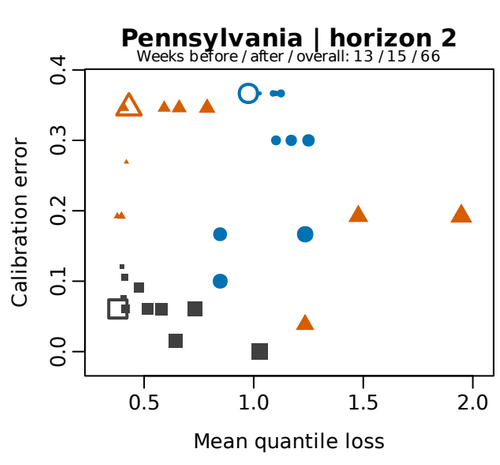


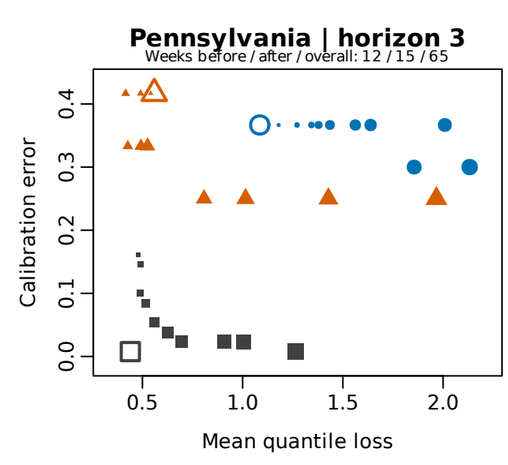


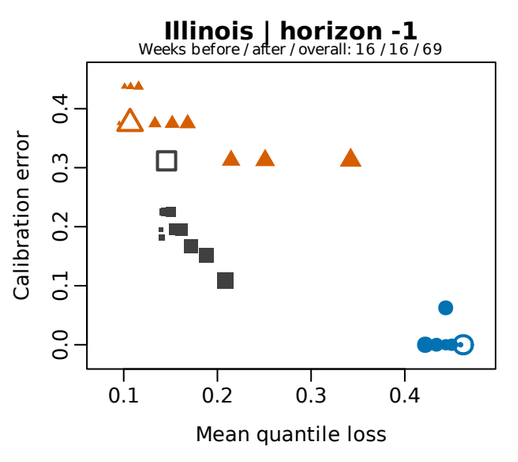


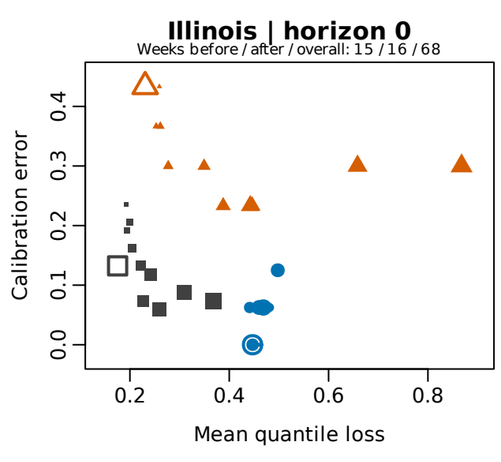


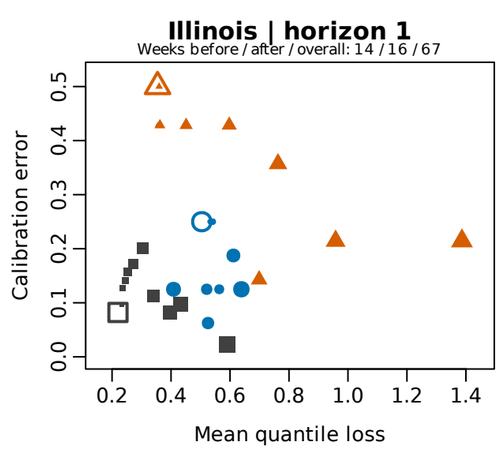


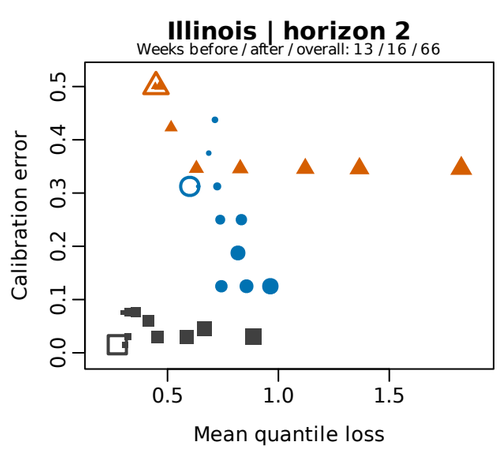


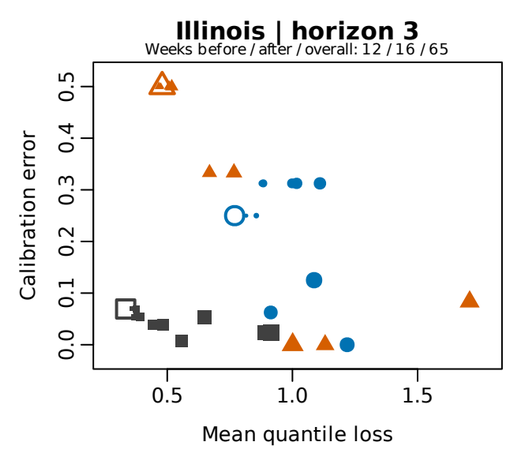












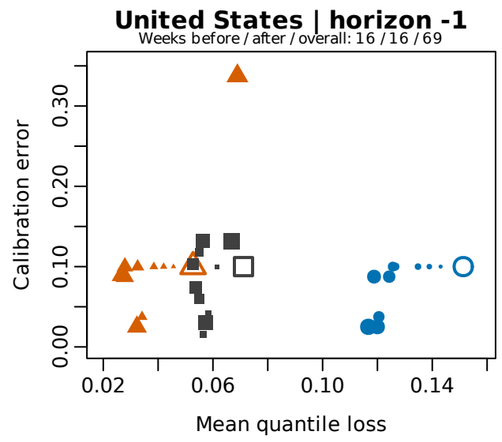
▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)

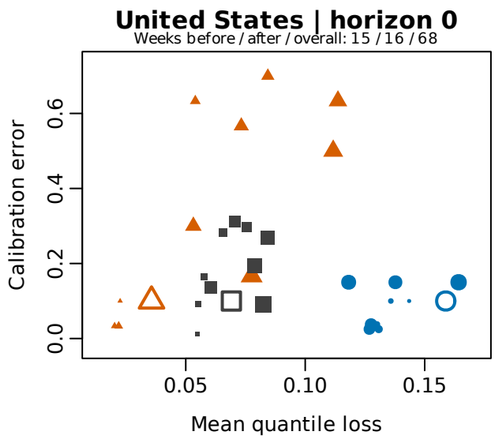
■ Overall (Jun 2025 to the end of season 2025/26)
○ Open symbol: m = 0 (original forecasts)

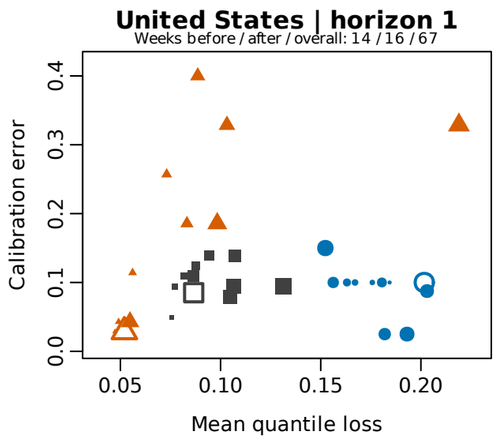
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

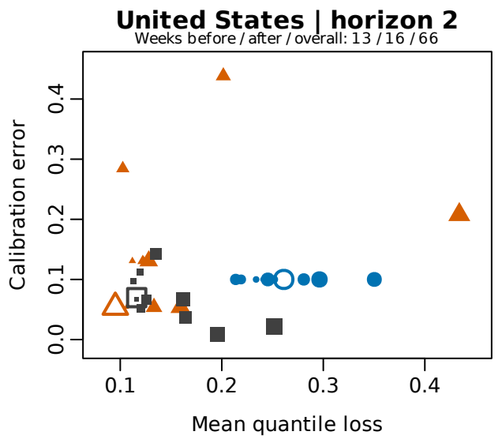
COVID | Fixed-rate MultiQT at quantile level alpha = 0.9: |coverage - alpha| vs quantile loss

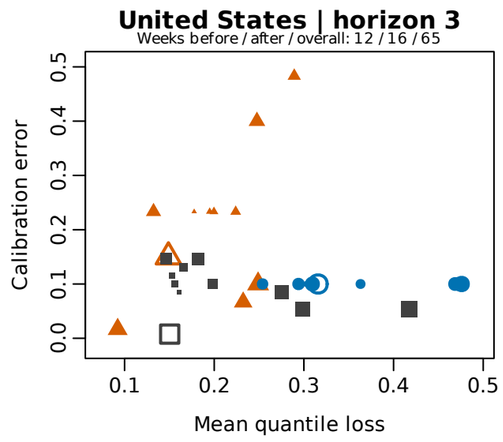
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

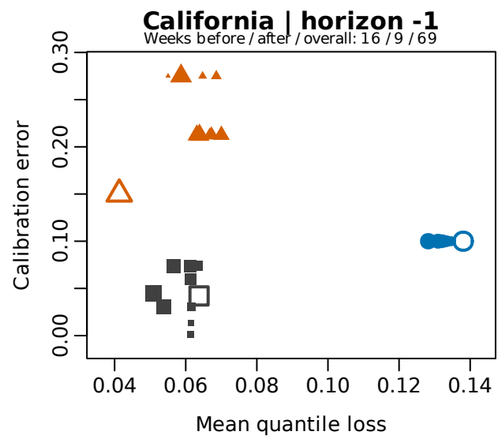


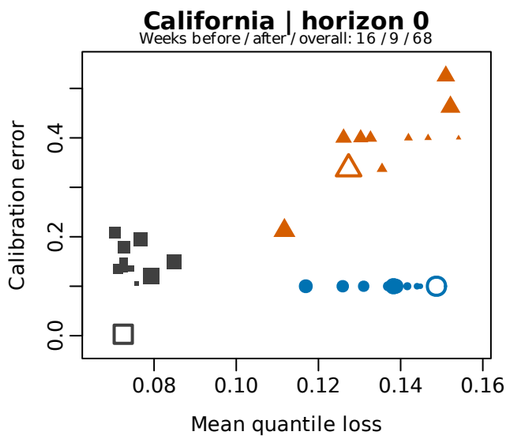


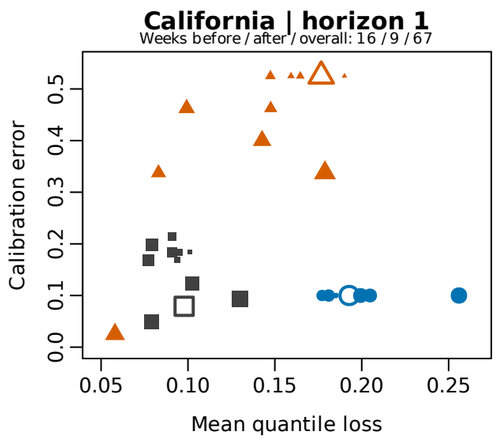


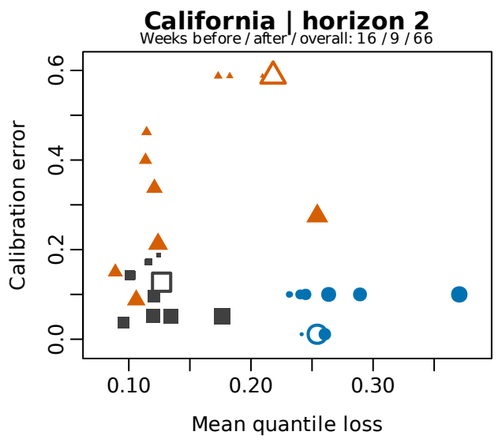


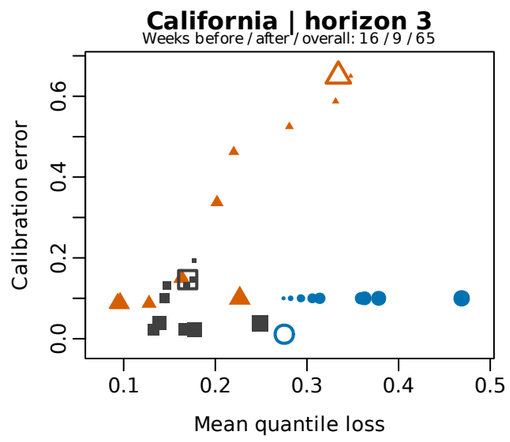


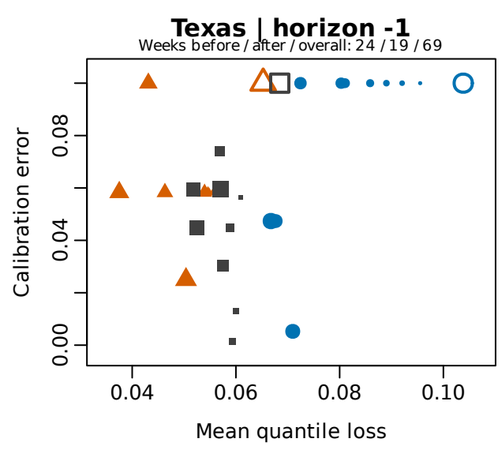


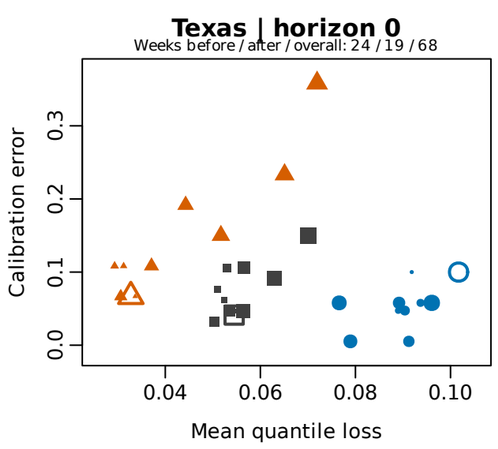


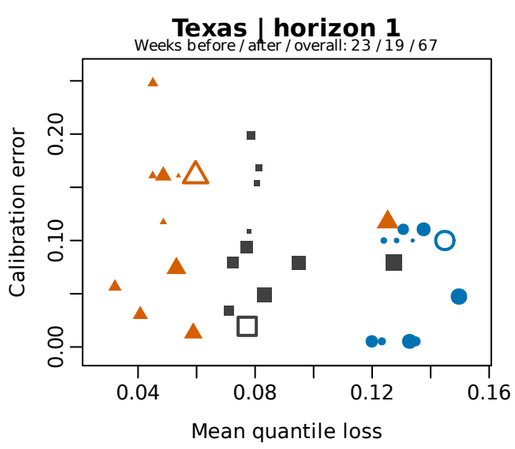


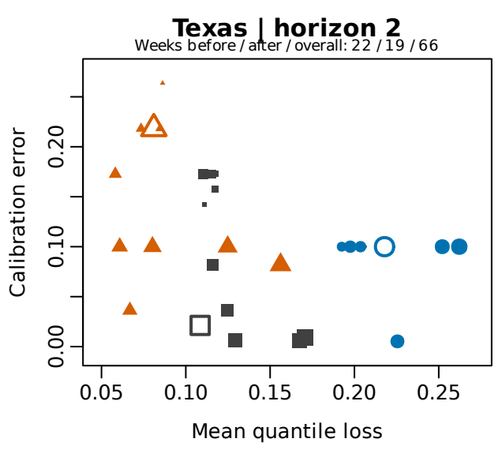


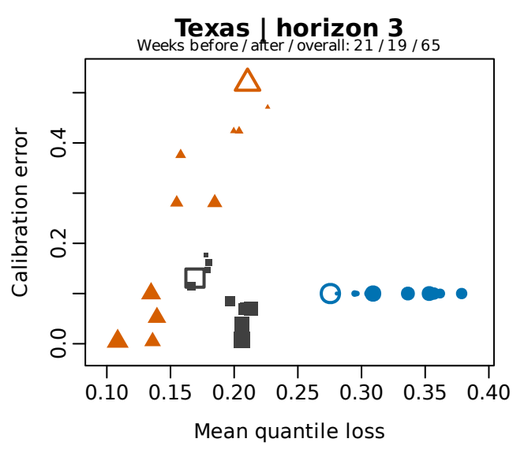


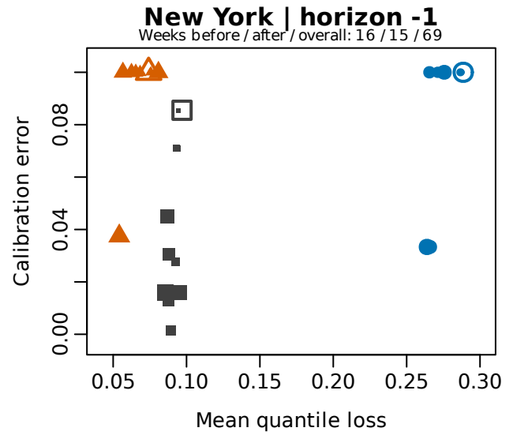


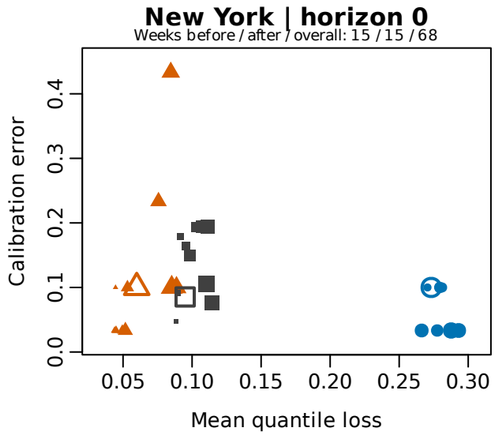


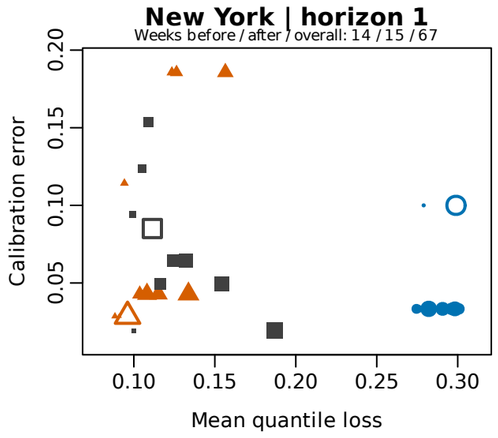


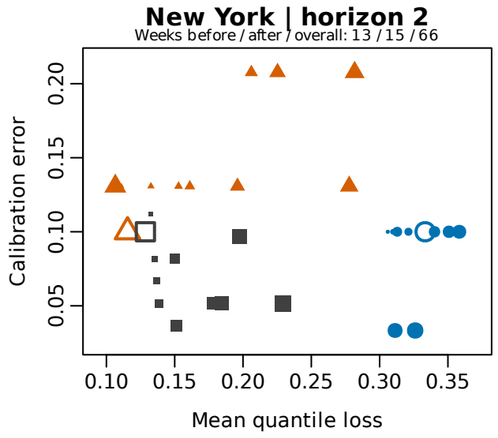


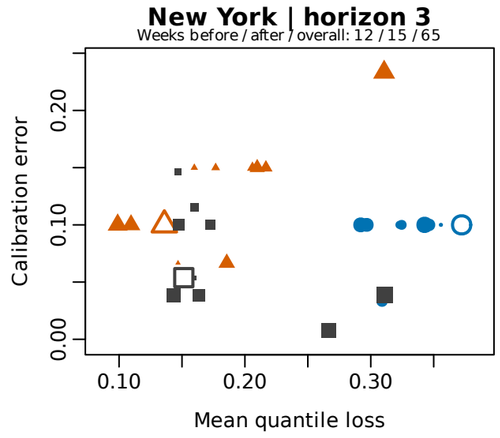


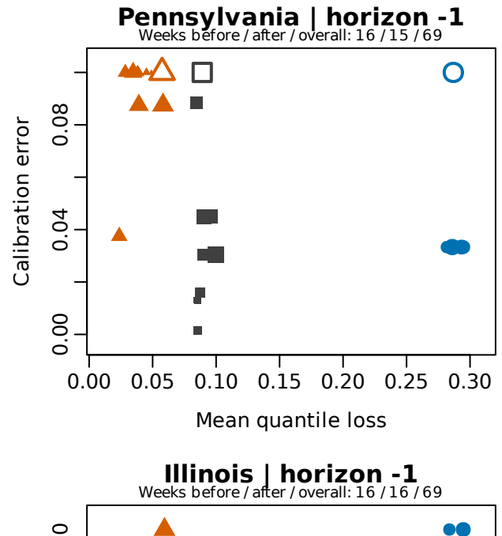


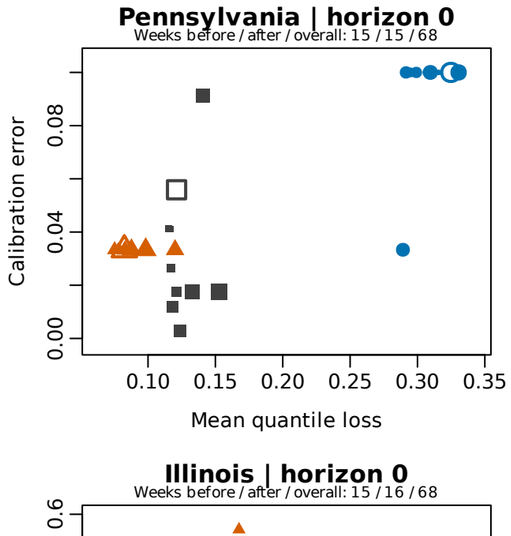


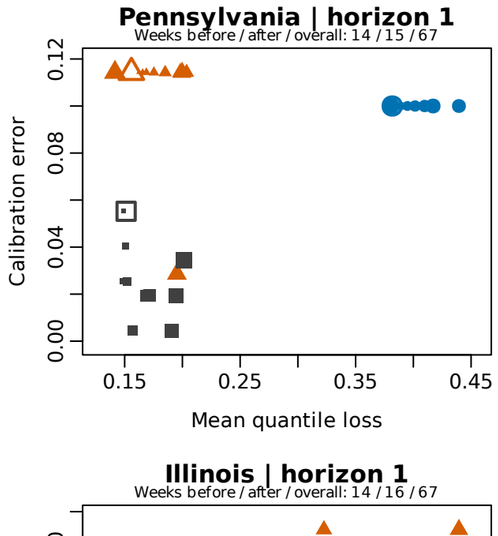


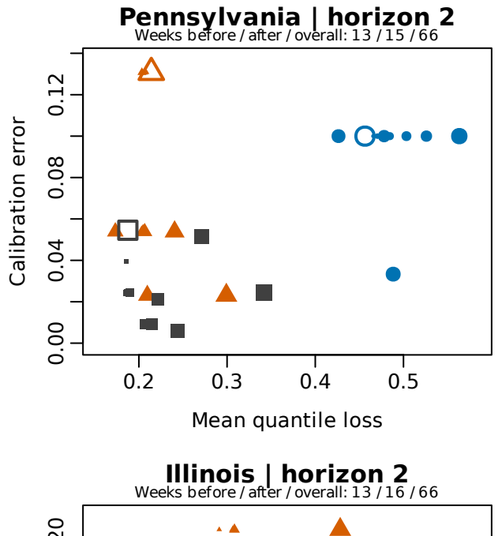


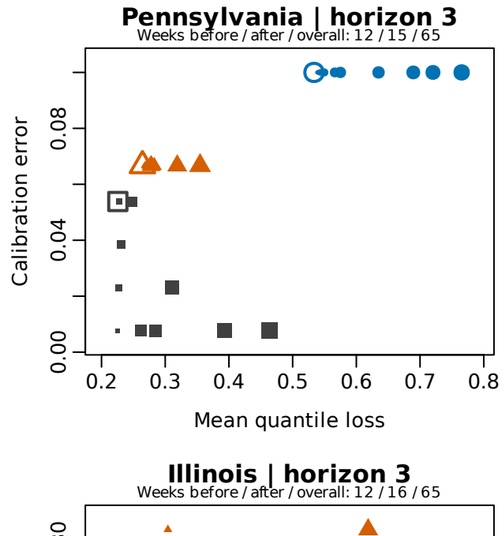


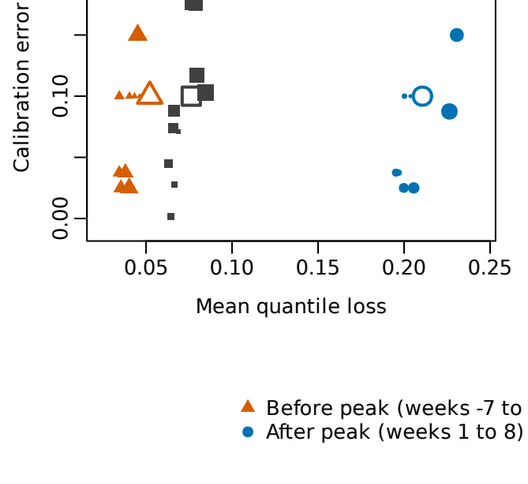


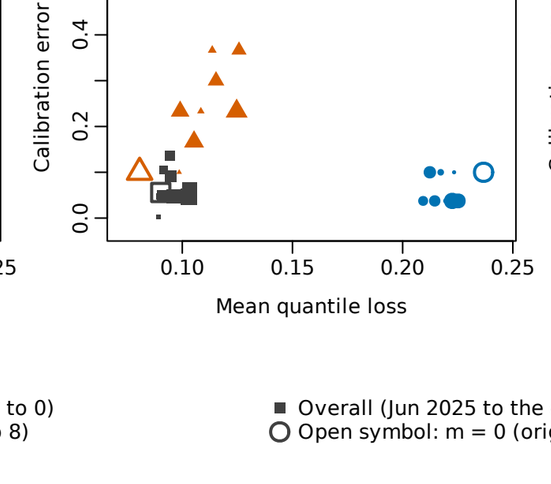


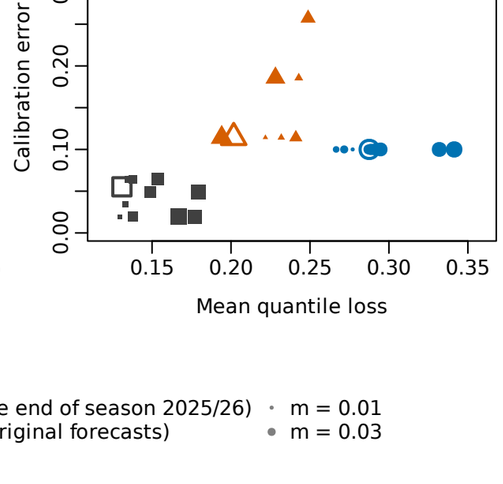


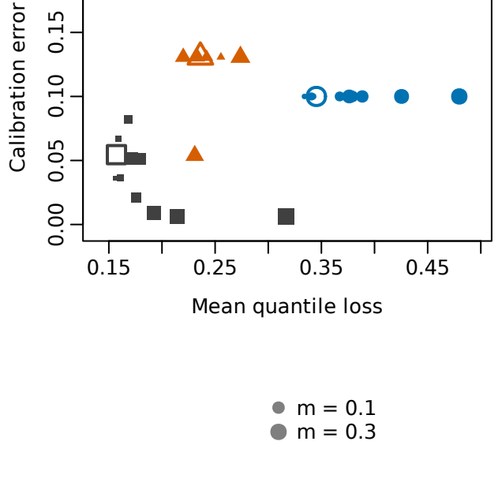


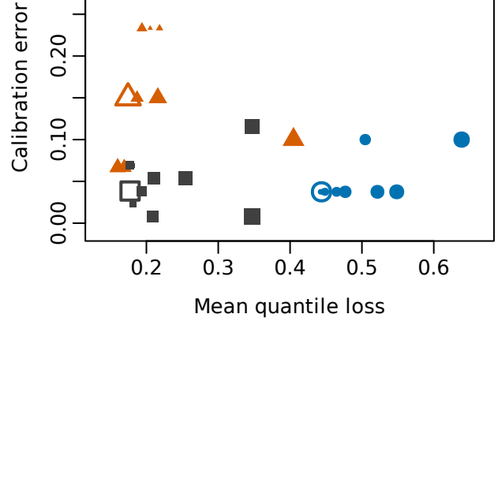












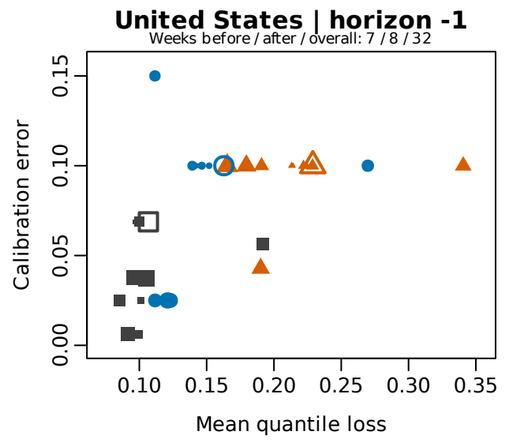
▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)

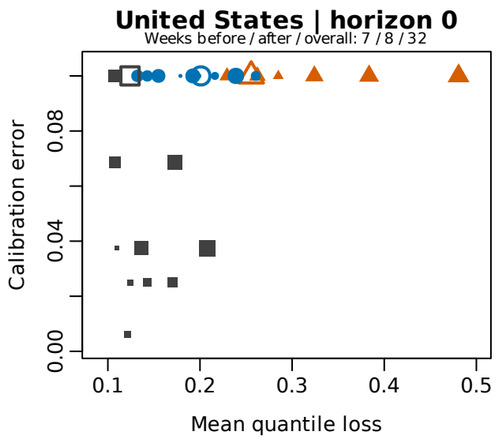
■ Overall (Jun 2025 to the end of season 2025/26)
○ Open symbol: m = 0 (original forecasts)

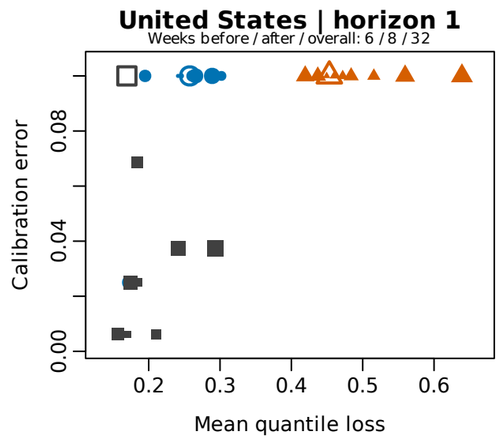
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

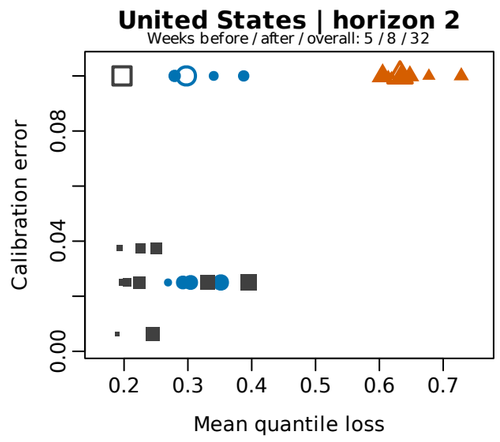
FLU | Fixed-rate MultiQT at quantile level alpha = 0.1: |coverage - alpha| vs quantile loss

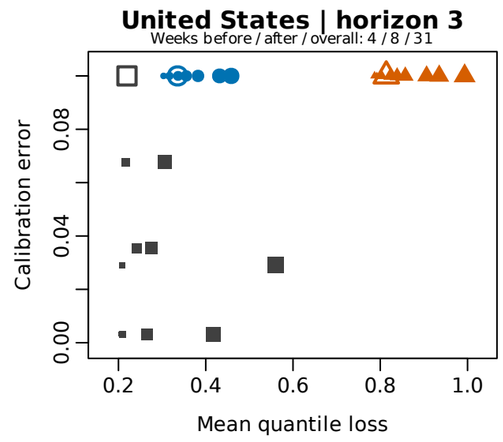
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

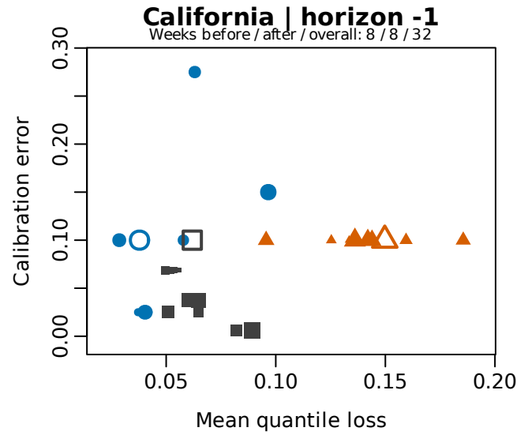


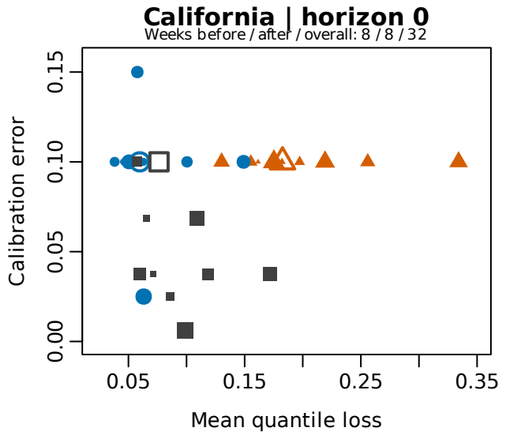


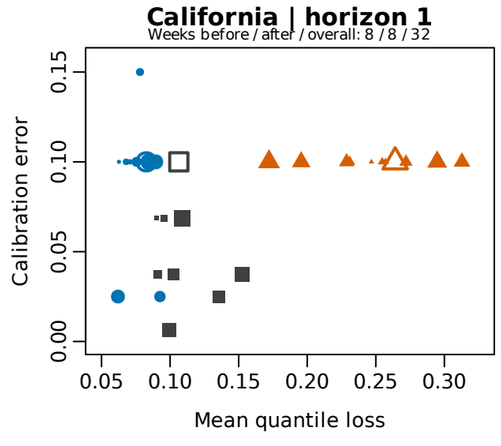


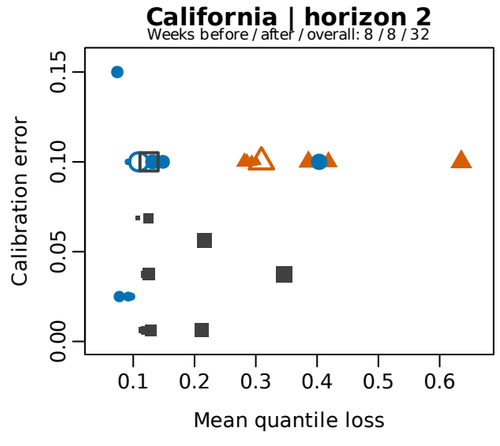


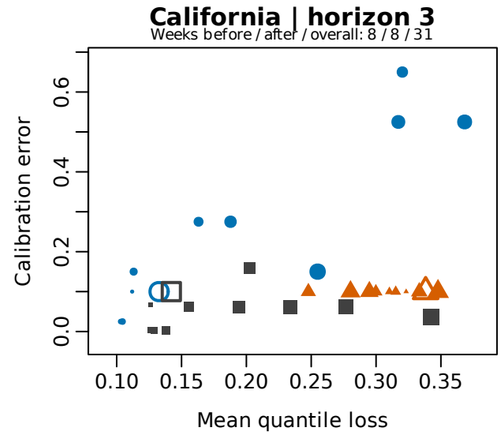


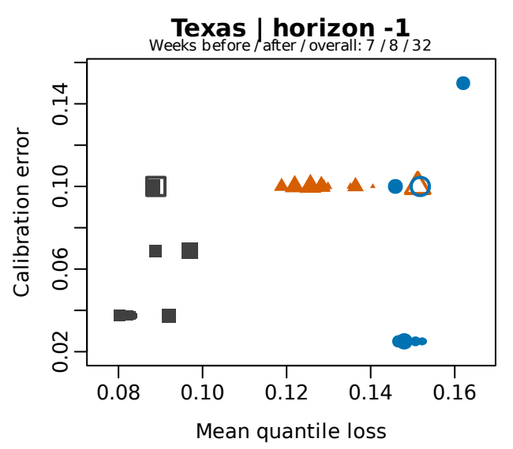


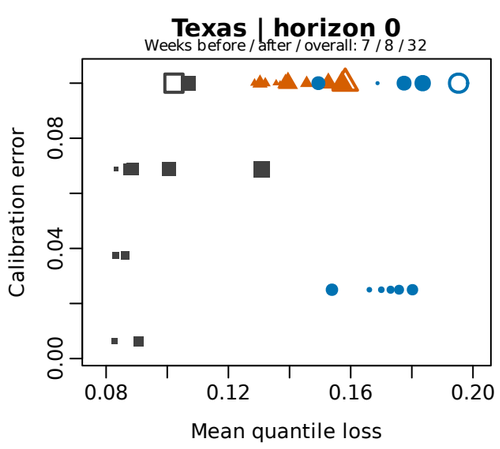


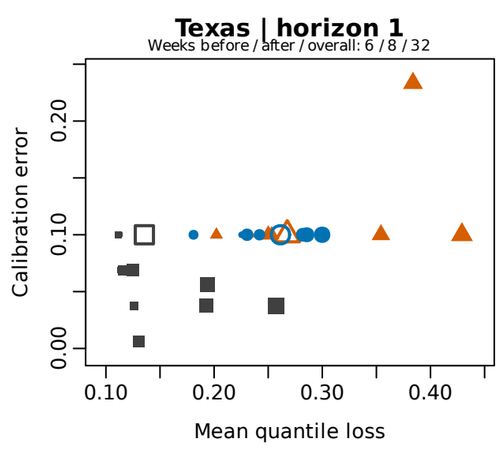


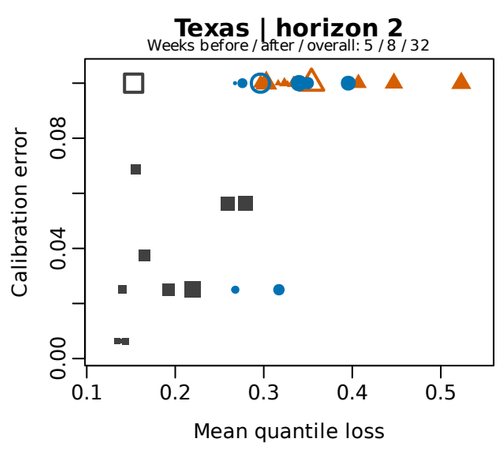


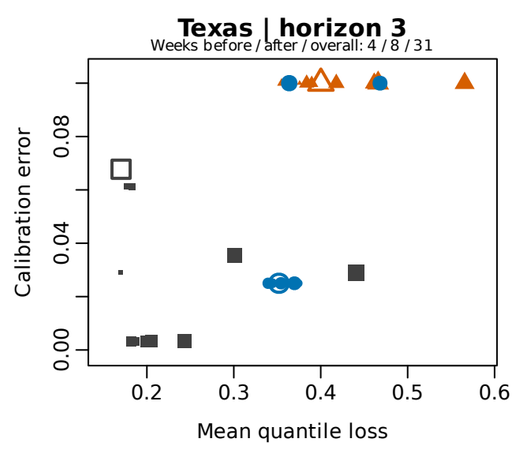


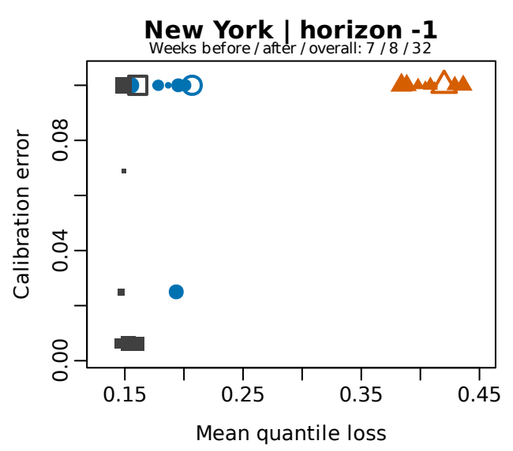


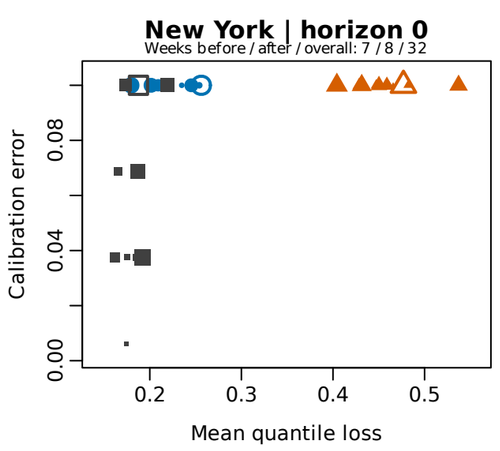


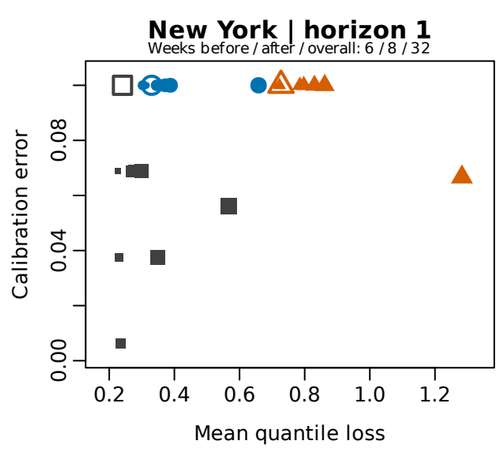


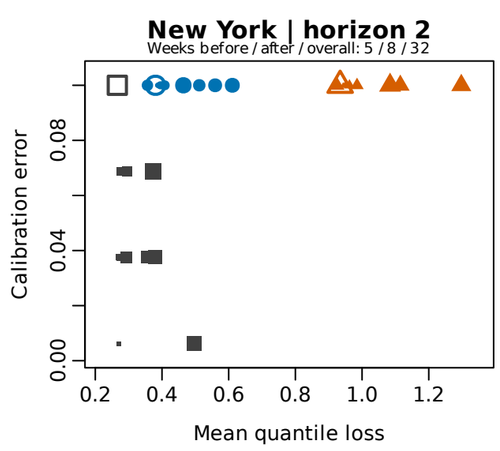


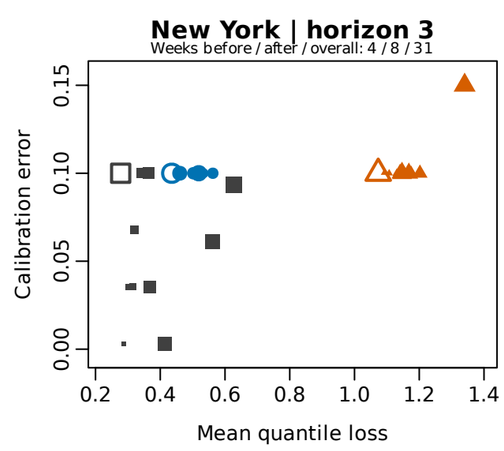


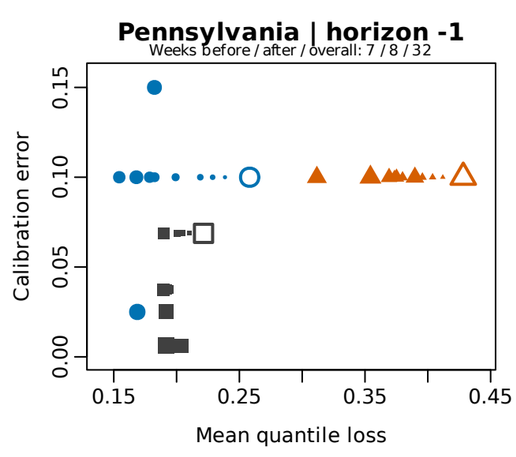


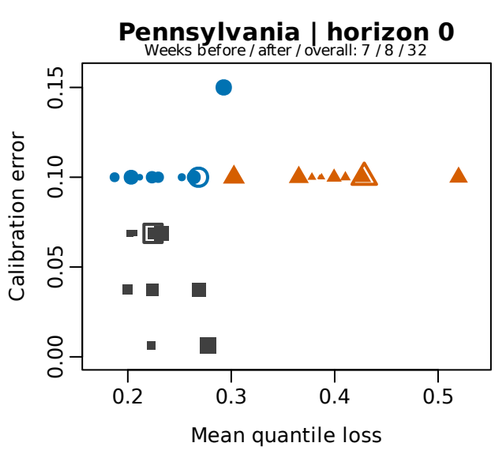


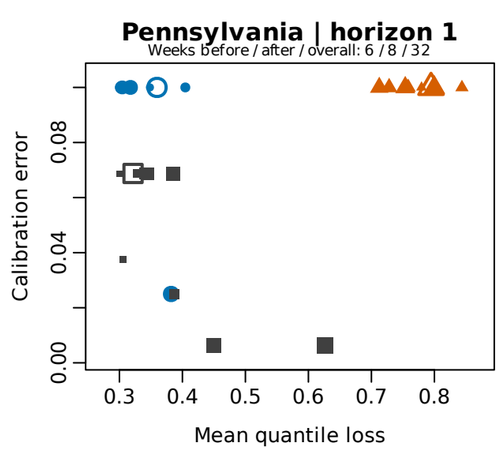


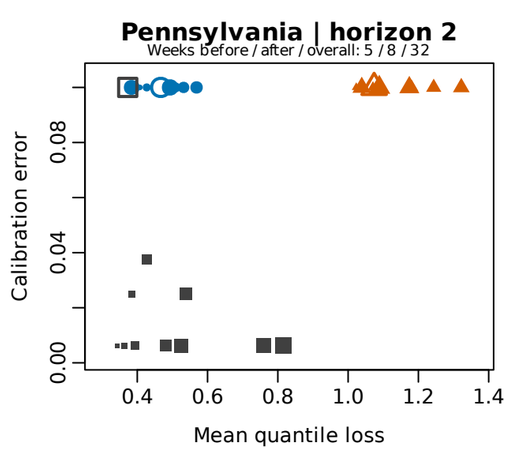


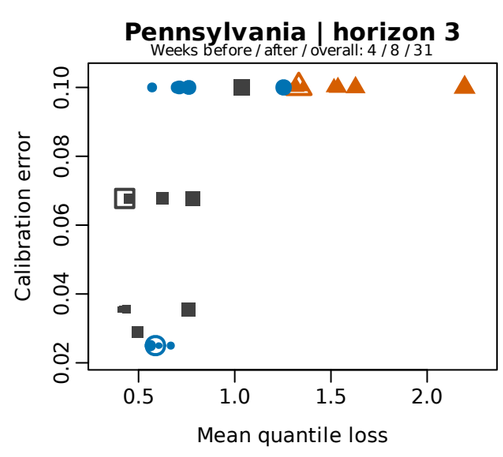


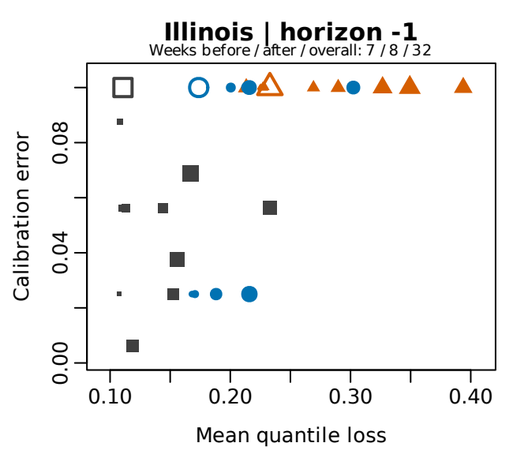


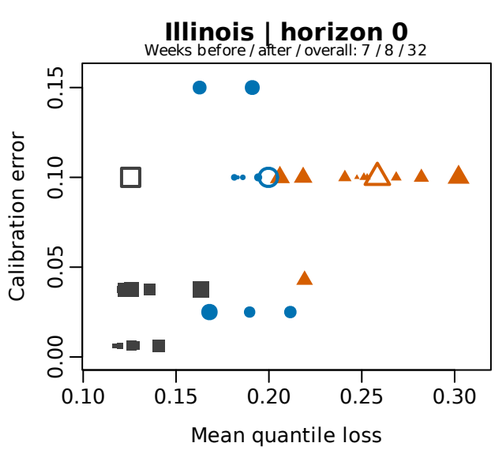


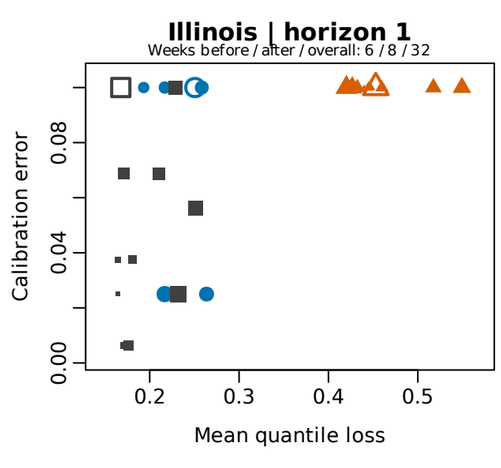


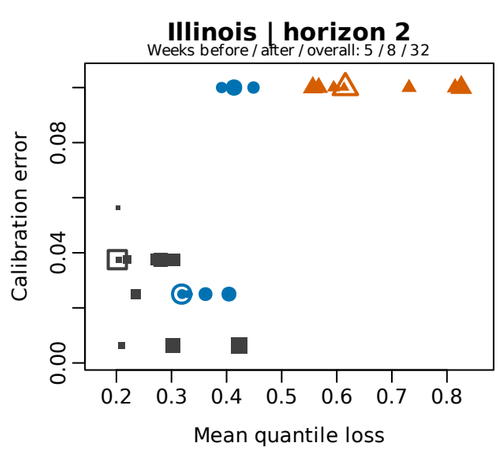


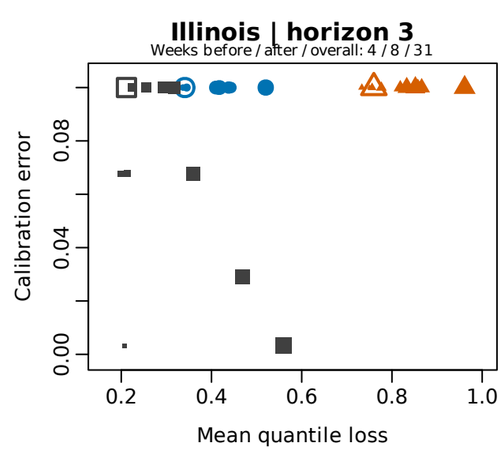












▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)

■ Overall (Jun 2025 to the end of season 2025/26)
○ Open symbol: m = 0 (original forecasts)

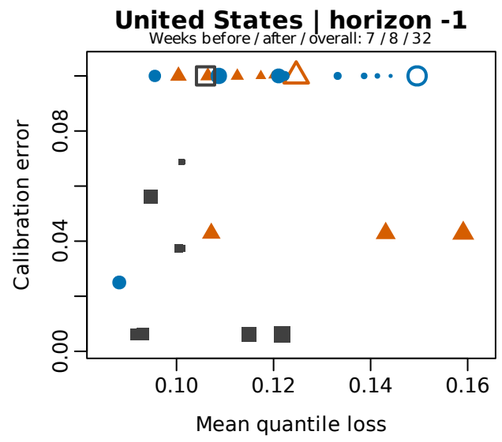
· m = 0.01
• m = 0.03
● m = 0.1
● m = 0.3

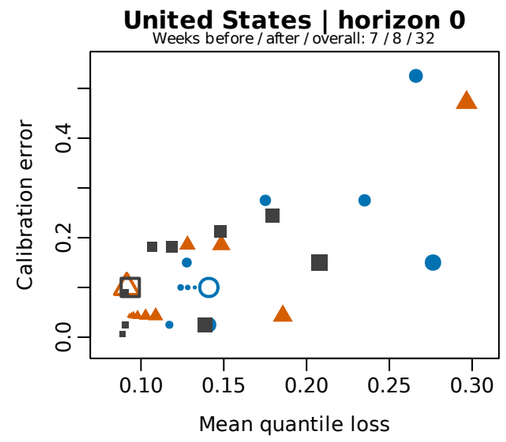
FLU | Fixed-rate MultiQT at quantile level alpha = 0.5: |coverage - alpha| vs quantile loss

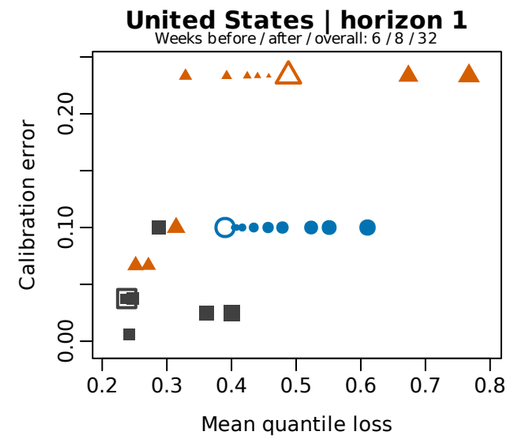
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

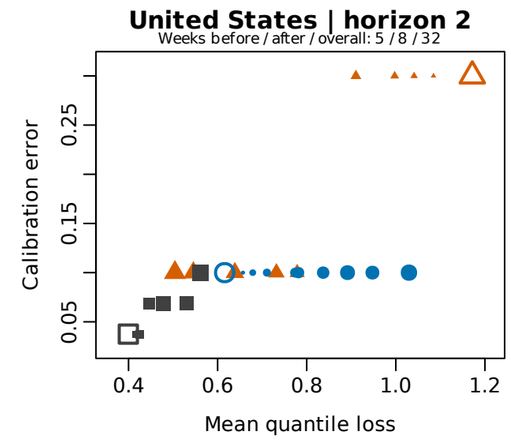
FLU | Fixed-rate MultiQT at quantile level alpha = 0.9: |coverage - alpha| vs quantile loss

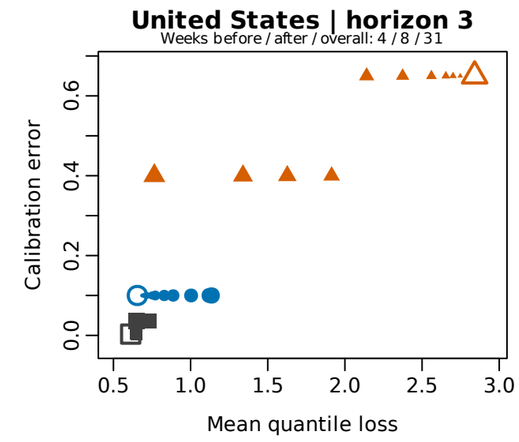
Count tracking | eta = m * S0, S0 frozen after 20 released rows (in hindsight: warmup replayed) | Feedback delay 0 week(s), protocol latest-at-use-v2 | Scored against latest reported counts in counts per 100k residents | Lower left is better.

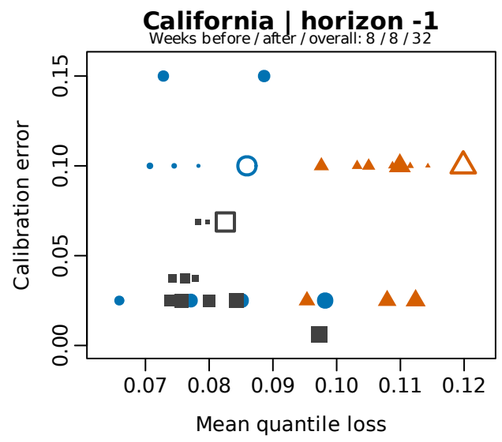


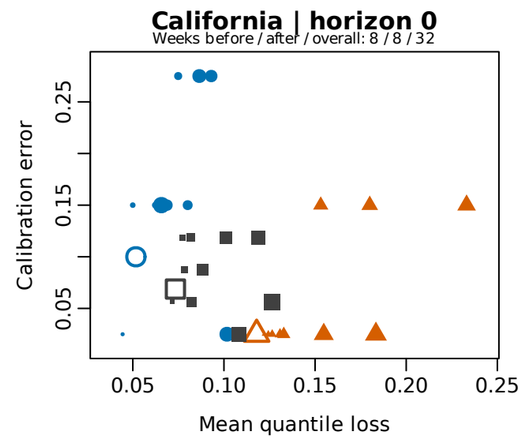


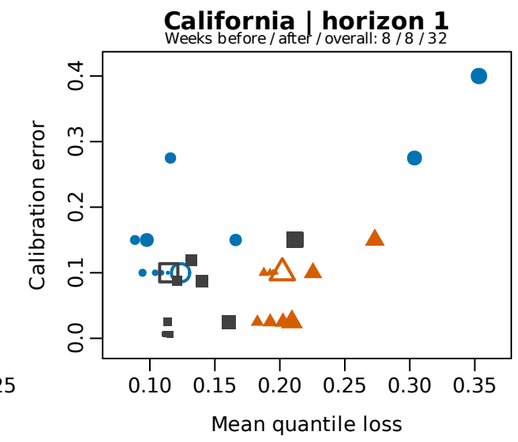


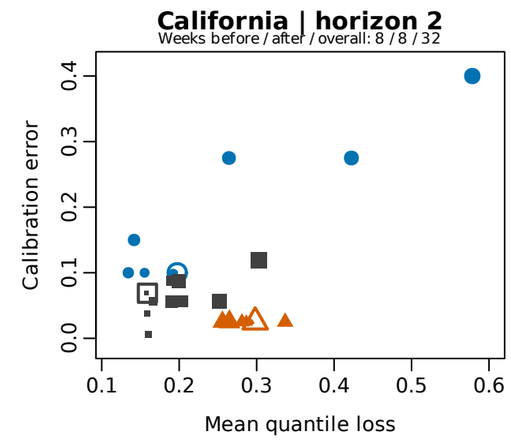


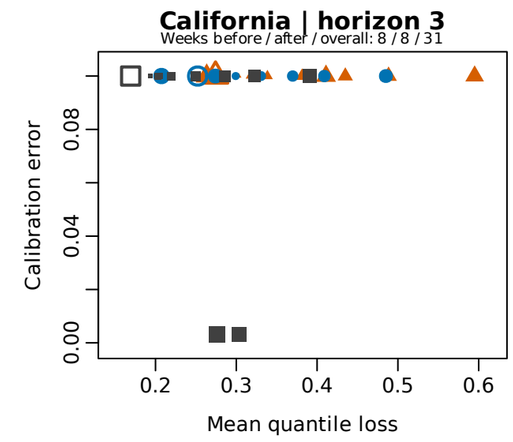


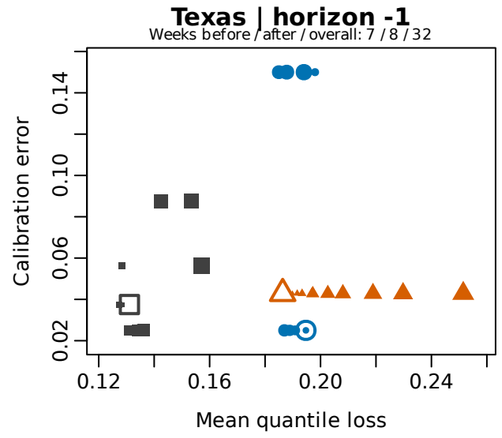


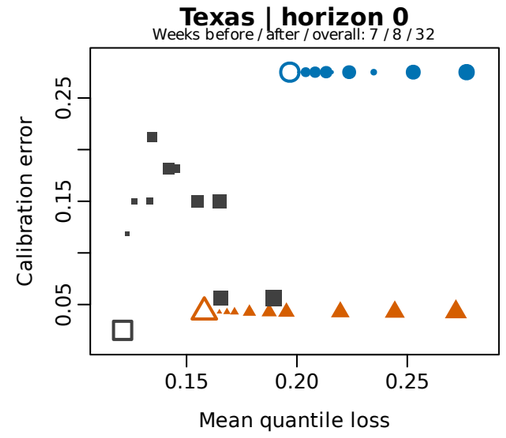


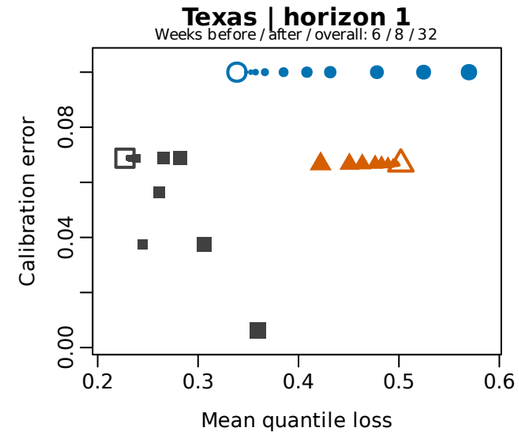


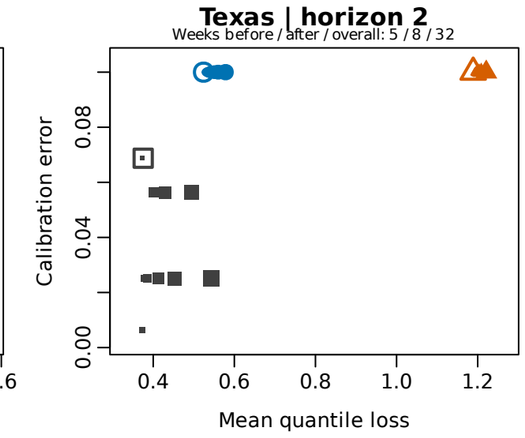


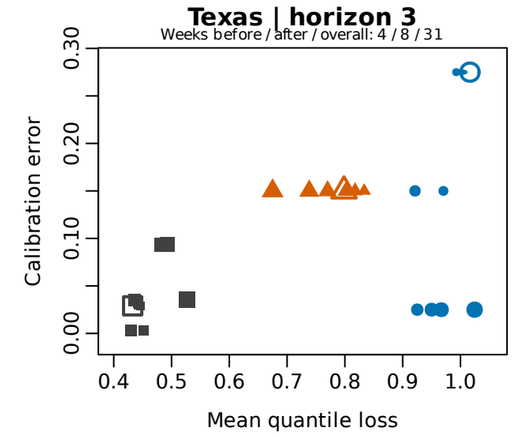


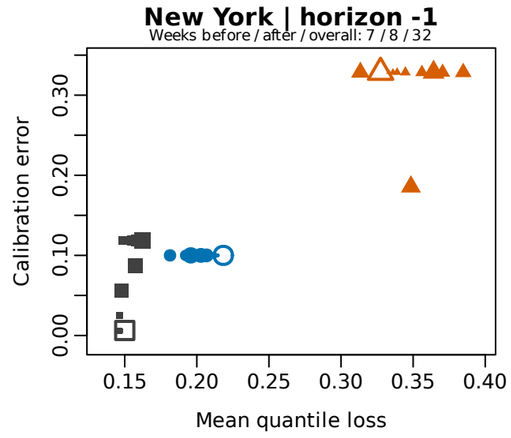


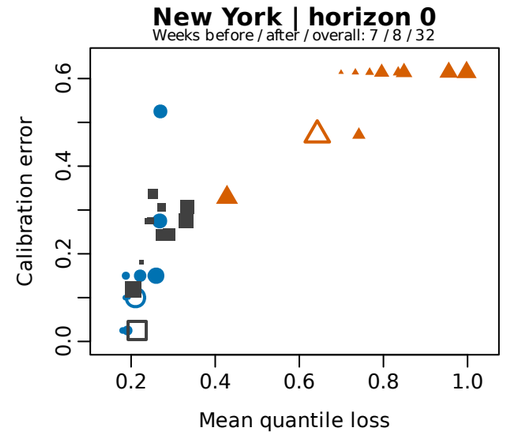


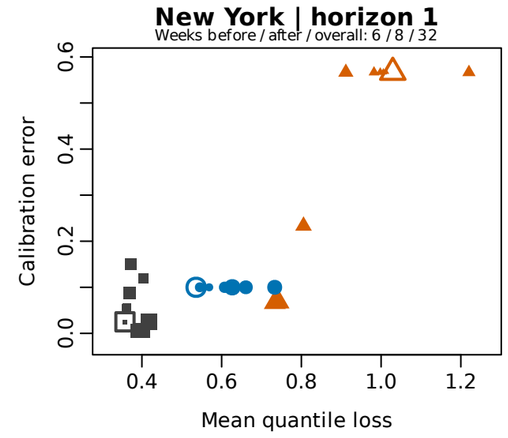


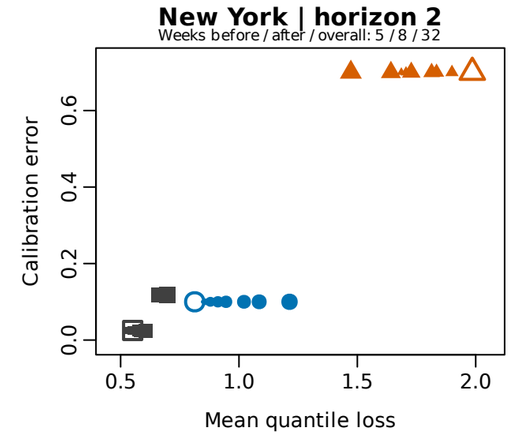


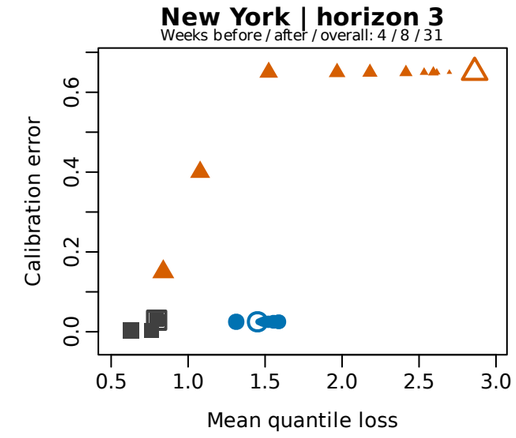


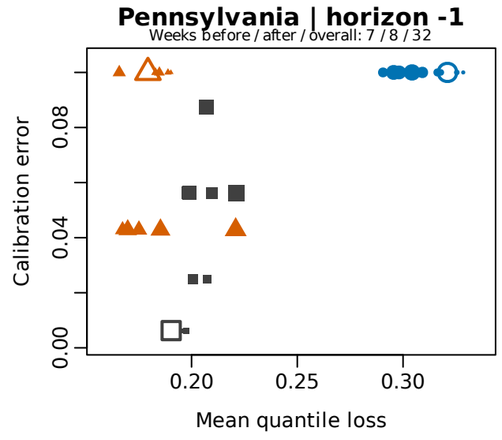


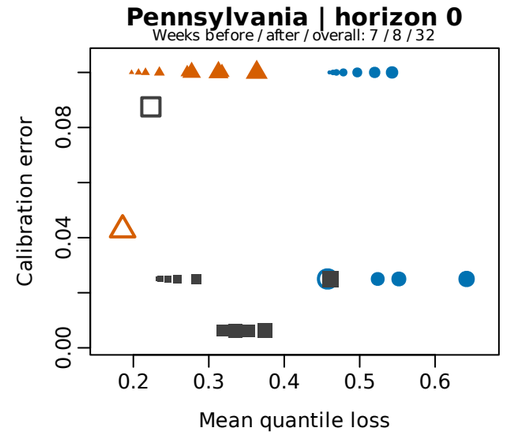


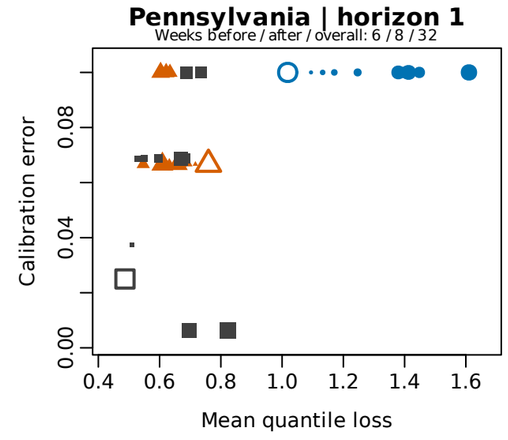


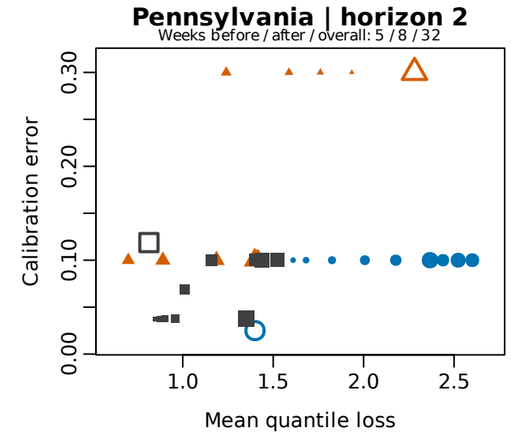


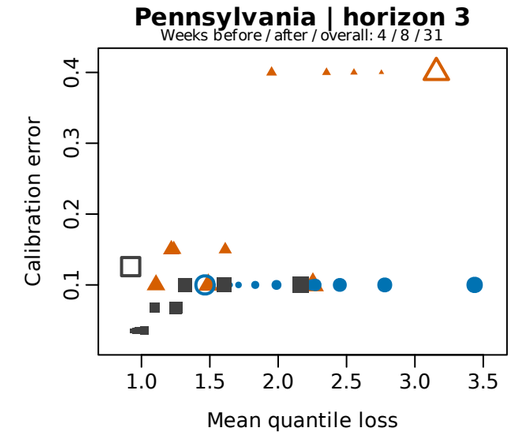


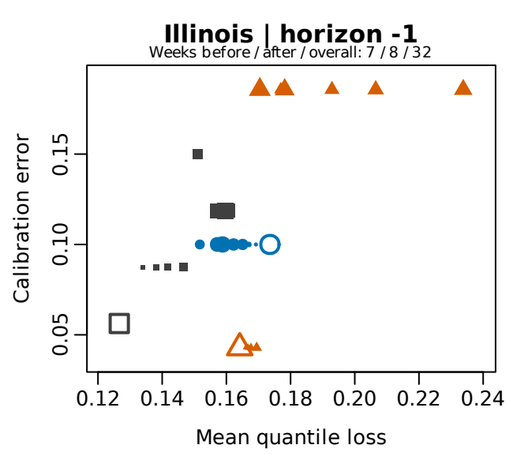


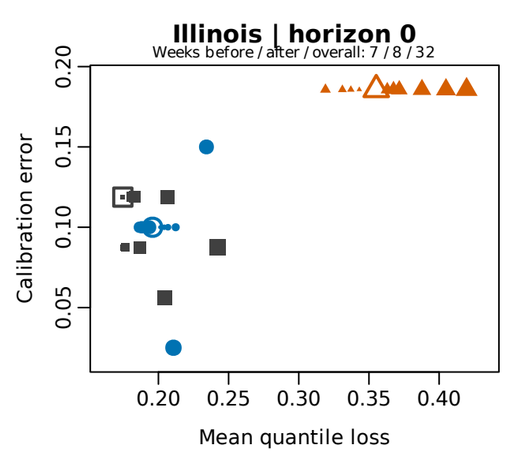


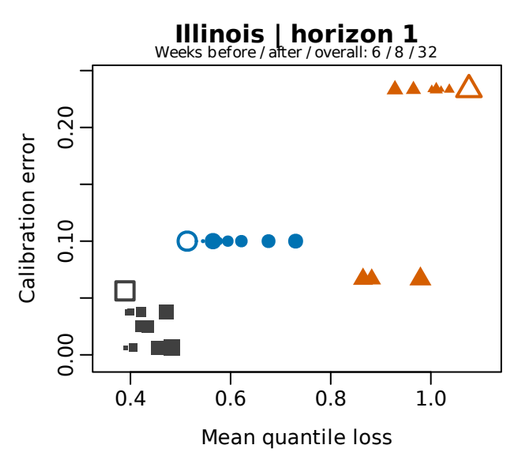


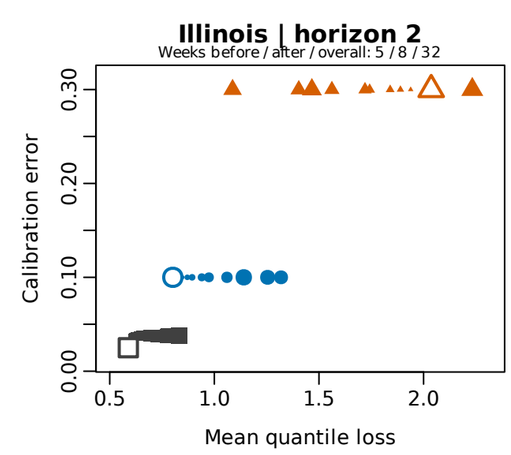


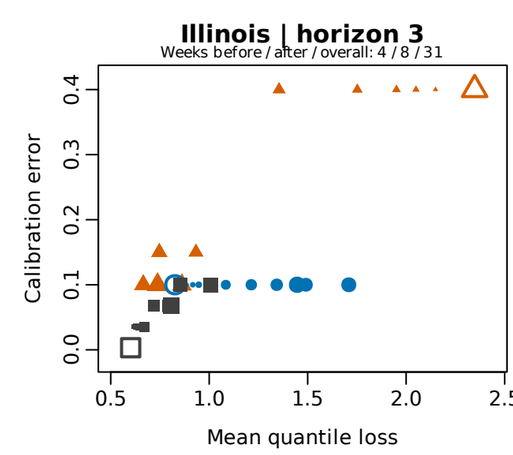












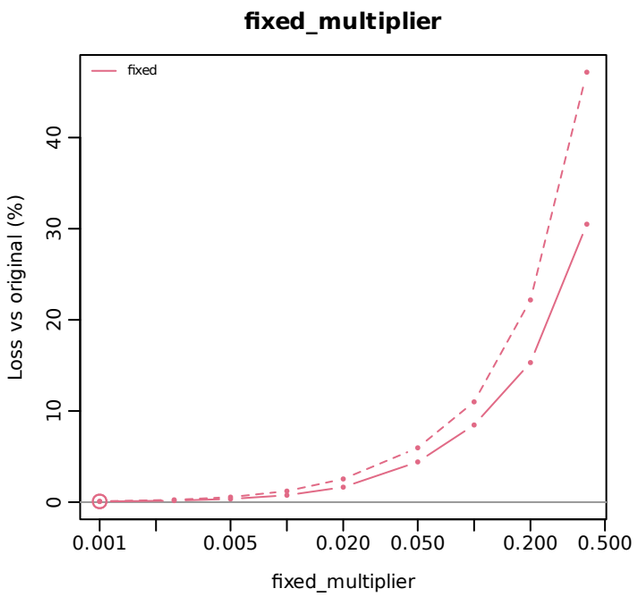
▲ Before peak (weeks -7 to 0)
● After peak (weeks 1 to 8)

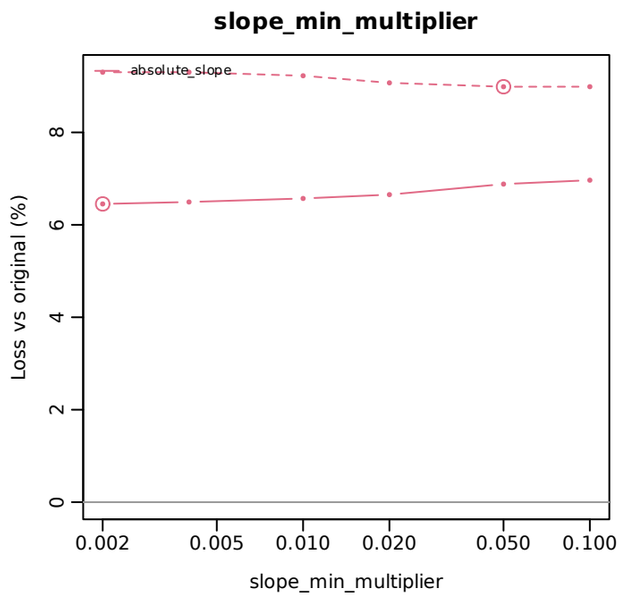
■ Overall (Jun 2025 to the end of season 2025/26)
○ Open symbol: m = 0 (original forecasts)

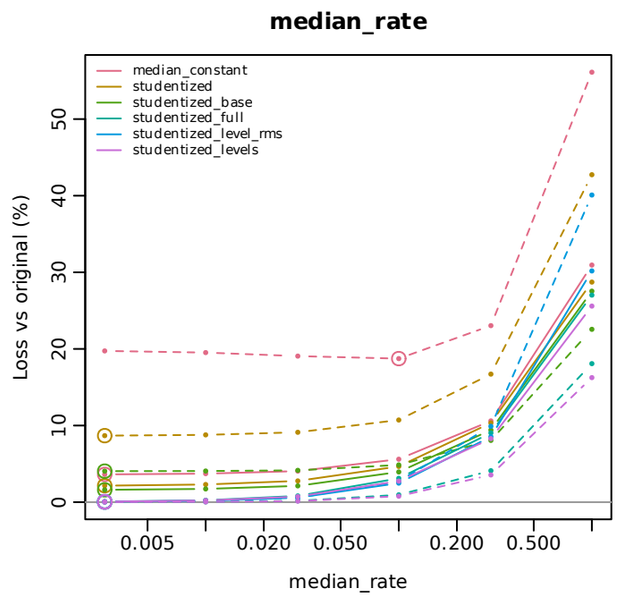
· m = 0.01
• m = 0.03

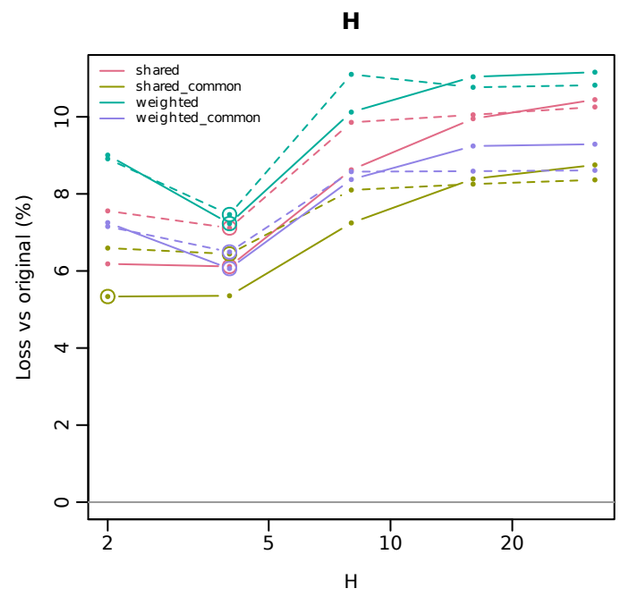
● m = 0.1
● m = 0.3

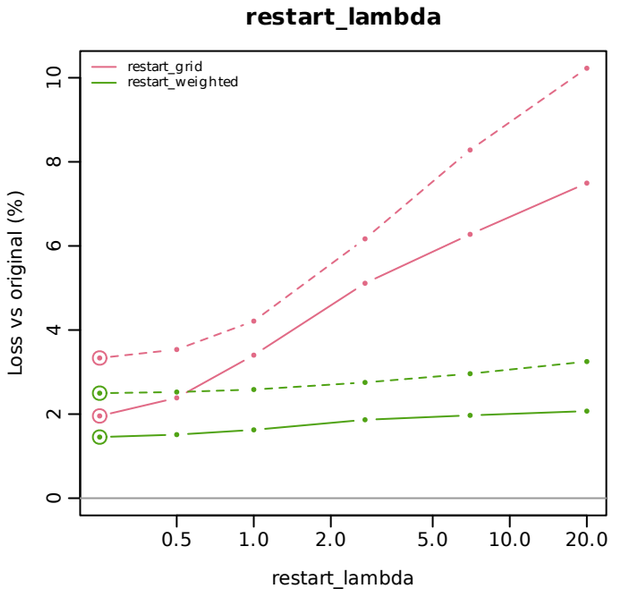
covid | design hub_first_months | feedback delay 0 (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations

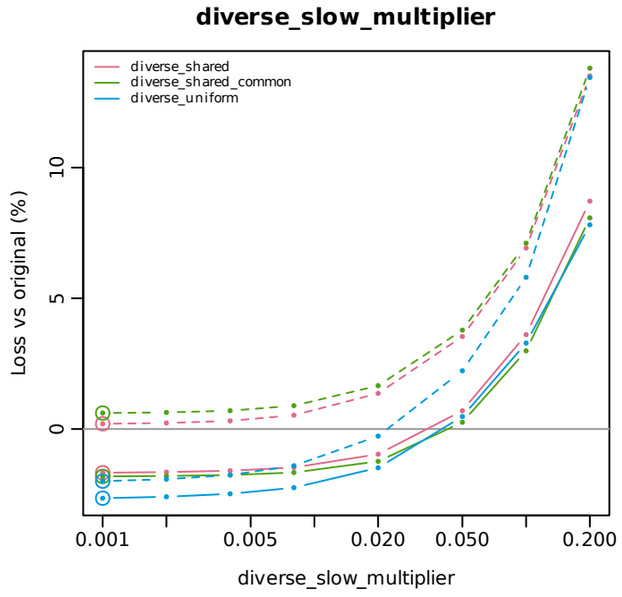


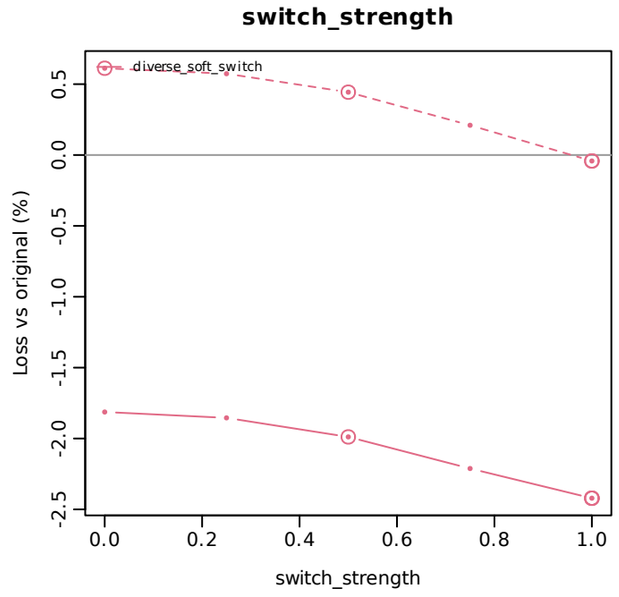


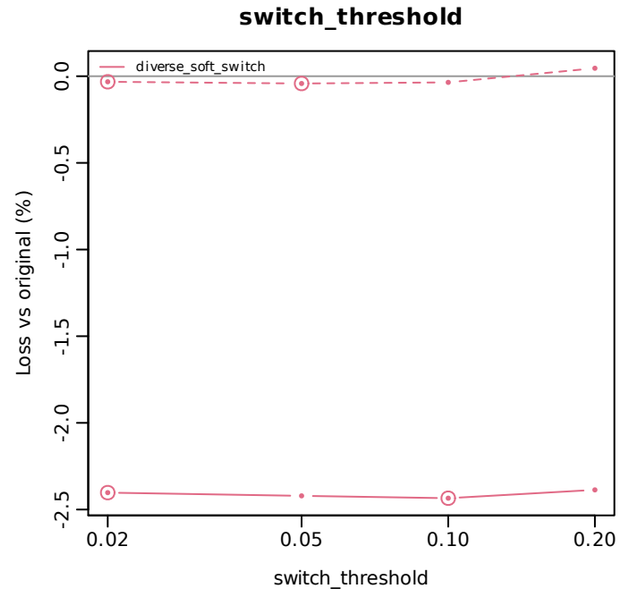


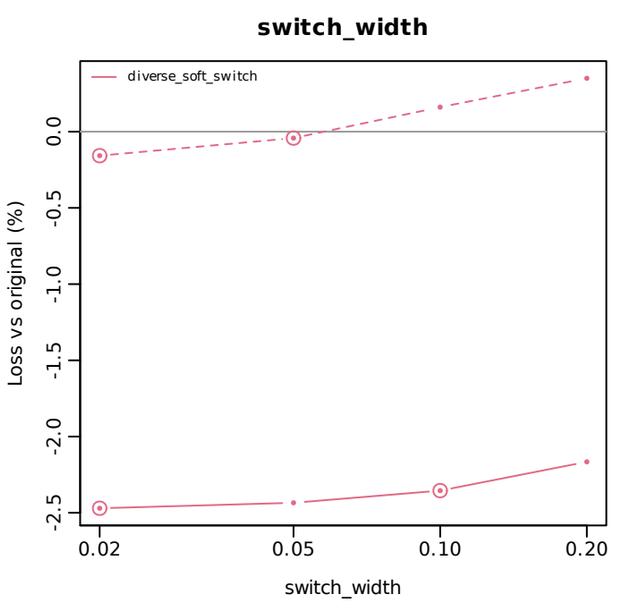


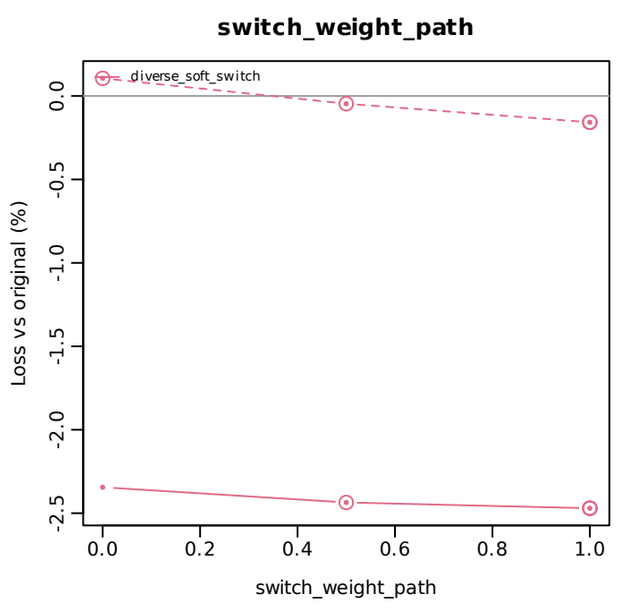


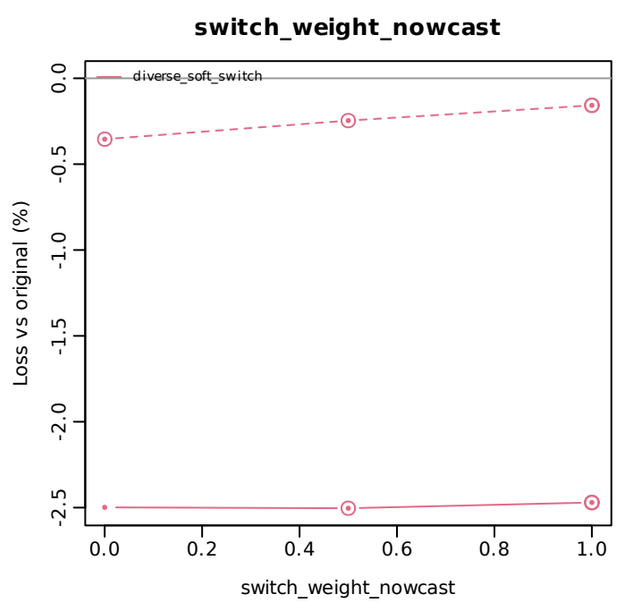


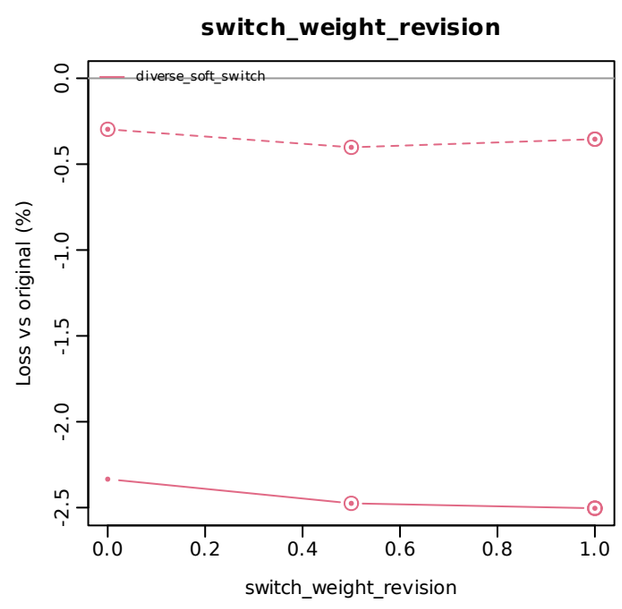


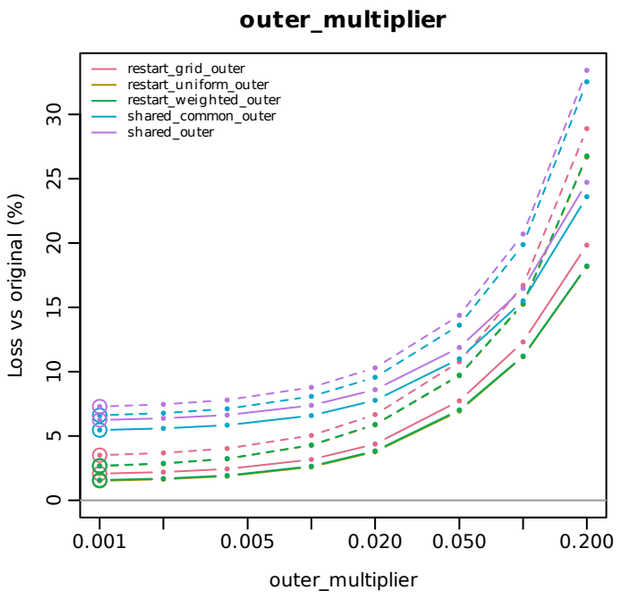












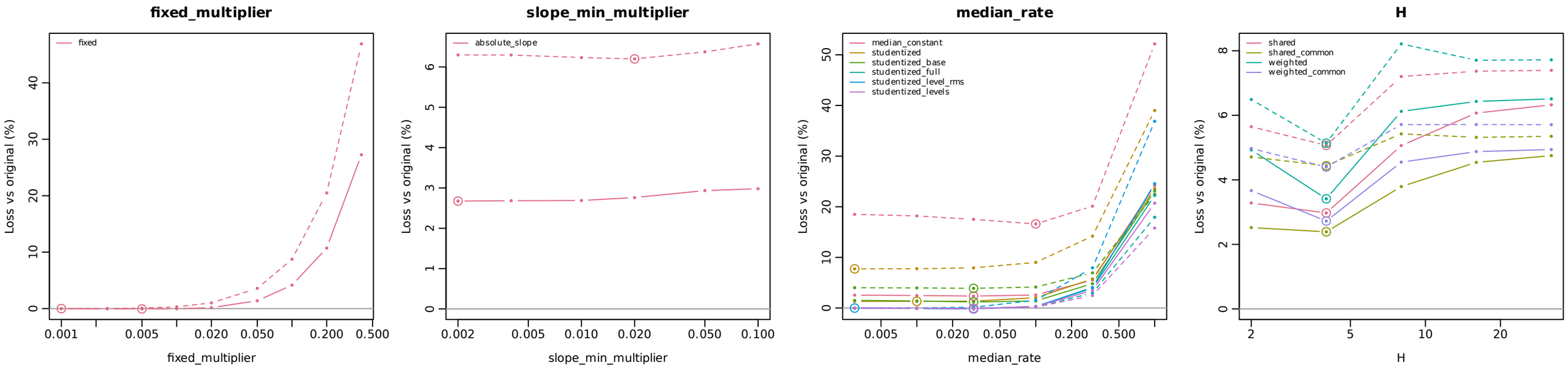
—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

covid | design hub_first_months | feedback delay 1 (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations

Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

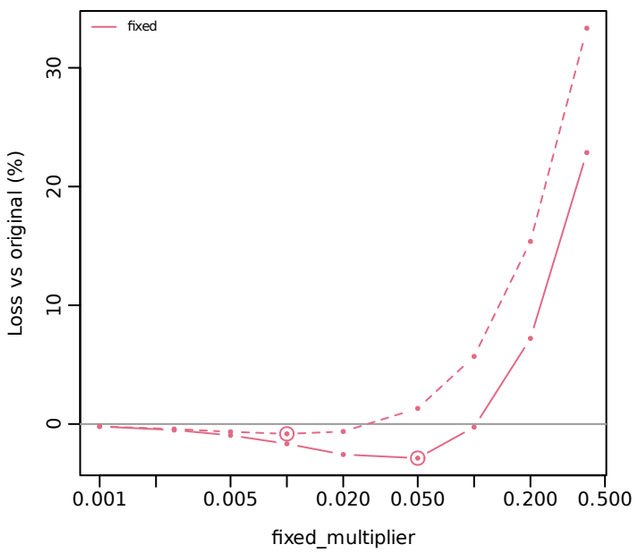
covid | design hub_first_months | feedback delay 0 + revisions (re-run) (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations



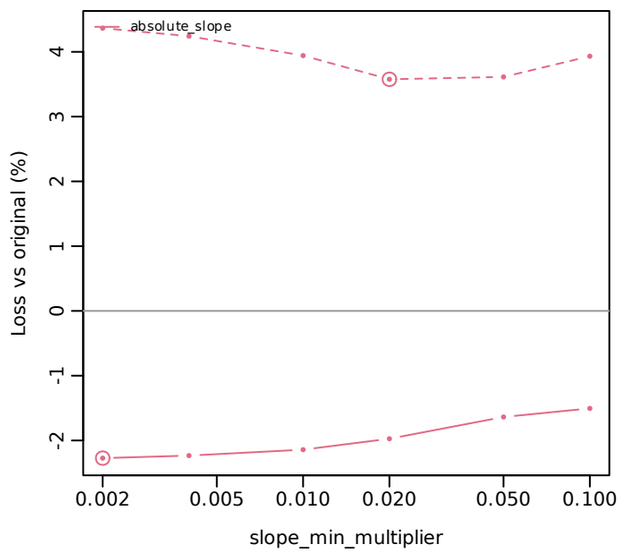
Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

flu | design hub_first_months | feedback delay 0 (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations

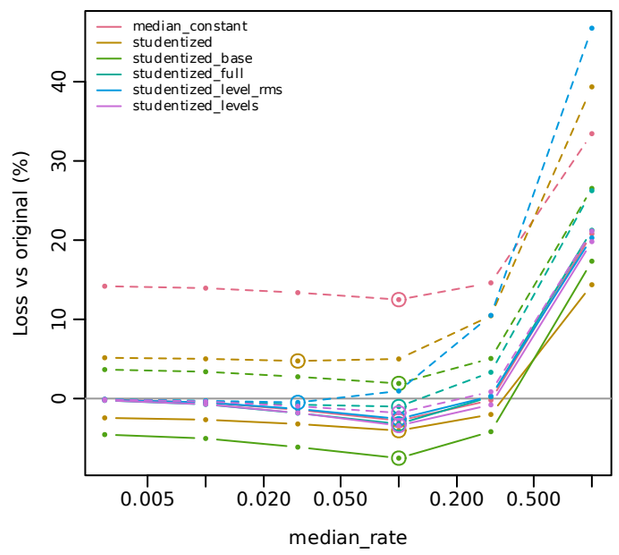
fixed_multiplier



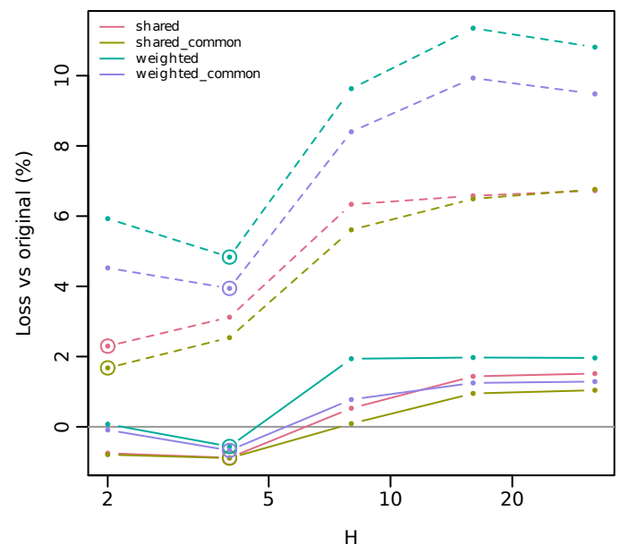
slope_min_multiplier



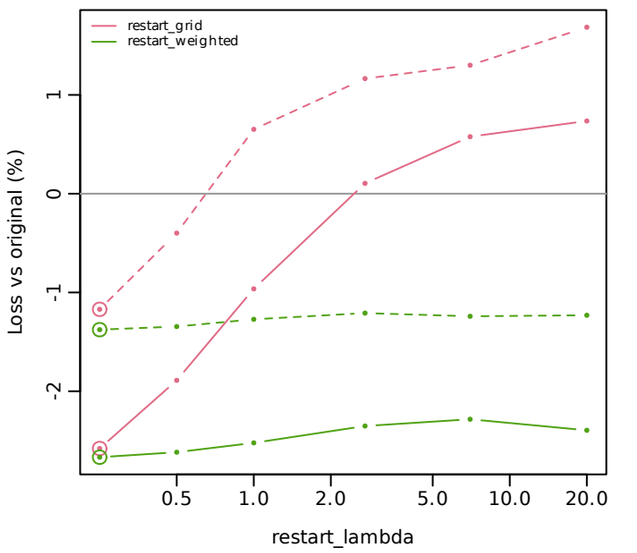
median_rate



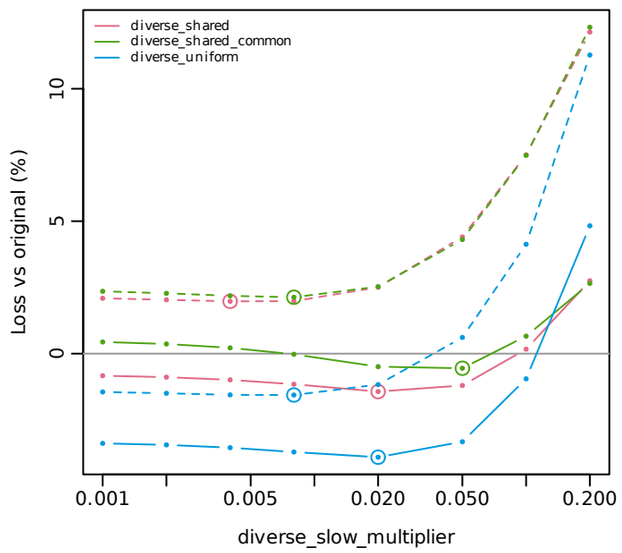
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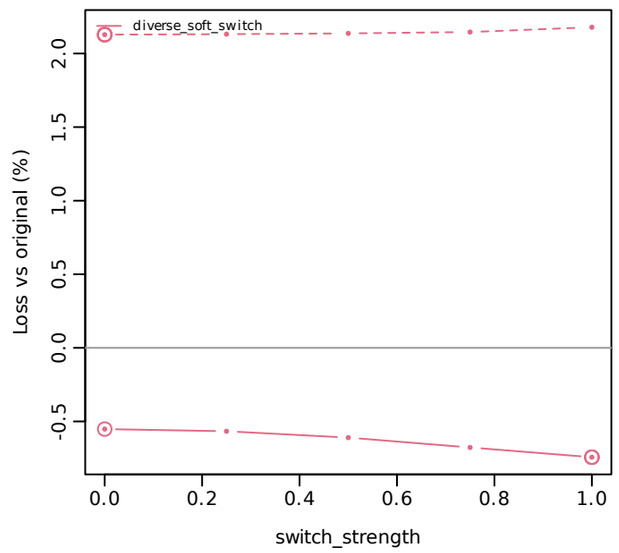
restart_lambda



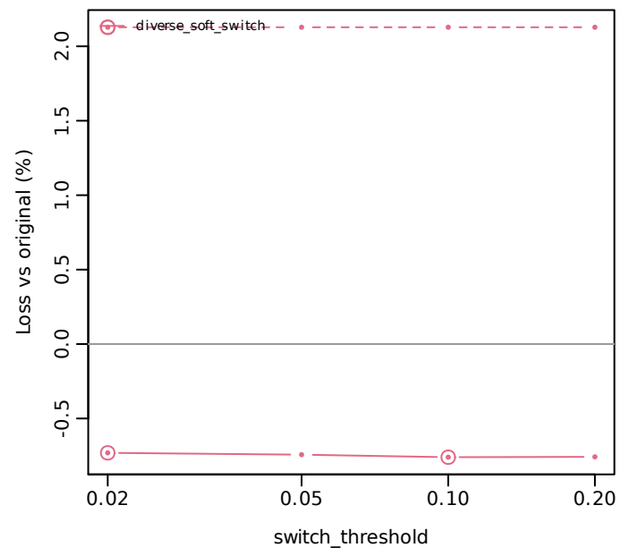
diverse_slow_multiplier



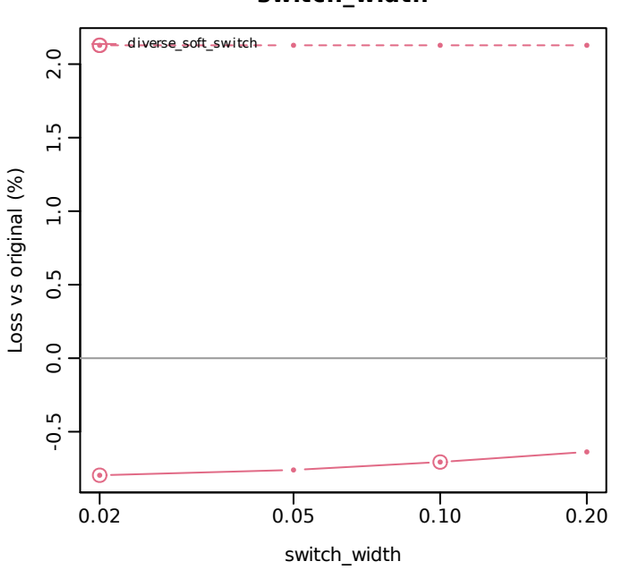
switch_strength



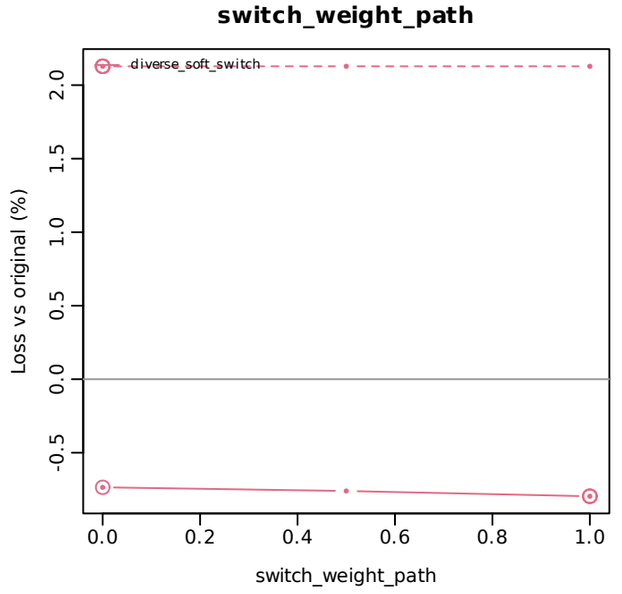
switch_threshold



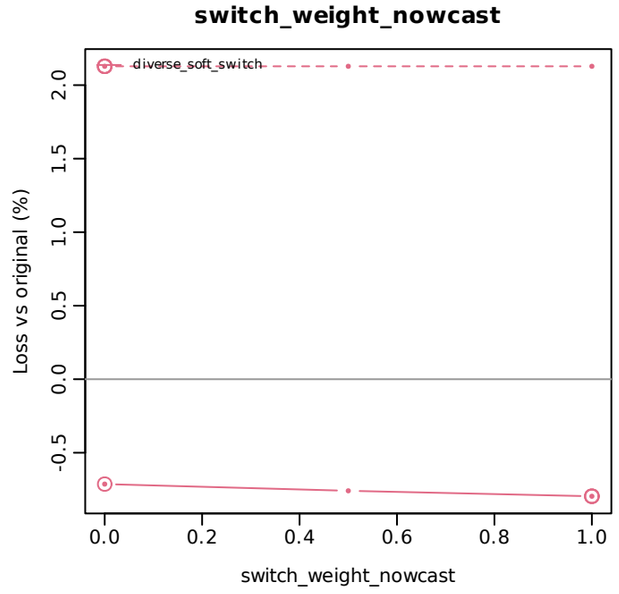
switch_width



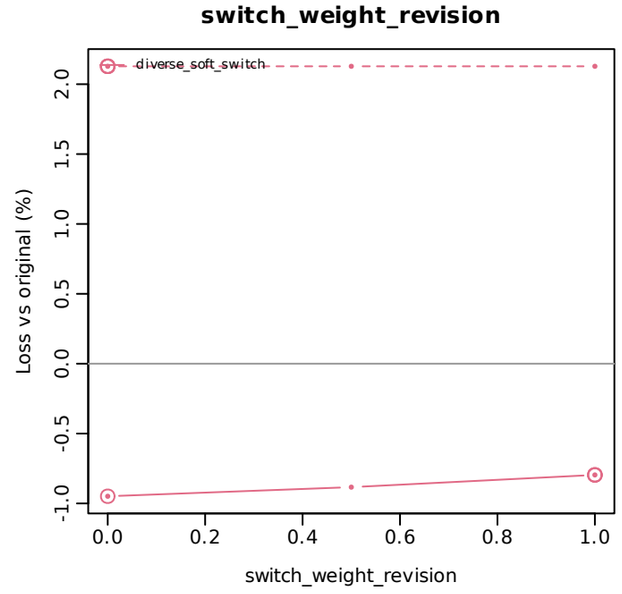
switch_weight_path



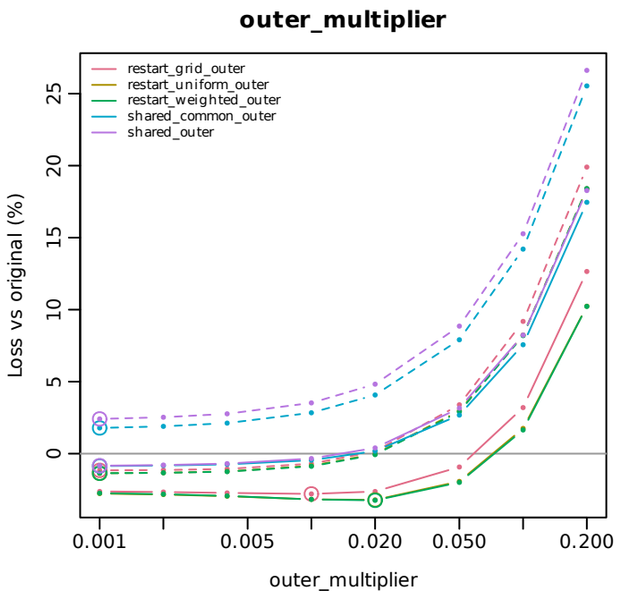
switch_weight_nowcast



switch_weight_revision



outer_multiplier

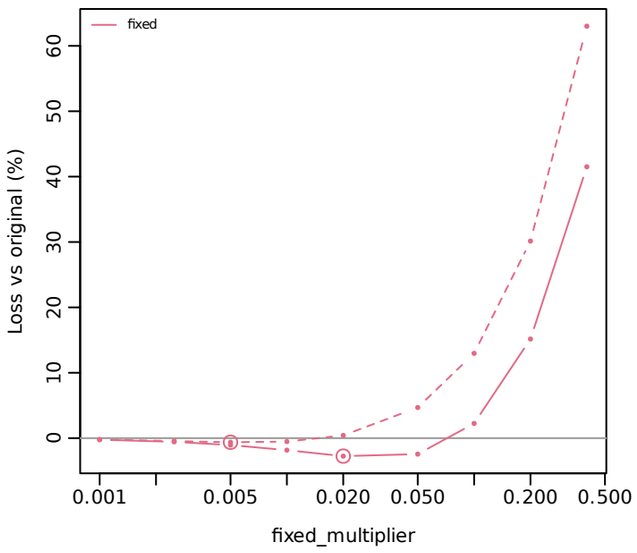


—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

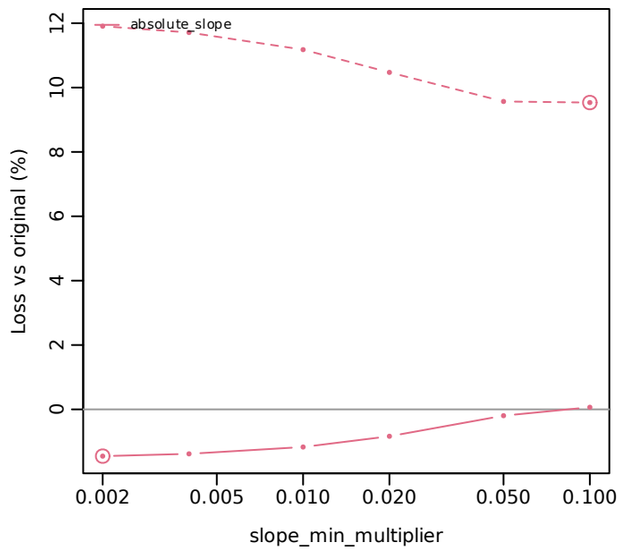
Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

flu | design hub_first_months | feedback delay 1 (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations

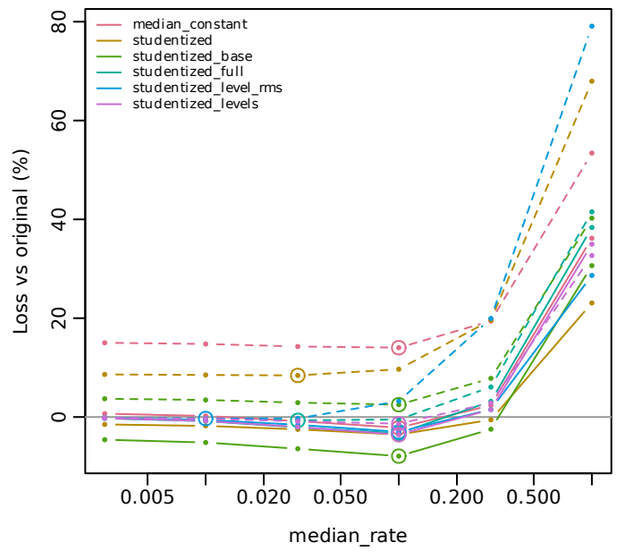
fixed_multiplier



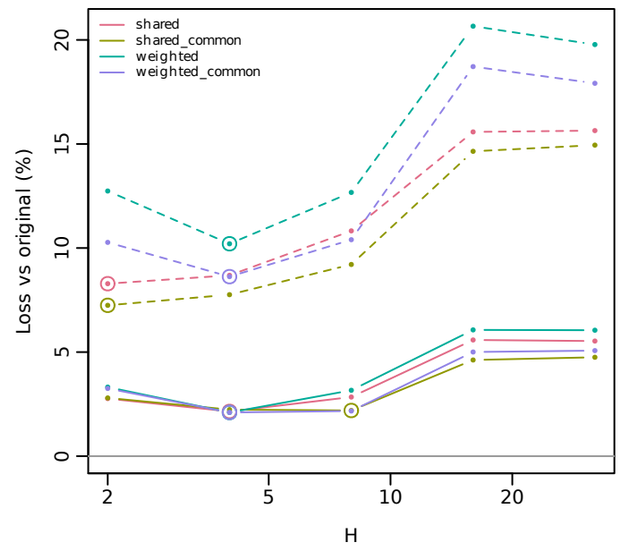
slope_min_multiplier



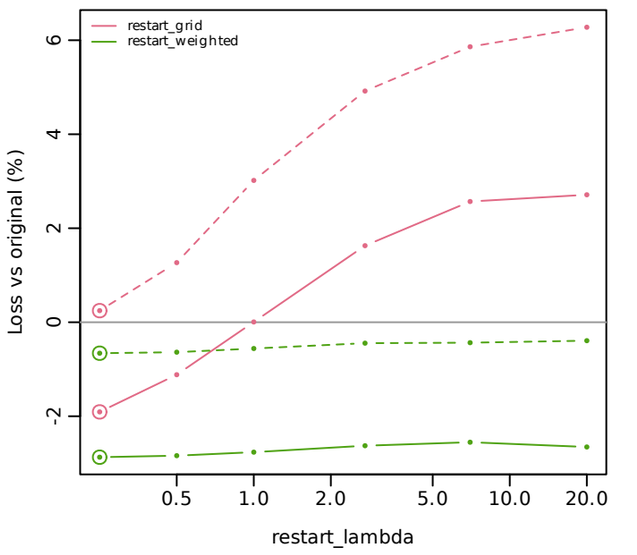
median_rate



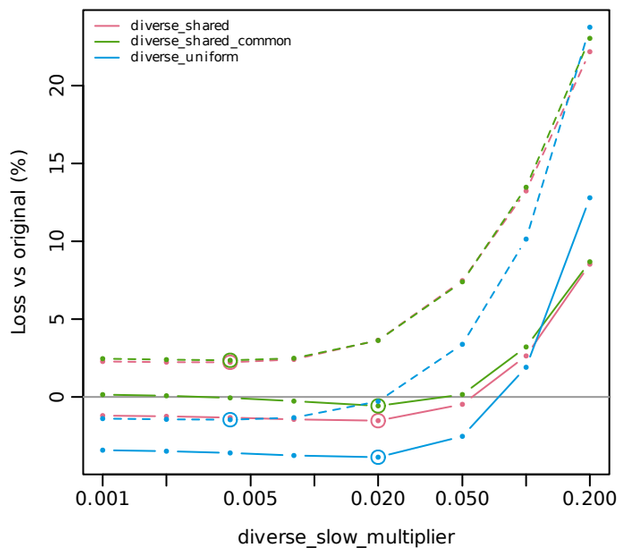
H



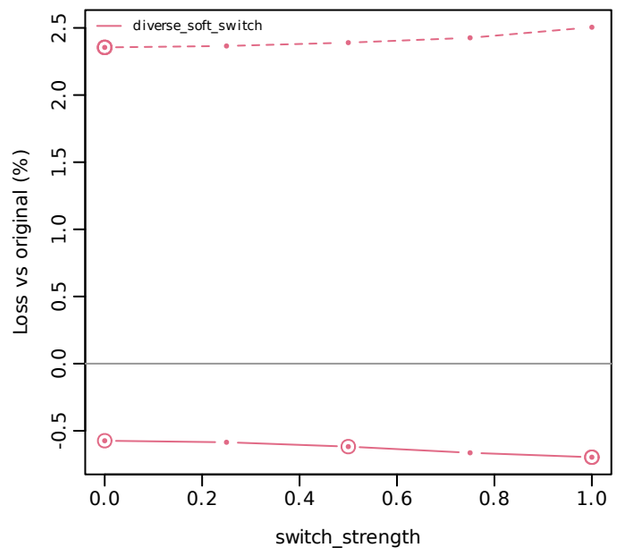
restart_lambda



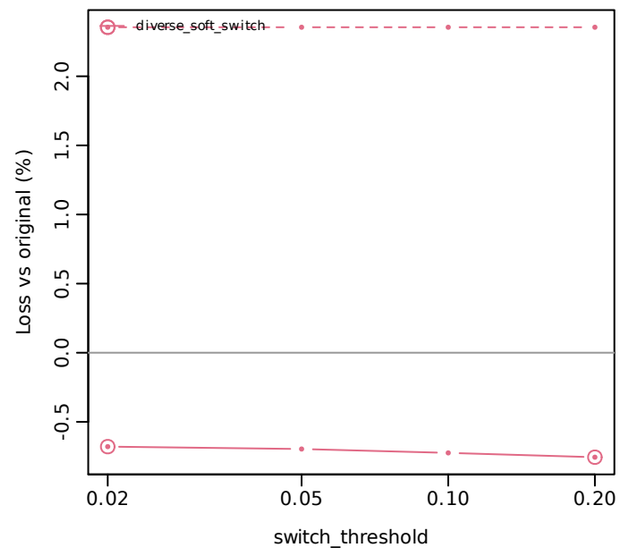
diverse_slow_multiplier



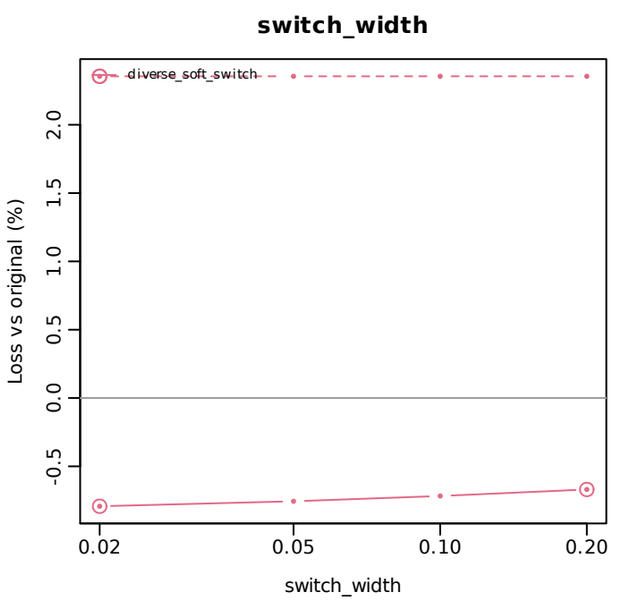
switch_strength



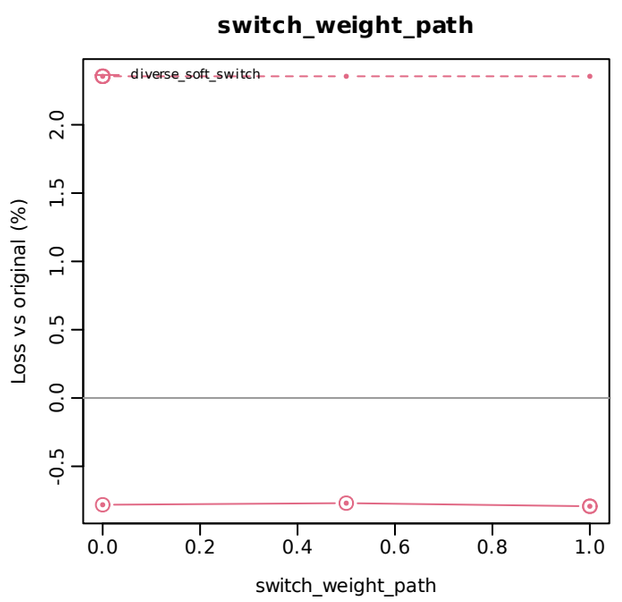
switch_threshold



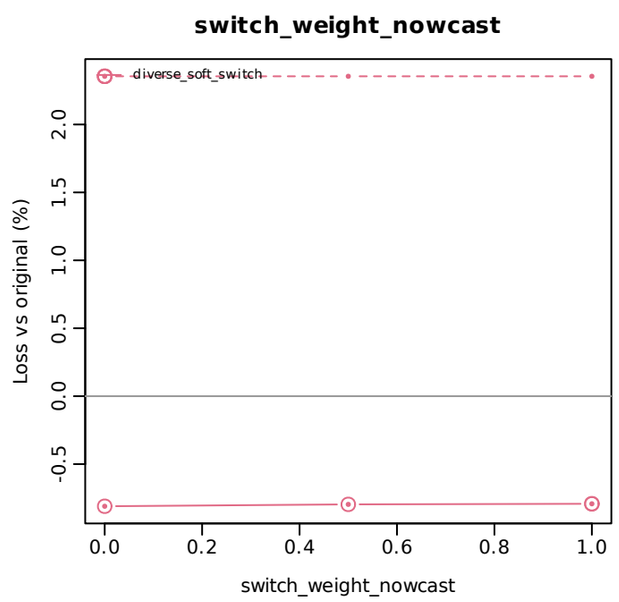
switch_width



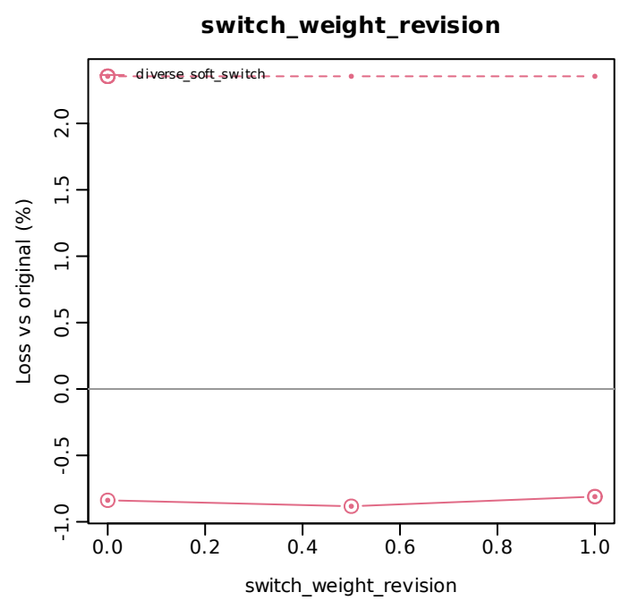
switch_weight_path



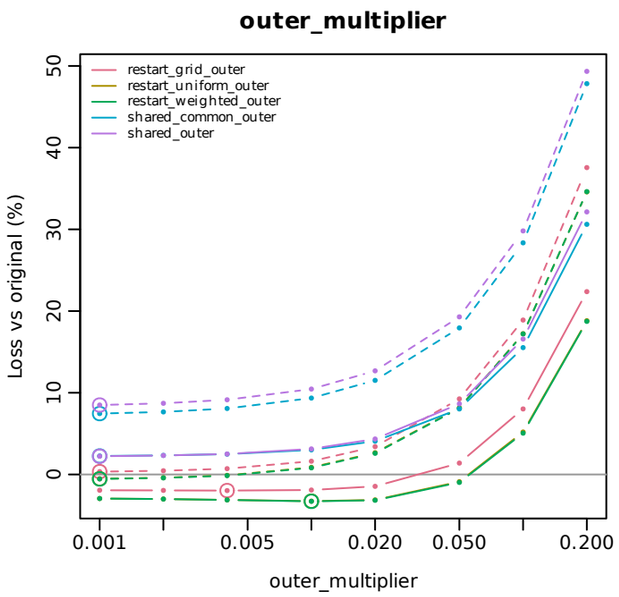
switch_weight_nowcast



switch_weight_revision



outer_multiplier

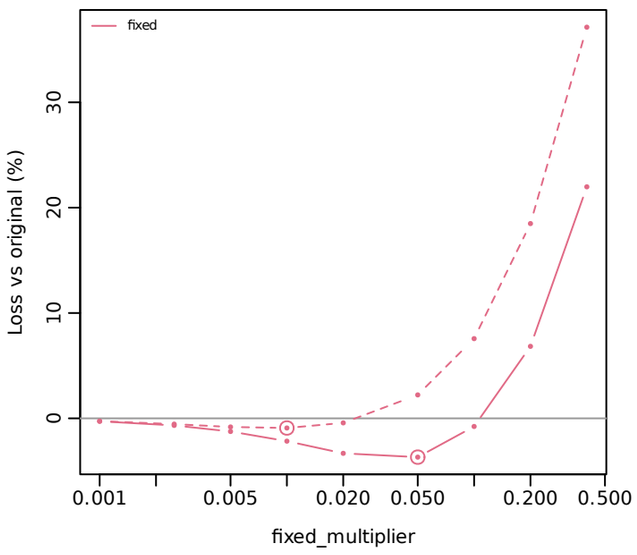


—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

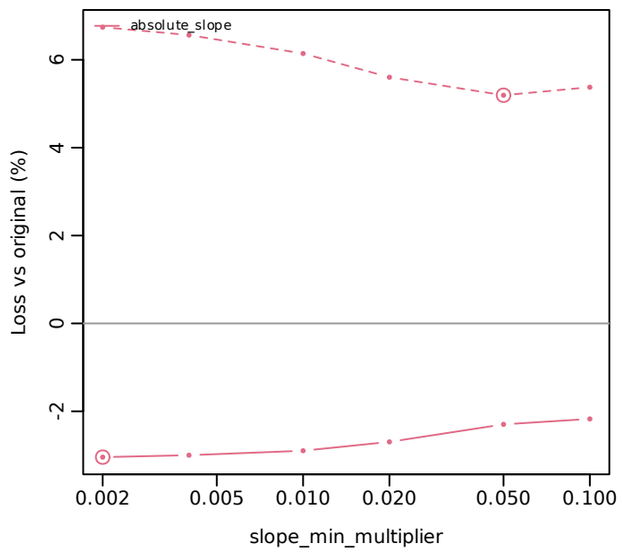
Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

flu | design hub_first_months | feedback delay 0 + revisions (re-run) (latest-at-use-v2) | tuning period Nov 2024 to May 2025 | horizons -1 to 3 | all locations

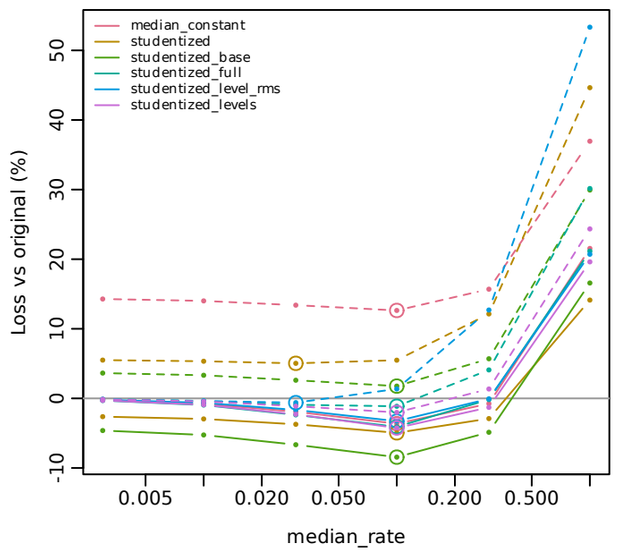
fixed_multiplier



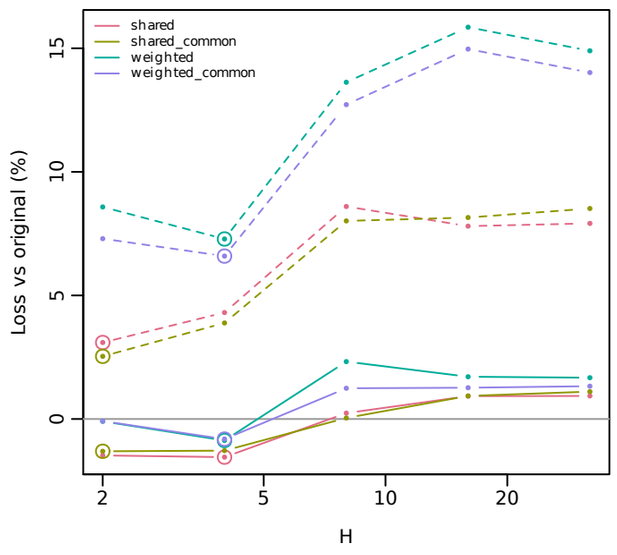
slope_min_multiplier



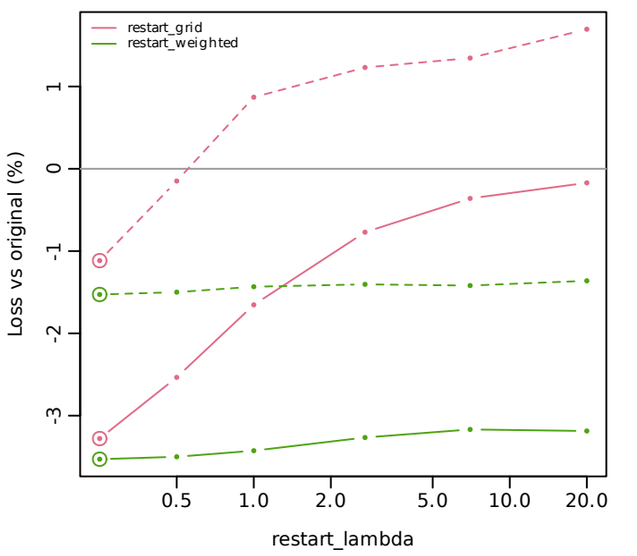
median_rate



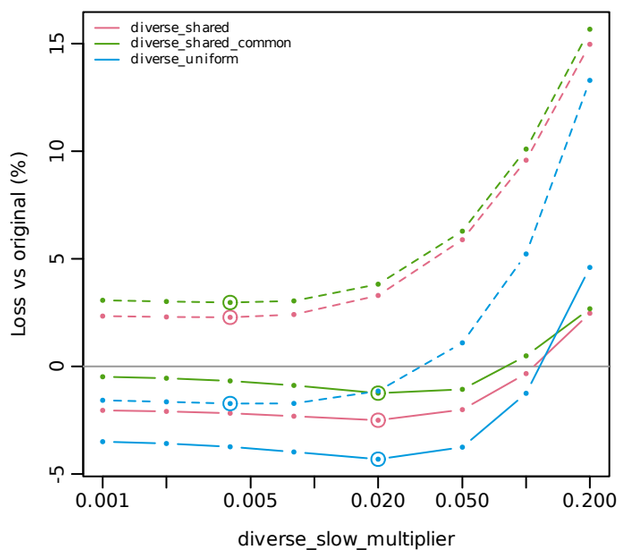
H



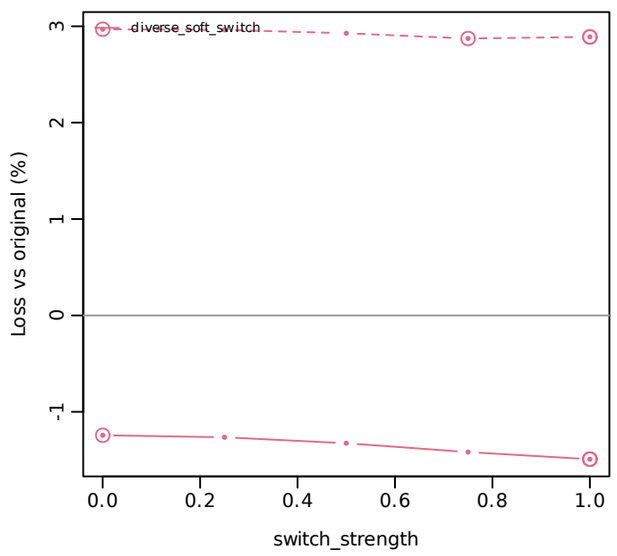
restart_lambda



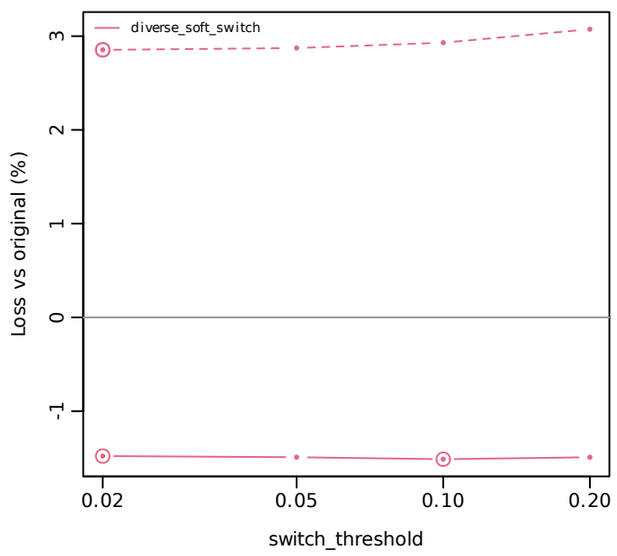
diverse_slow_multiplier



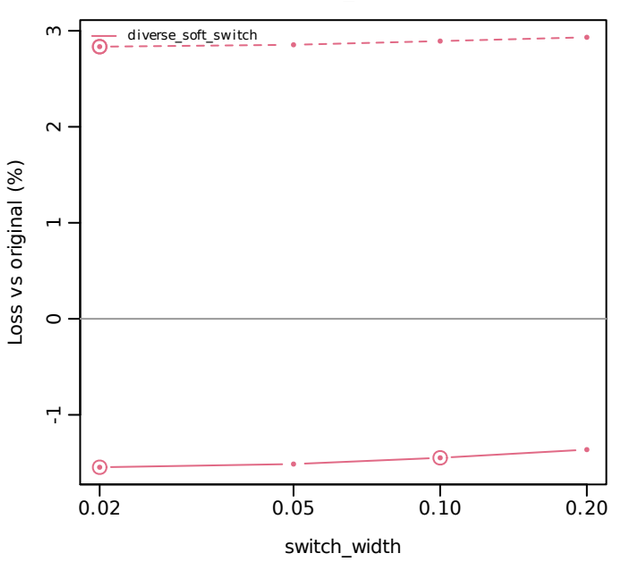
switch_strength



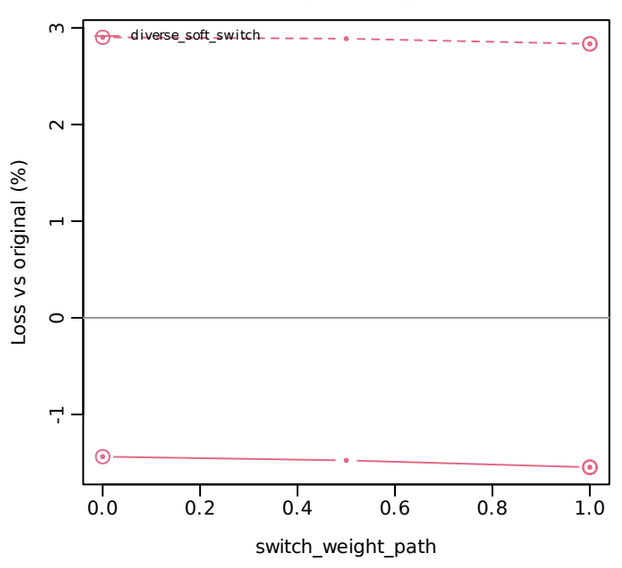
switch_threshold



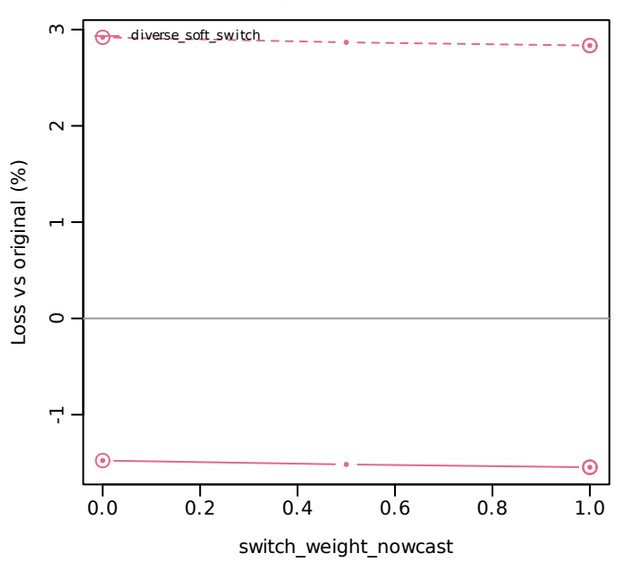
switch_width



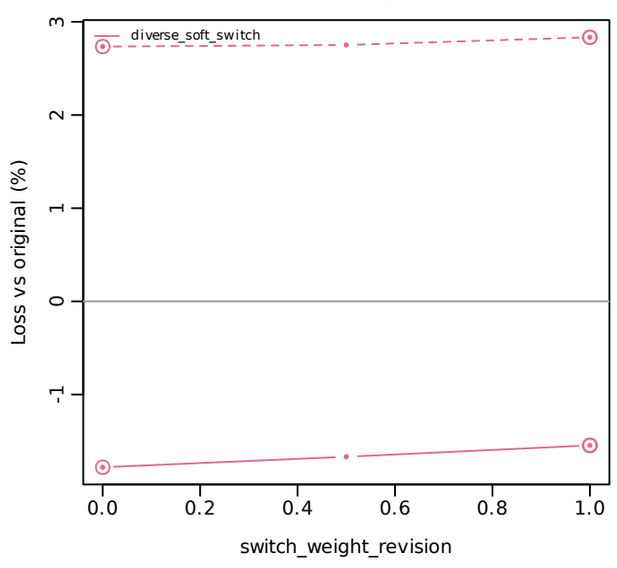
switch_weight_path



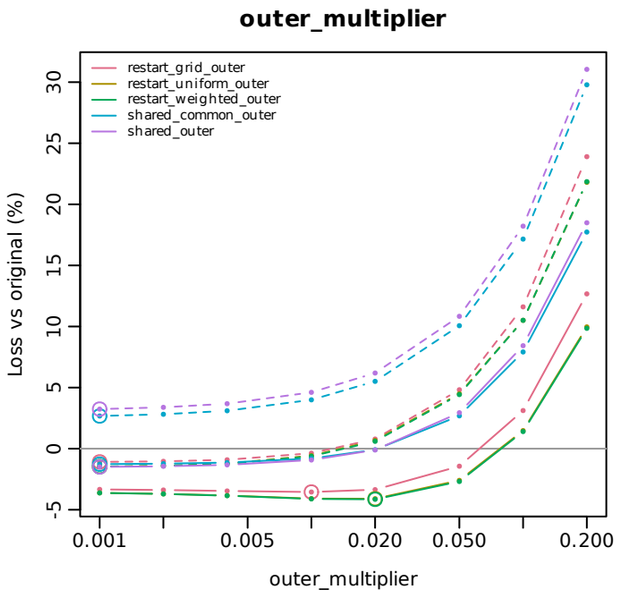
switch_weight_nowcast



switch_weight_revision



outer_multiplier



—— log1p tracker (loss of log(counts + 1)) - - - identity tracker (loss in counts per 100k residents) o chosen value

Mean quantile loss relative to the original forecasts (mean over locations and horizons), on the scale of each tracker

covid | all 53 locations | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.
Small: single locations; large: means over locations; axes: 1st to 99th percentile of both.

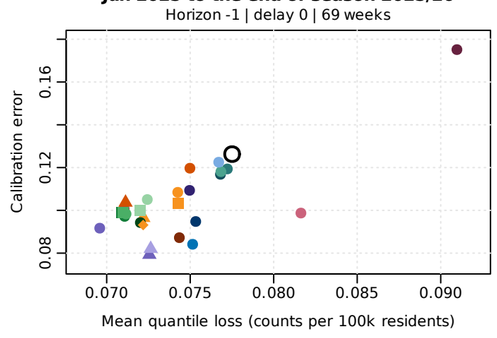
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection).

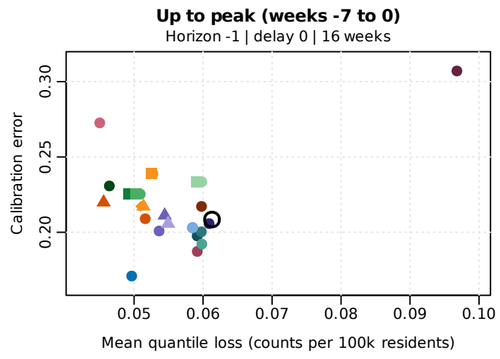
covid | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates, loss in counts per 100k residents

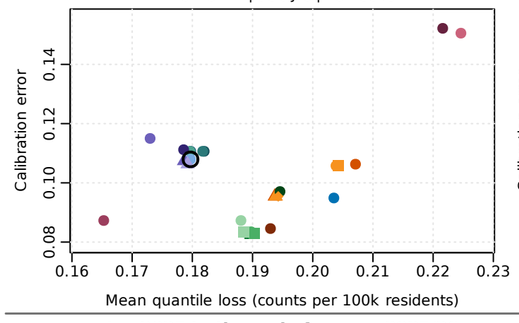
Rows: feedback design (delay d: learn from the report d weeks after the first one; re-run: re-run every week on the reports known then) and window; columns: horizon.

Horizon -1

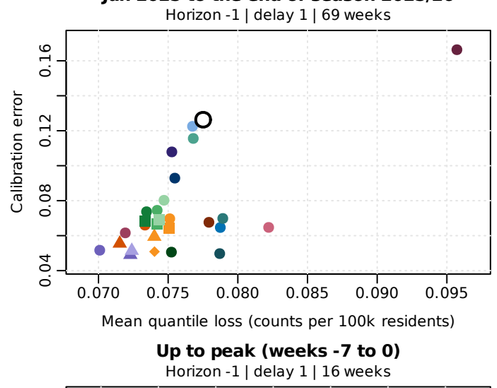
Feedback delay 0

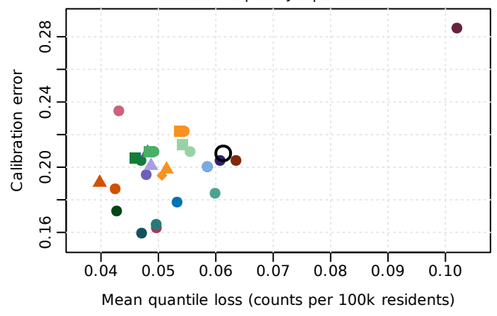


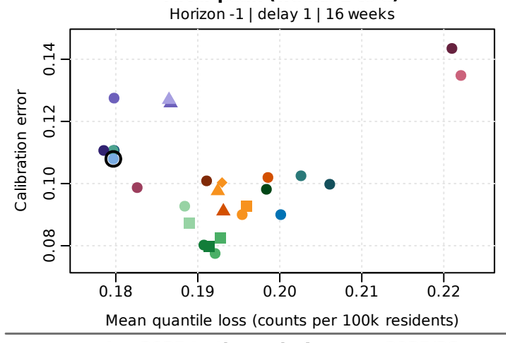




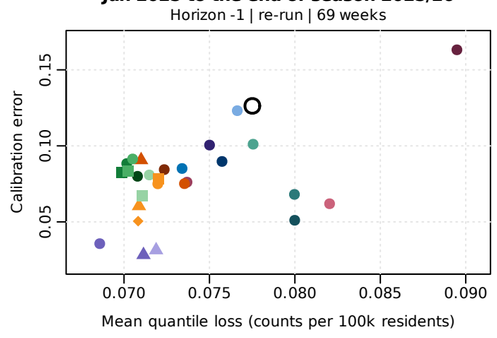
Feedback delay 1

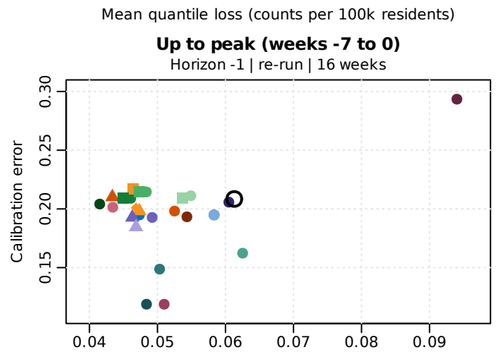


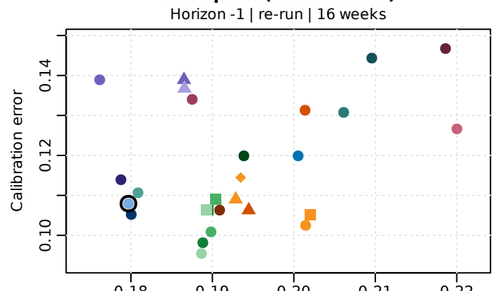




Feedback delay 0 + revisions (re-run)

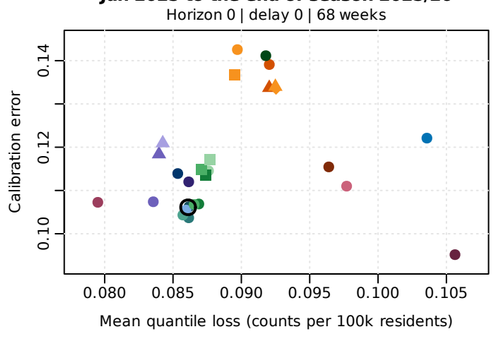


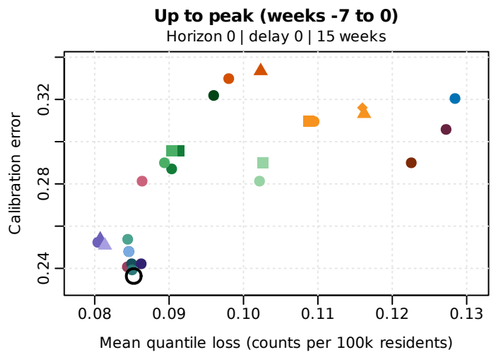


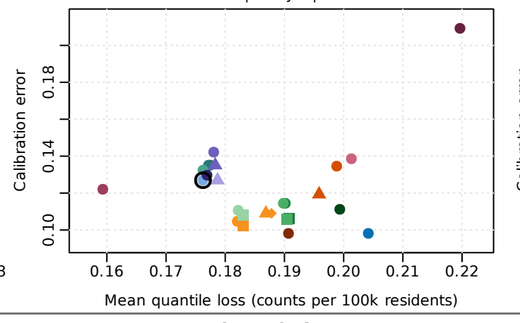


Horizon 0

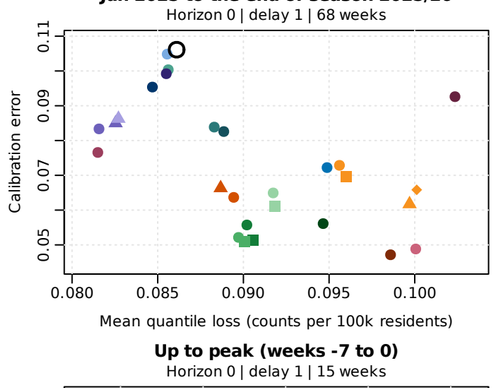
Feedback delay 0

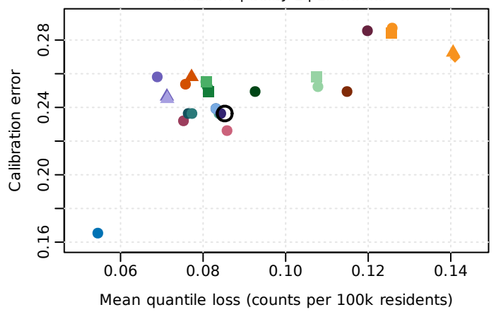


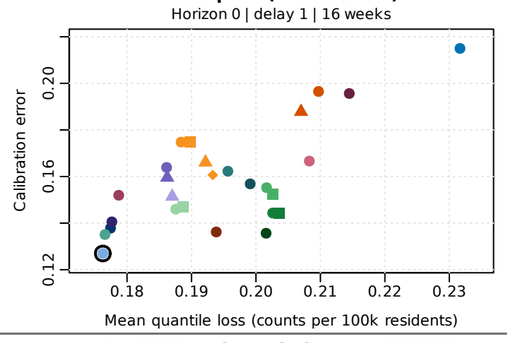




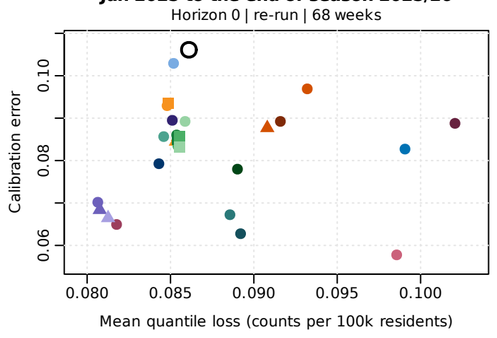
Feedback delay 1

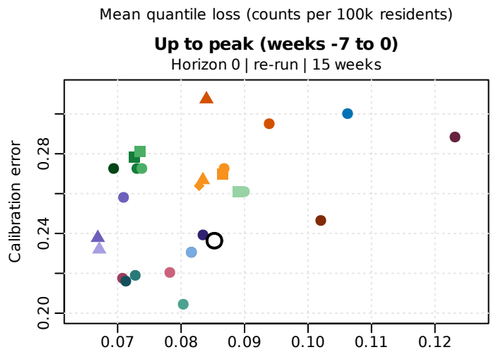


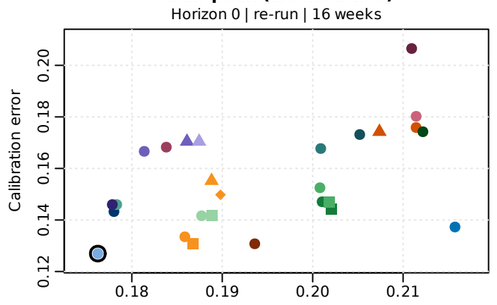




Feedback delay 0 + revisions (re-run)

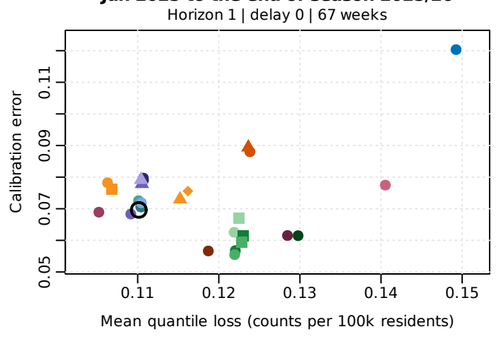


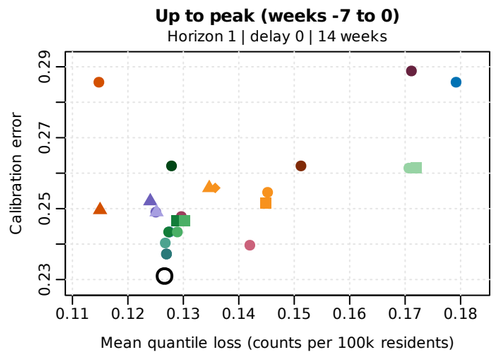


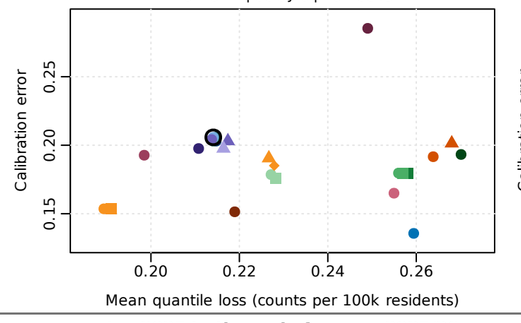


Horizon 1

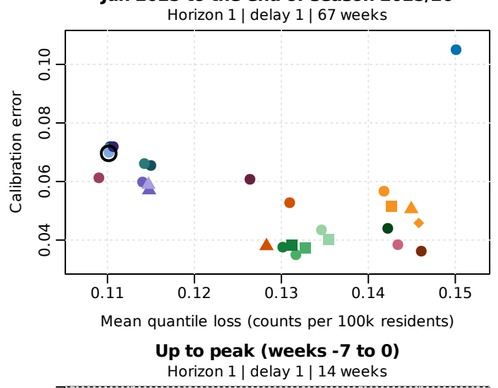
Feedback delay 0

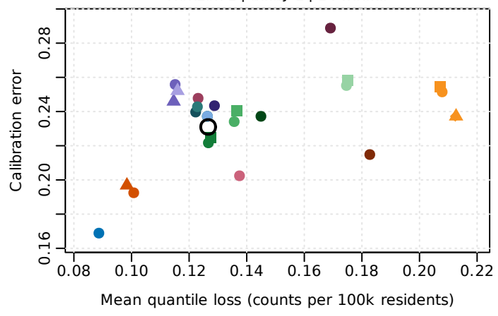


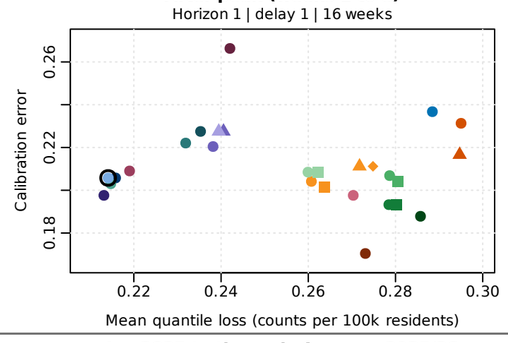




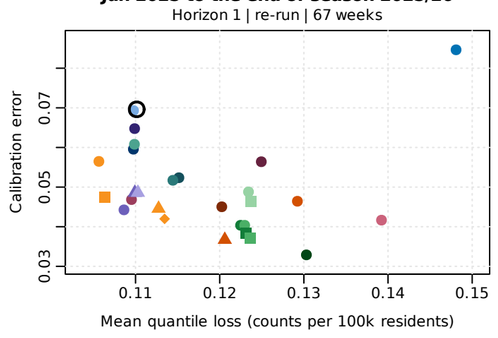
Feedback delay 1

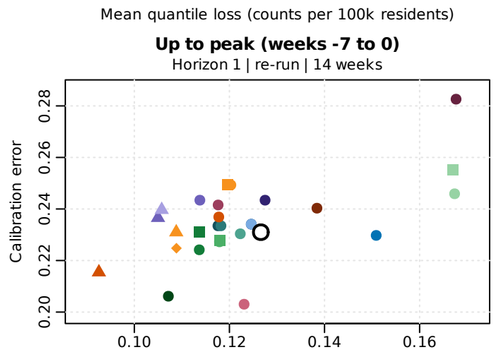


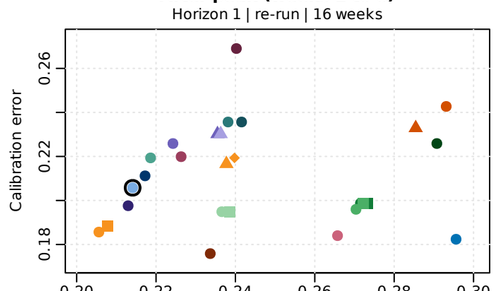




Feedback delay 0 + revisions (re-run)

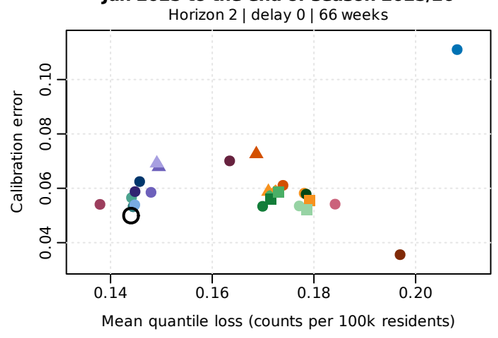


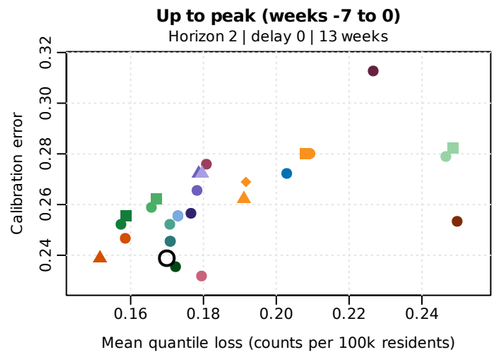


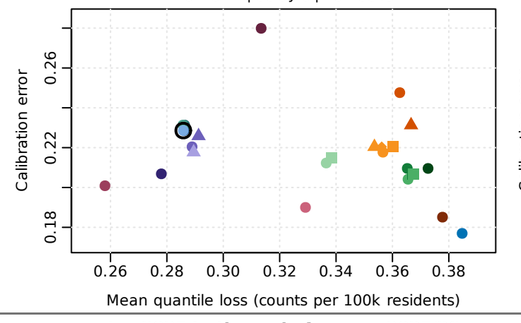


Horizon 2

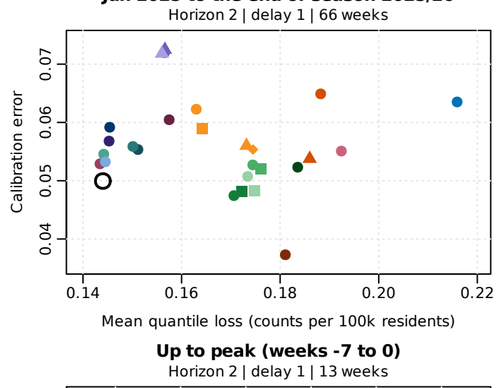
Feedback delay 0

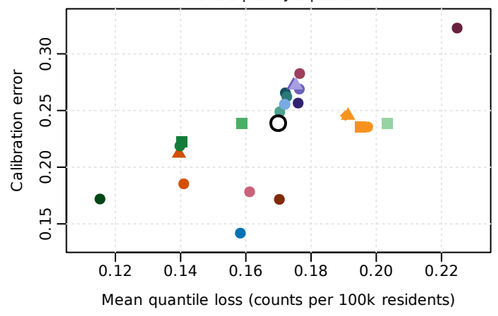


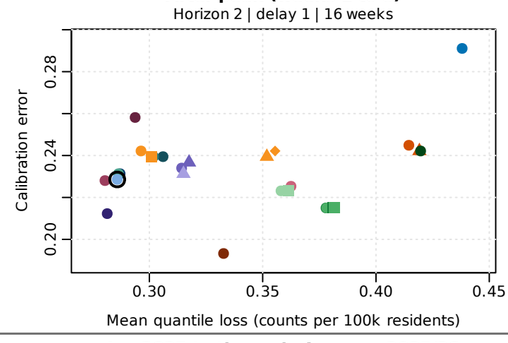




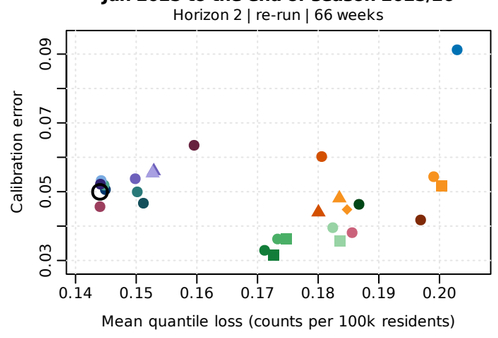
Feedback delay 1

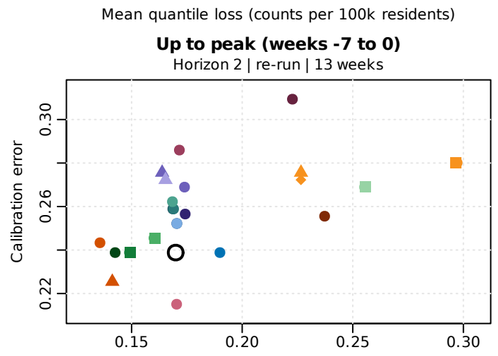


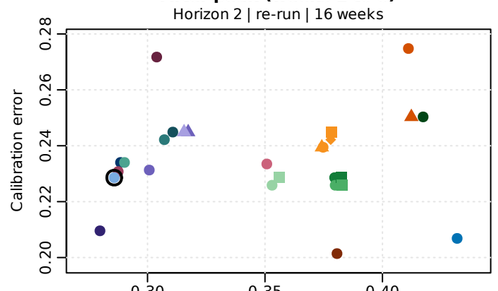




Feedback delay 0 + revisions (re-run)

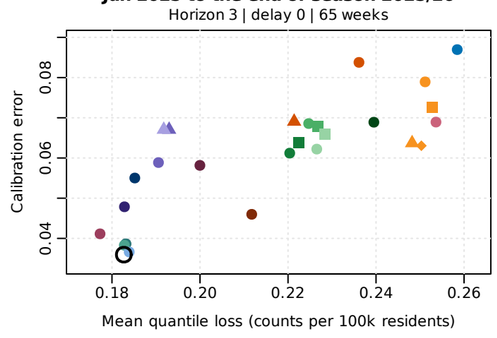


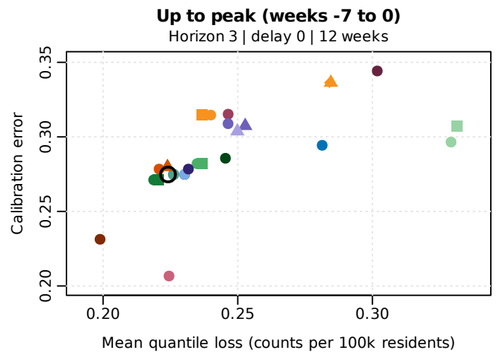


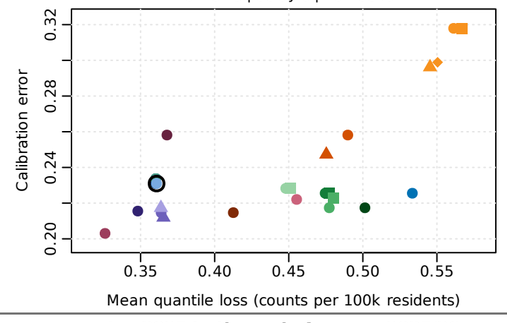


Horizon 3

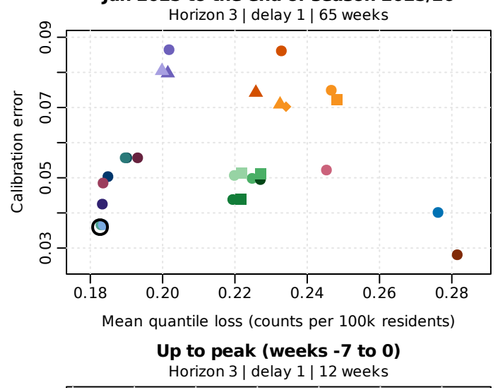
Feedback delay 0

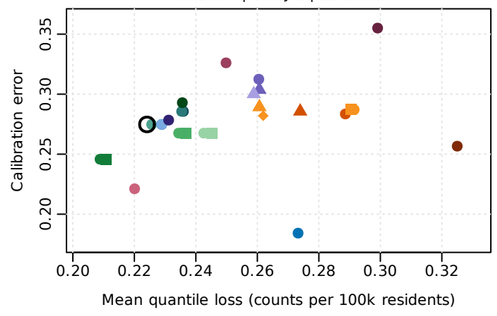


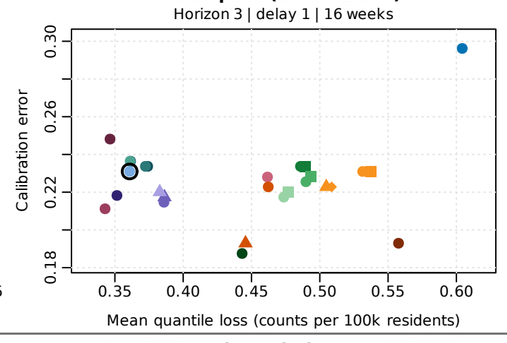




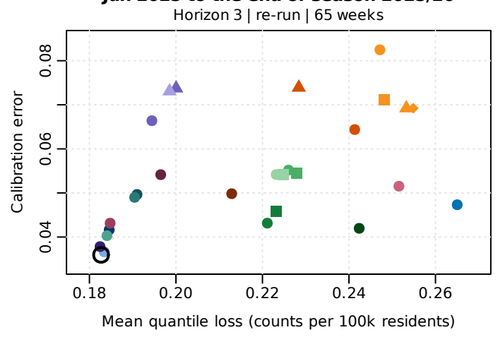
Feedback delay 1

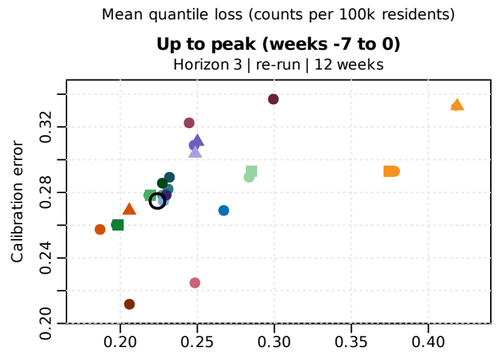


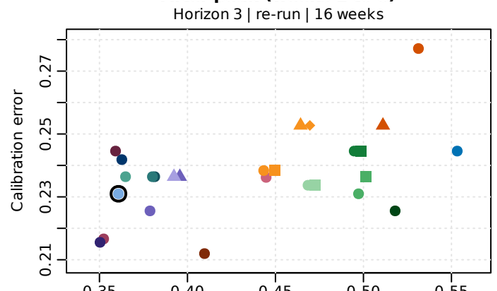




Feedback delay 0 + revisions (re-run)







Single trackers
- Original
- Fixed MultiQT (eta=0.001*S0)
- Count-slope rate (min m=0.02)
- Target oracle (hindsight)

Rate-grid experts
- Rate grid: equal
- Rate grid: learned, no sharing (H=4)
- Rate grid: learned, no sharing, common weights (H=4)
- Rate grid: learned + sharing (H=4)
- Rate grid: learned + sharing, common weights (H=4)
- Rate grid: sharing + outer (outer m=0.001)
- Rate grid: sharing, common weights + outer (outer m=0.001)

Weekly starts: immediate admission
- First start only (m=0.05)
- Weekly starts: equal (m=0.05)
- Weekly starts: learned (m=0.05, lambda=0.25)
- Starts x rate grid: learned (lambda=0.25)
- Starts equal + outer (outer m=0.001)
- Starts learned + outer (outer m=0.001)
- Starts x rate grid + outer (outer m=0.001)

Diverse experts
- Diverse experts: equal (slow m=0.001)
- Diverse experts: learned + sharing (slow m=0.001, H=8)
- Diverse experts: learned + sharing, common weights (slow m=0.002)
- Diverse experts: soft switching (strength=1)

Studentized median
- Studentized median (RMS of 5, normal start, eta=0.003)
- Studentized median (base 80% width, normal start, eta=0.01)
- Median, constant sigma (RMS of 5, normal start, eta=0.1)

Studentized full base
- Studentized full base (base 80% width, eta=0.003)
- Studentized full base (level scales, eta=0.003)
- Studentized full base (level RMS of 5, eta=0.003)

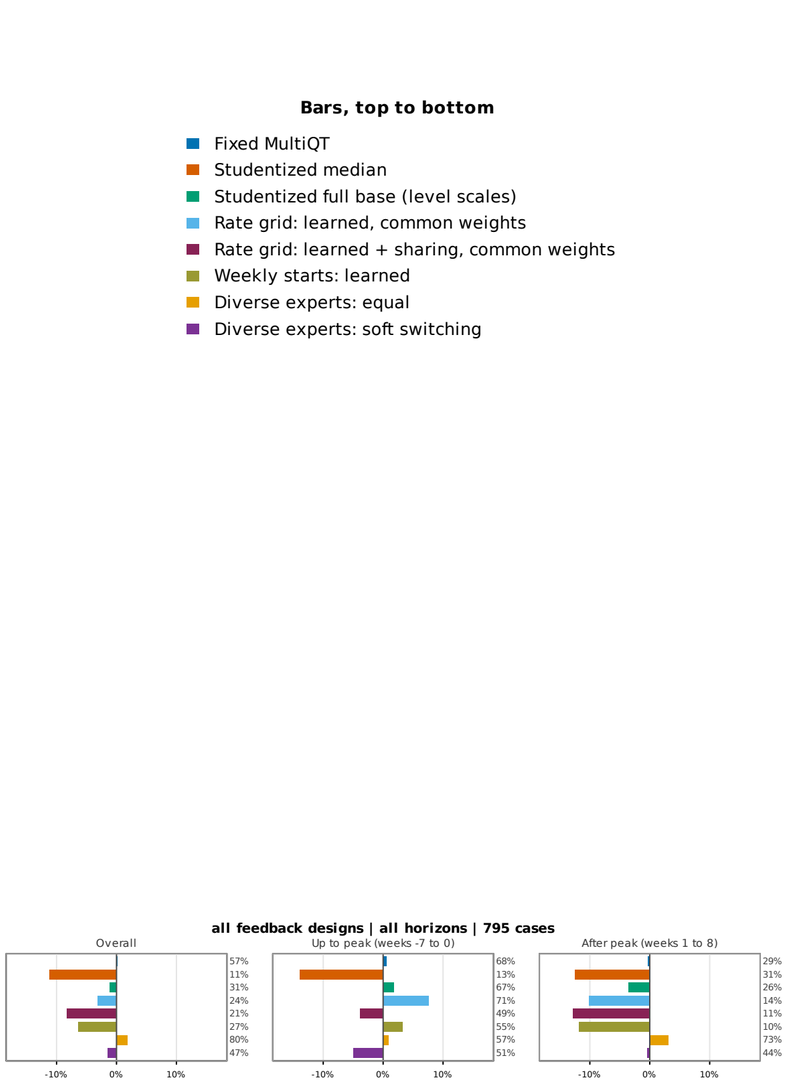
Calibration error (CE): mean absolute coverage residual after averaging weeks within each quantile level.
Open circle: original forecasts. Filled circle: weights per quantile level; triangle: one weight vector shared by all levels; square / diamond: the same + outer MultiQT.
Each feedback design uses the settings tuned with it; the values in the legend are those of feedback delay 0 (all: output/results/tuning).
Studentized median methods: sigma_t = RMS of the last 5 released median residuals, or the 80% interval width of the base / 2.563 (a normal sd).
With residuals, the original base is issued until sigma_t exists; these weeks are scored as the base.
Studentized full base: offsets are initially zero; level scales = normal sd of the central interval of the base at each level, or the RMS of the last 5 released residuals at each level (weighted isotonic projection).

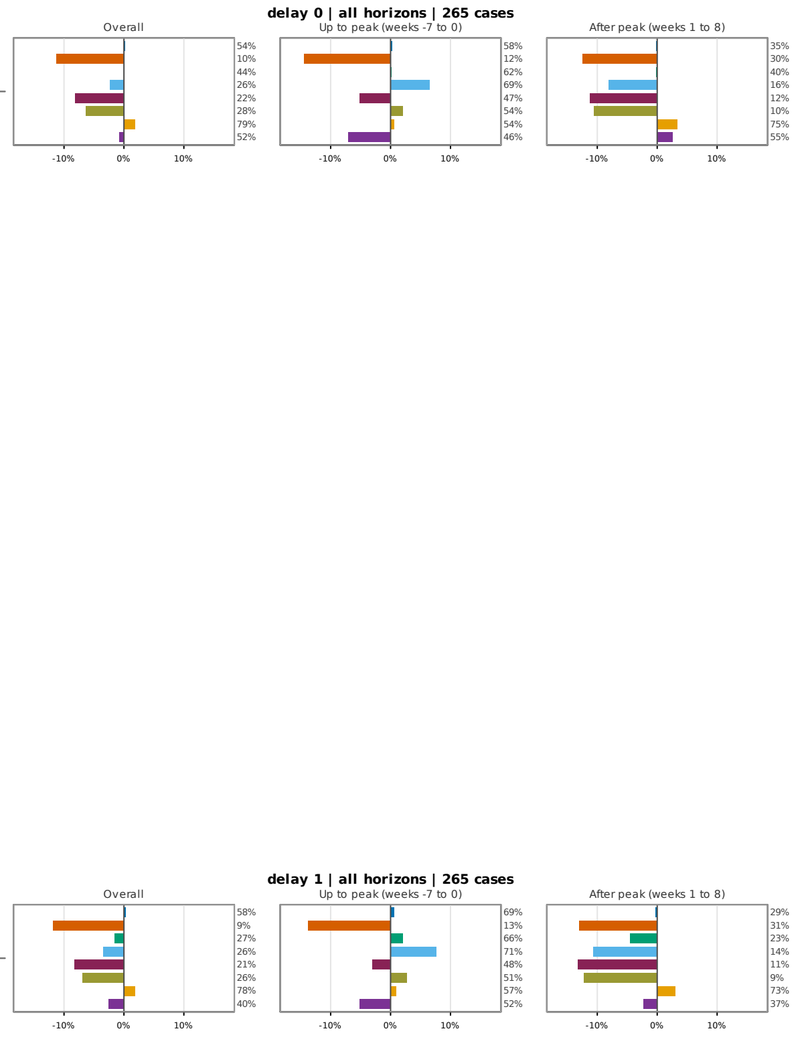
covid | all 53 locations pooled | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

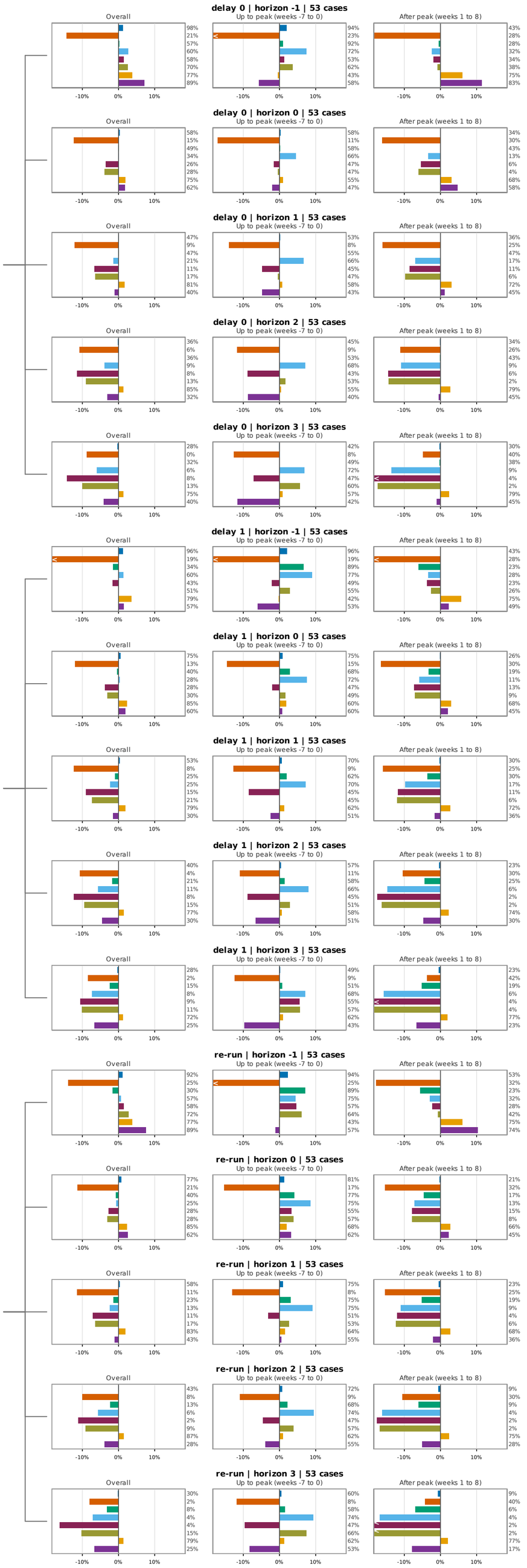
Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.

Bars, top to bottom

- Fixed MultiQT
- Studentized median
- Studentized full base (level scales)
- Rate grid: learned, common weights
- Rate grid: learned + sharing, common weights
- Weekly starts: learned
- Diverse experts: equal
- Diverse experts: soft switching







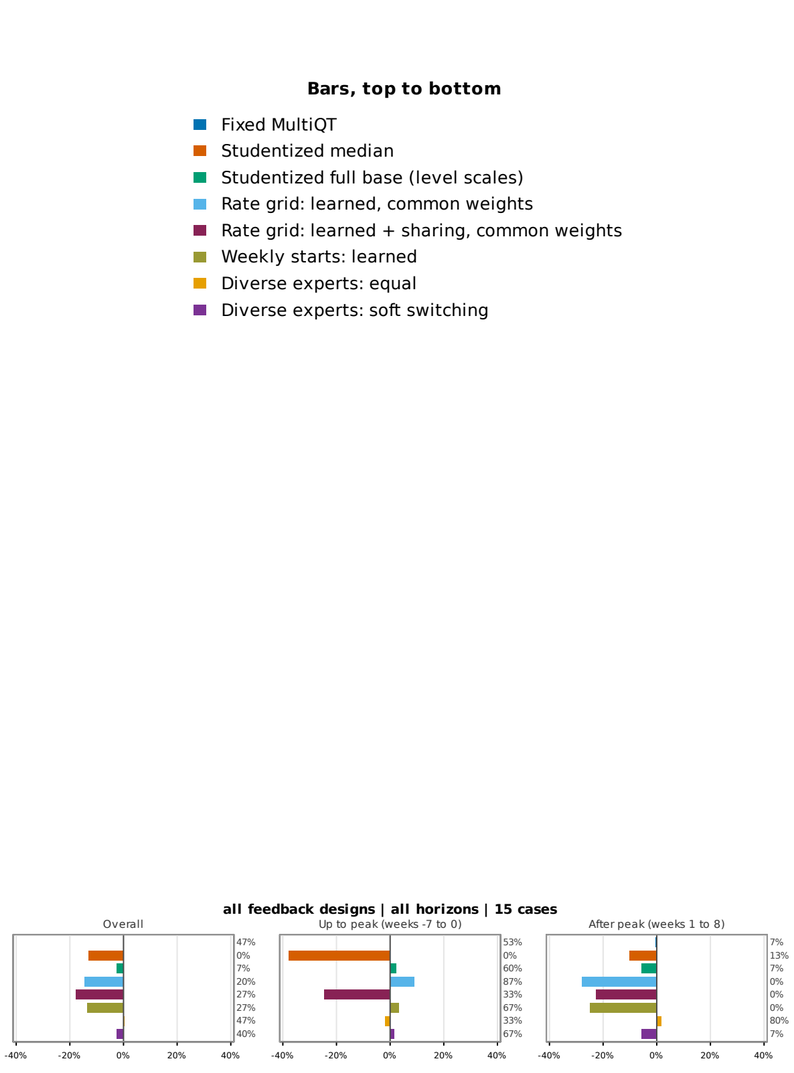
Bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Each disease and feedback design uses the settings tuned with it (output/results/tuning); every method: output/results/peak_reasoning/evaluation_tree.csv.

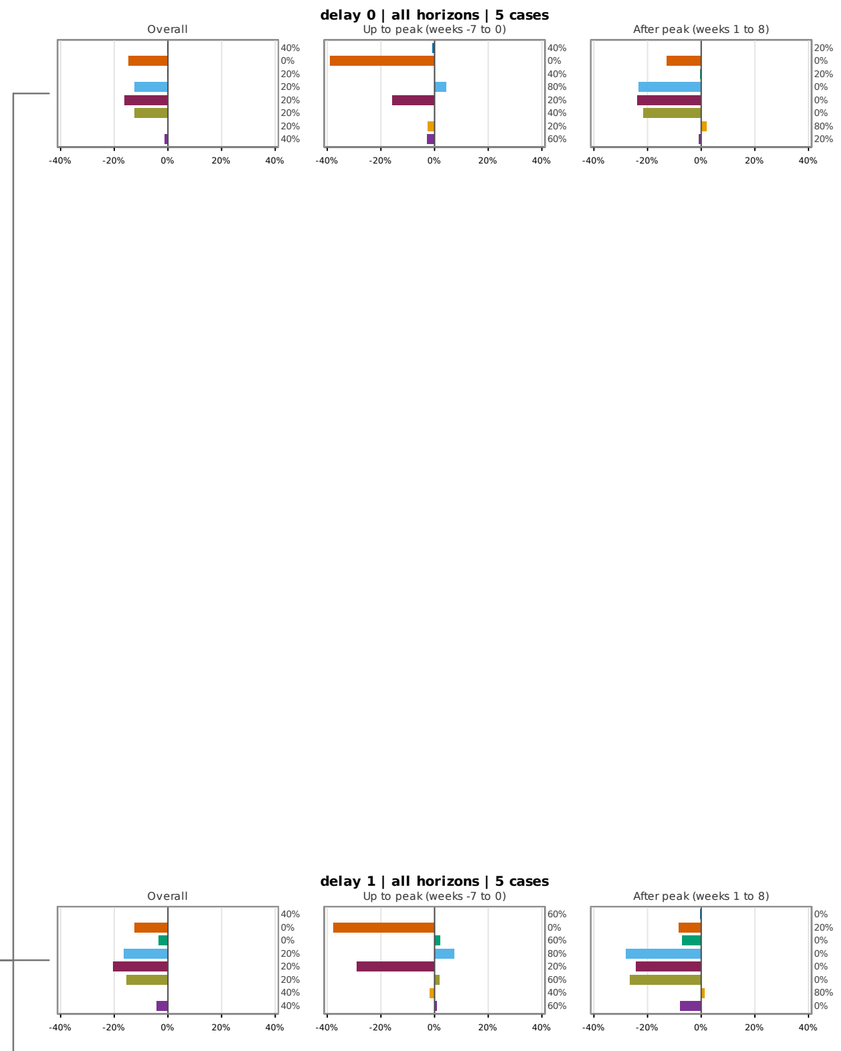
covid | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

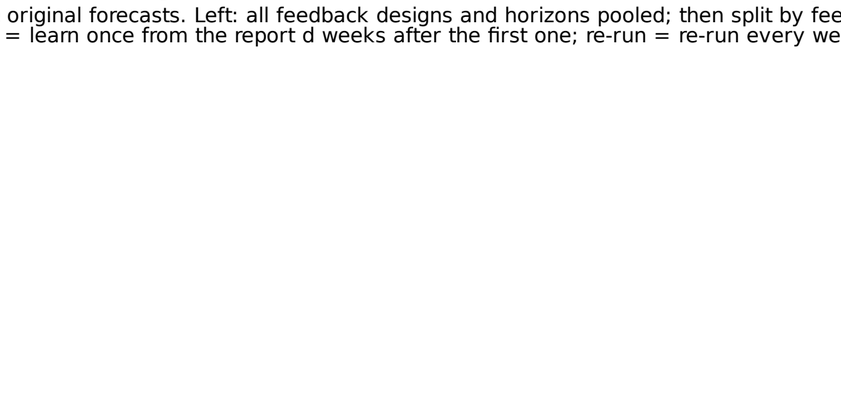
Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.

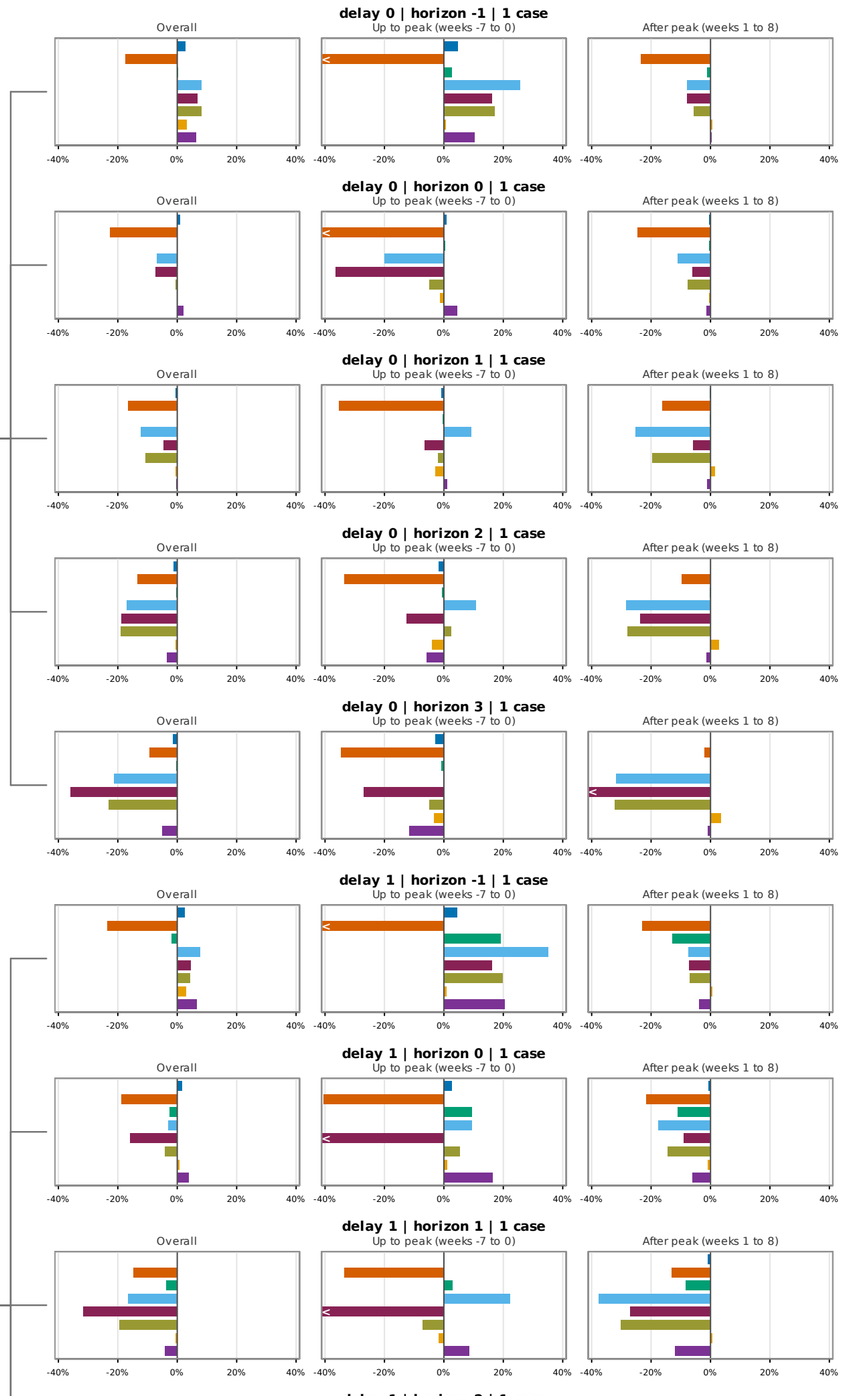
Bars, top to bottom

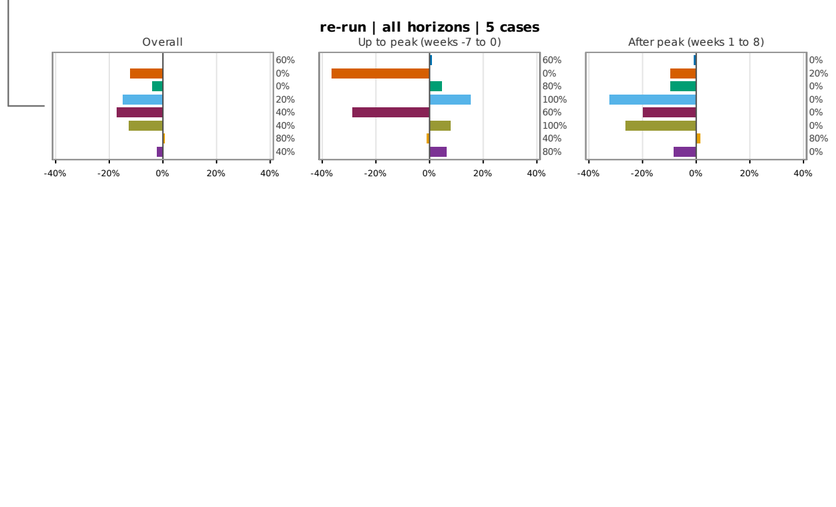
- Fixed MultiQT
- Studentized median
- Studentized full base (level scales)
- Rate grid: learned, common weights
- Rate grid: learned + sharing, common weights
- Weekly starts: learned
- Diverse experts: equal
- Diverse experts: soft switching











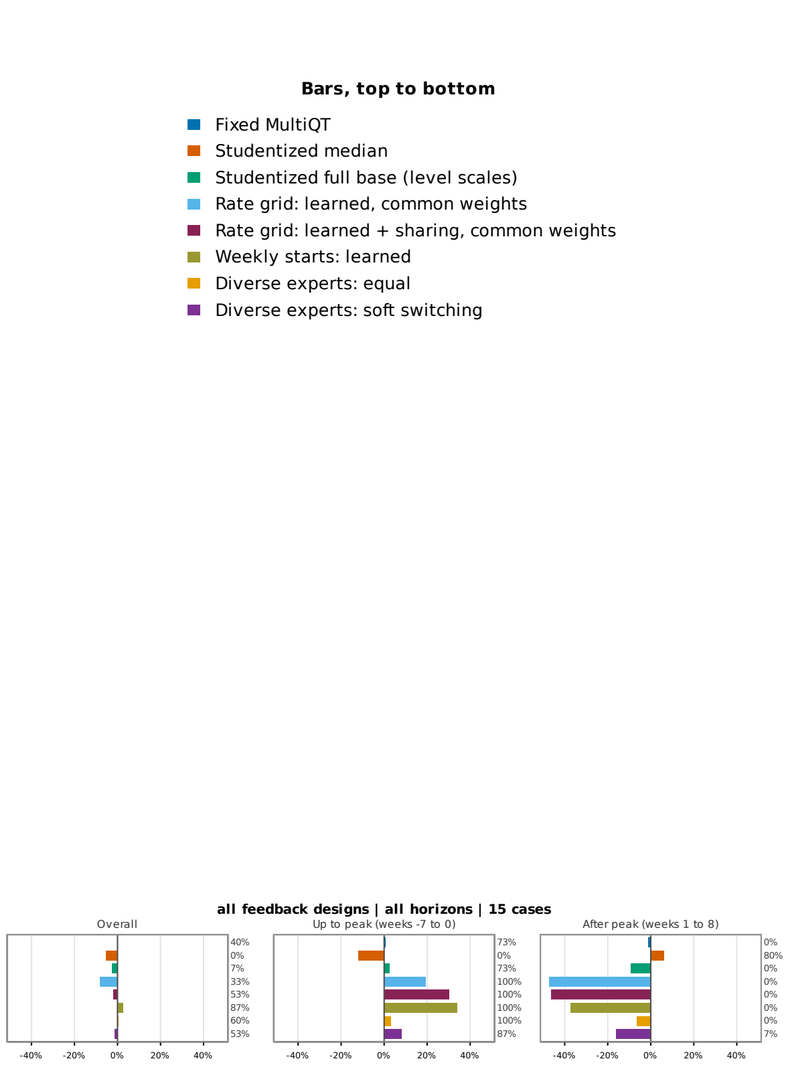
Bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Each disease and feedback design uses the settings tuned with it (output/results/tuning); every method: output/results/peak_reasoning/evaluation_tree.csv.

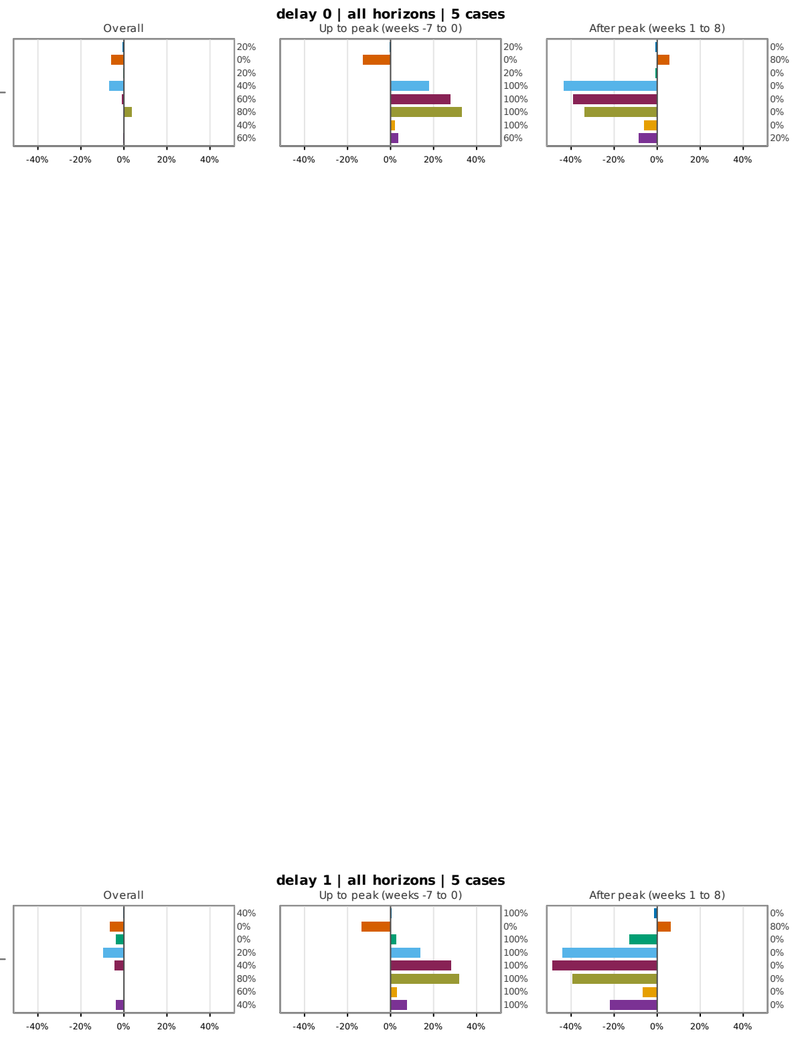
covid | California (06) | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

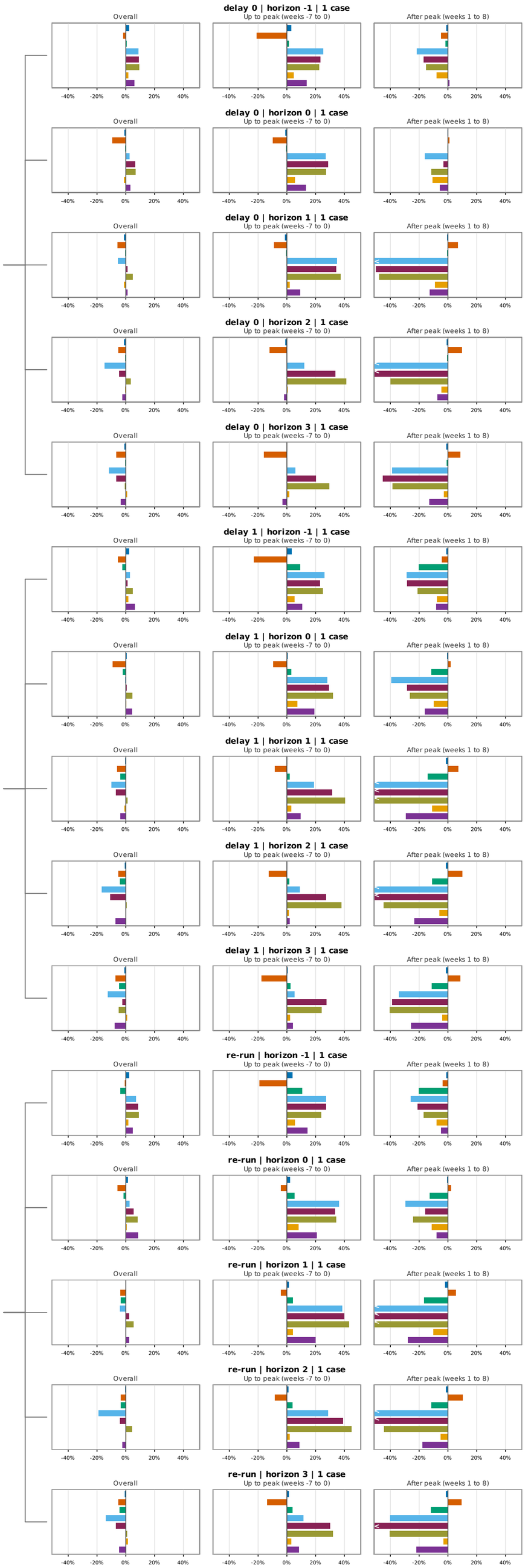
Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.

Bars, top to bottom

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- Studentized median
- Studentized full base (level scales)
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- Weekly starts: learned
- Diverse experts: equal
- Diverse experts: soft switching



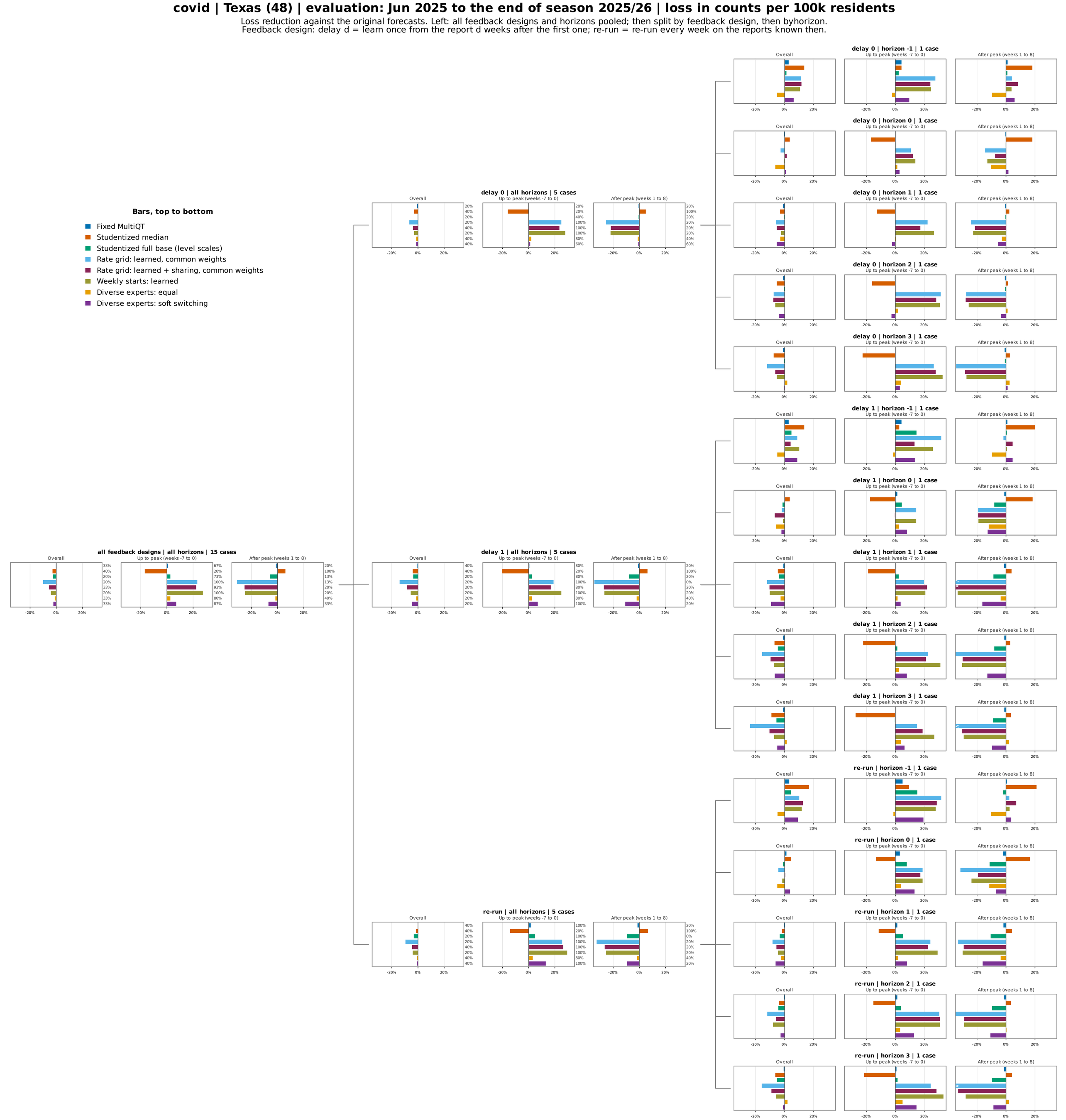




Bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Each disease and feedback design uses the settings tuned with it (output/results/tuning); every method: output/results/peak_reasoning/evaluation_tree.csv.

covid | Texas (48) | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.



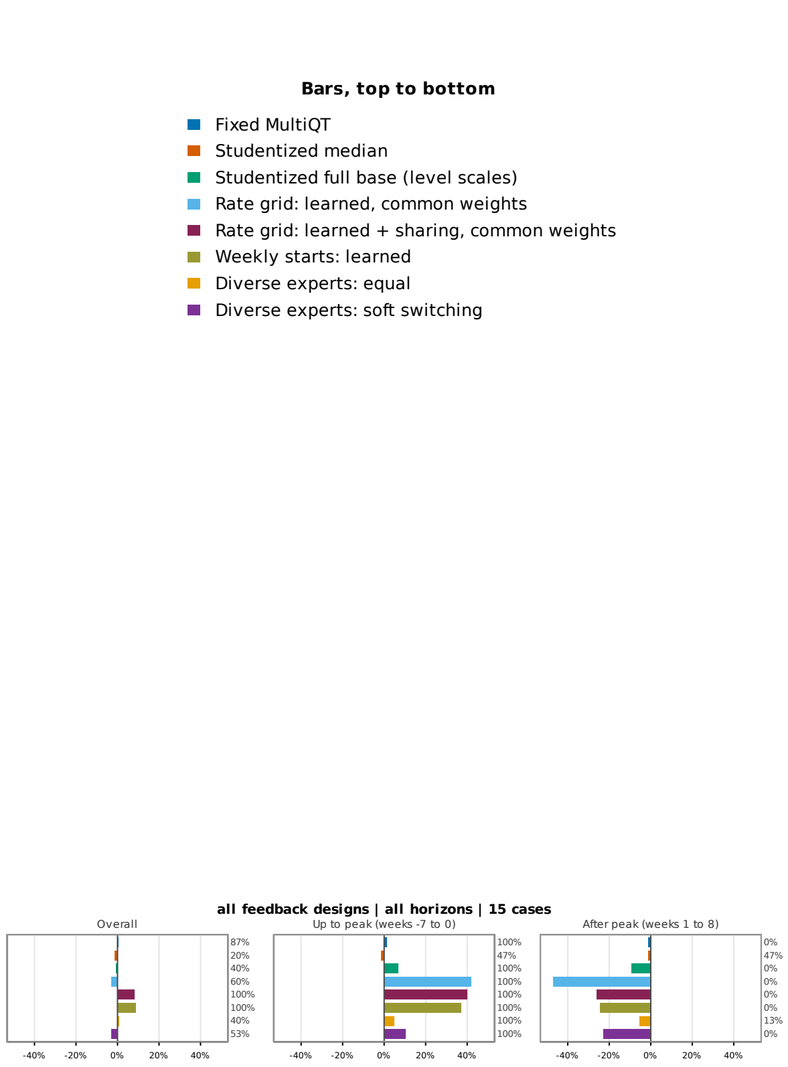
Bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
Numbers right of each panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
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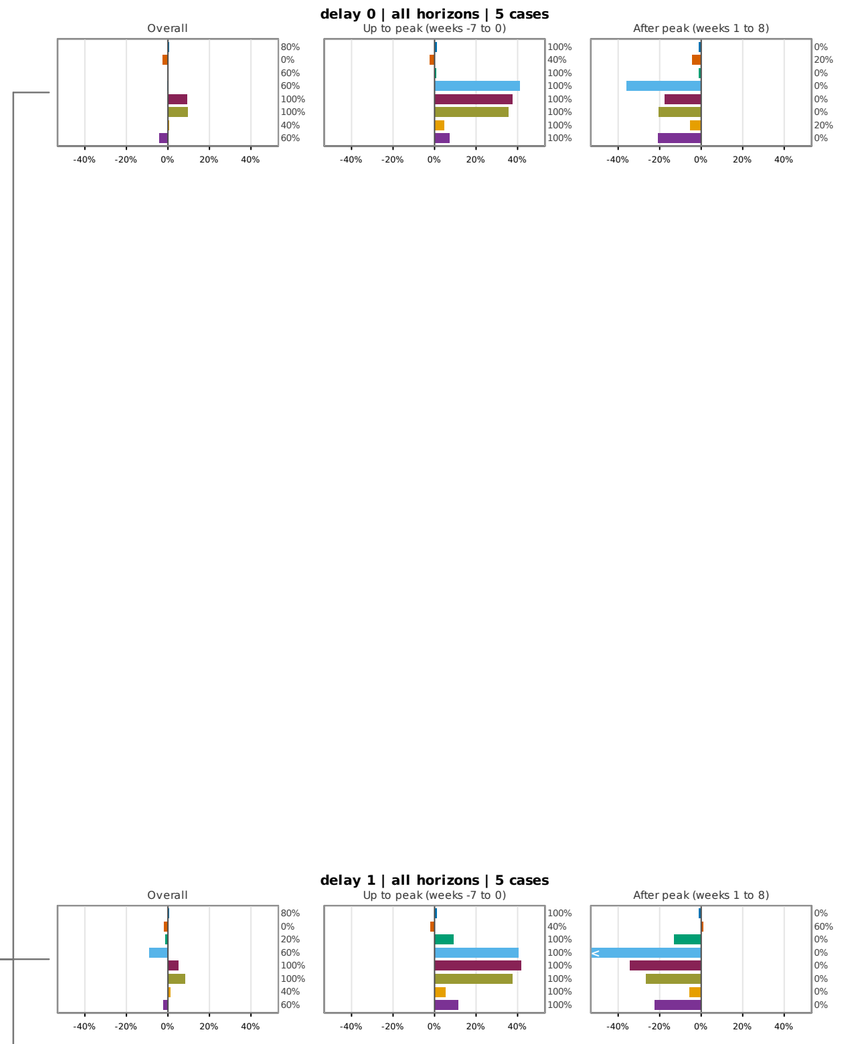
covid | Florida (12) | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

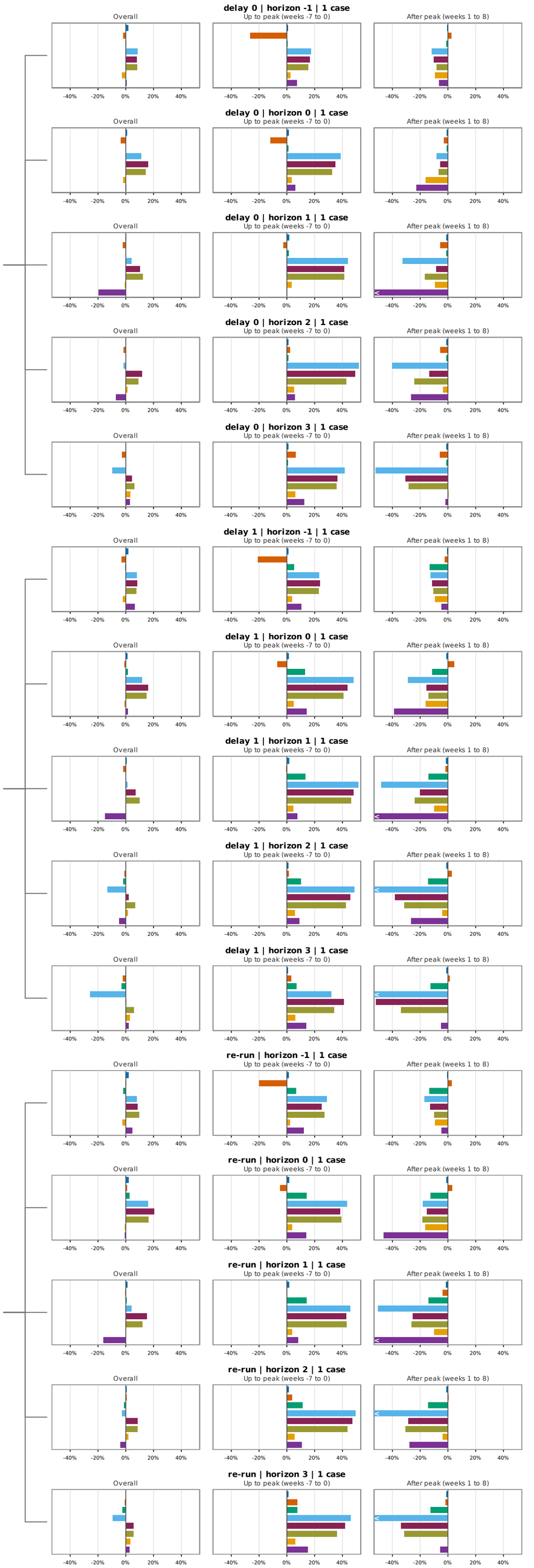
Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.

Bars, top to bottom

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- Studentized median
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- Weekly starts: learned
- Diverse experts: equal
- Diverse experts: soft switching







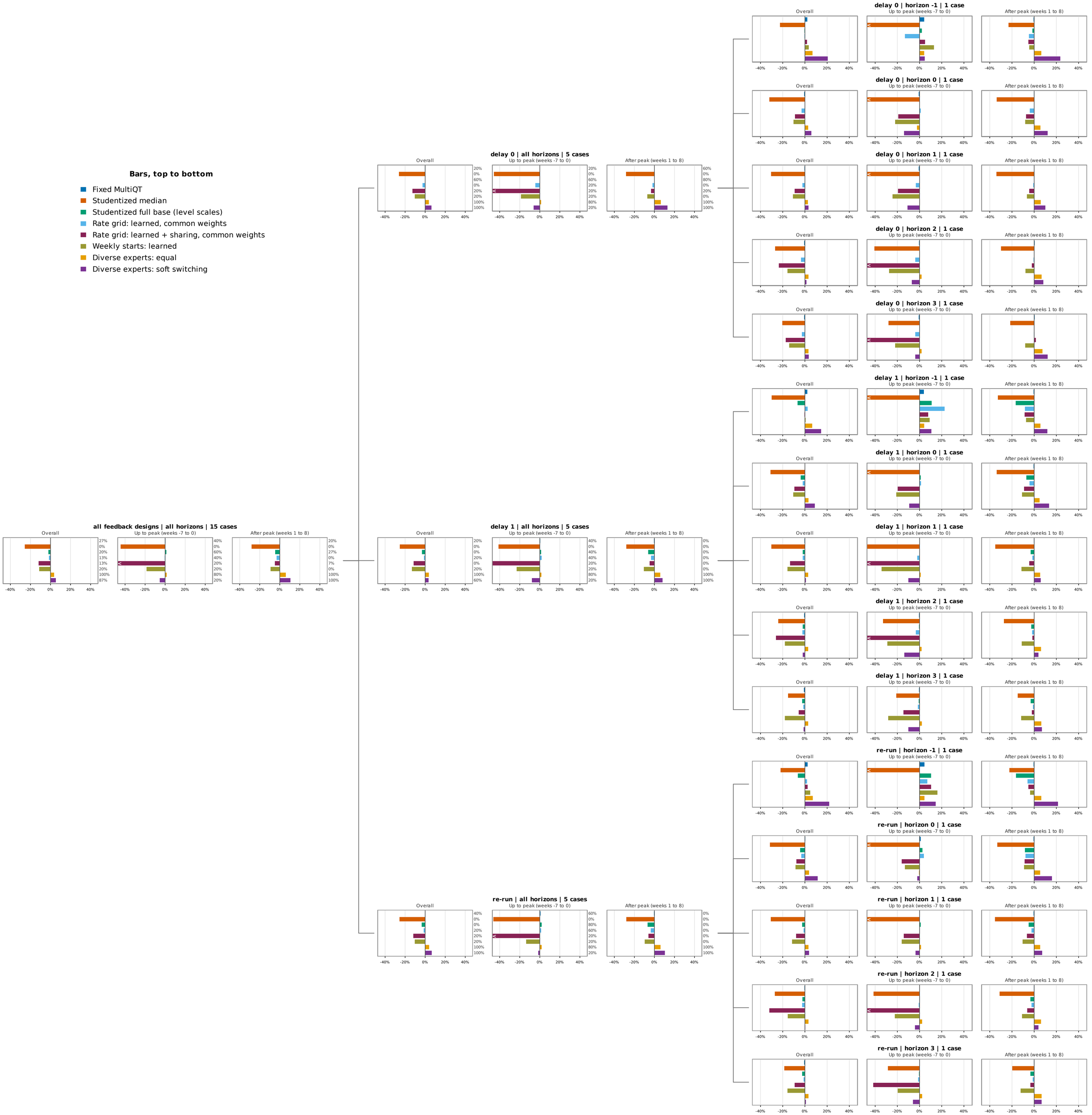
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Numbers right of each panel: share of the cases in the node in which the loss of the method is below the loss of the original (nodes with at least 5 cases). Up to peak / after peak: the weeks around every local peak of the latest counts.
Each disease and feedback design uses the settings tuned with it (output/results/tuning); every method: output/results/peak_reasoning/evaluation_tree.csv.

covid | New York (36) | evaluation: Jun 2025 to the end of season 2025/26 | loss in counts per 100k residents

Loss reduction against the original forecasts. Left: all feedback designs and horizons pooled; then split by feedback design, then byhorizon.
Feedback design: delay d = learn once from the report d weeks after the first one; re-run = re-run every week on the reports known then.

Bars, top to bottom

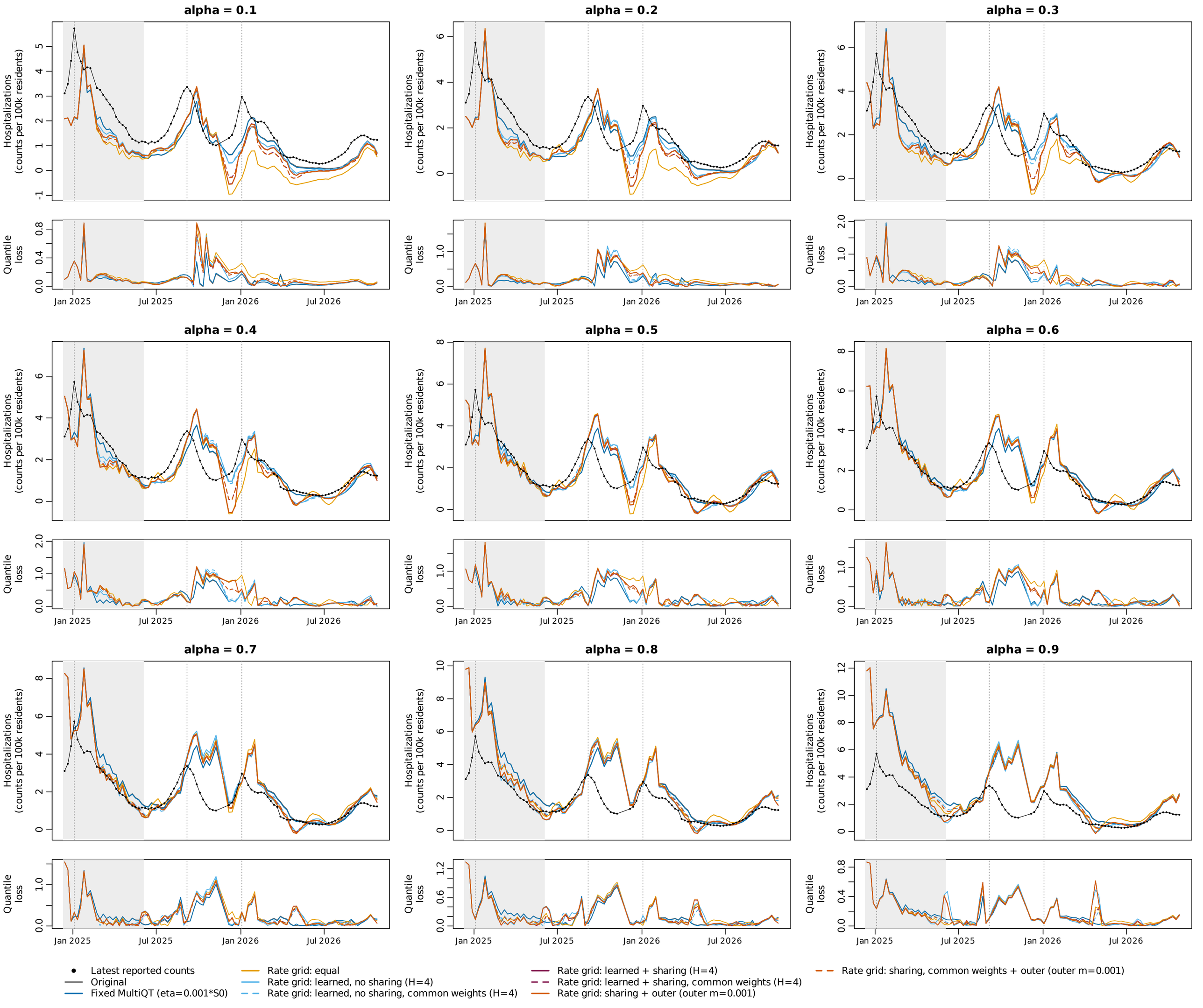
- Fixed MultiQT
- Studentized median
- Studentized full base (level scales)
- Rate grid: learned, common weights
- Rate grid: learned + sharing, common weights
- Weekly starts: learned
- Diverse experts: equal
- Diverse experts: soft switching



Bar: 100 * (1 - pooled loss / pooled loss of the original), pooled over the cases in the node (location x horizon x feedback design), the mean loss of each case weighted by its number of weeks; to the right: positive = smaller loss than the original.
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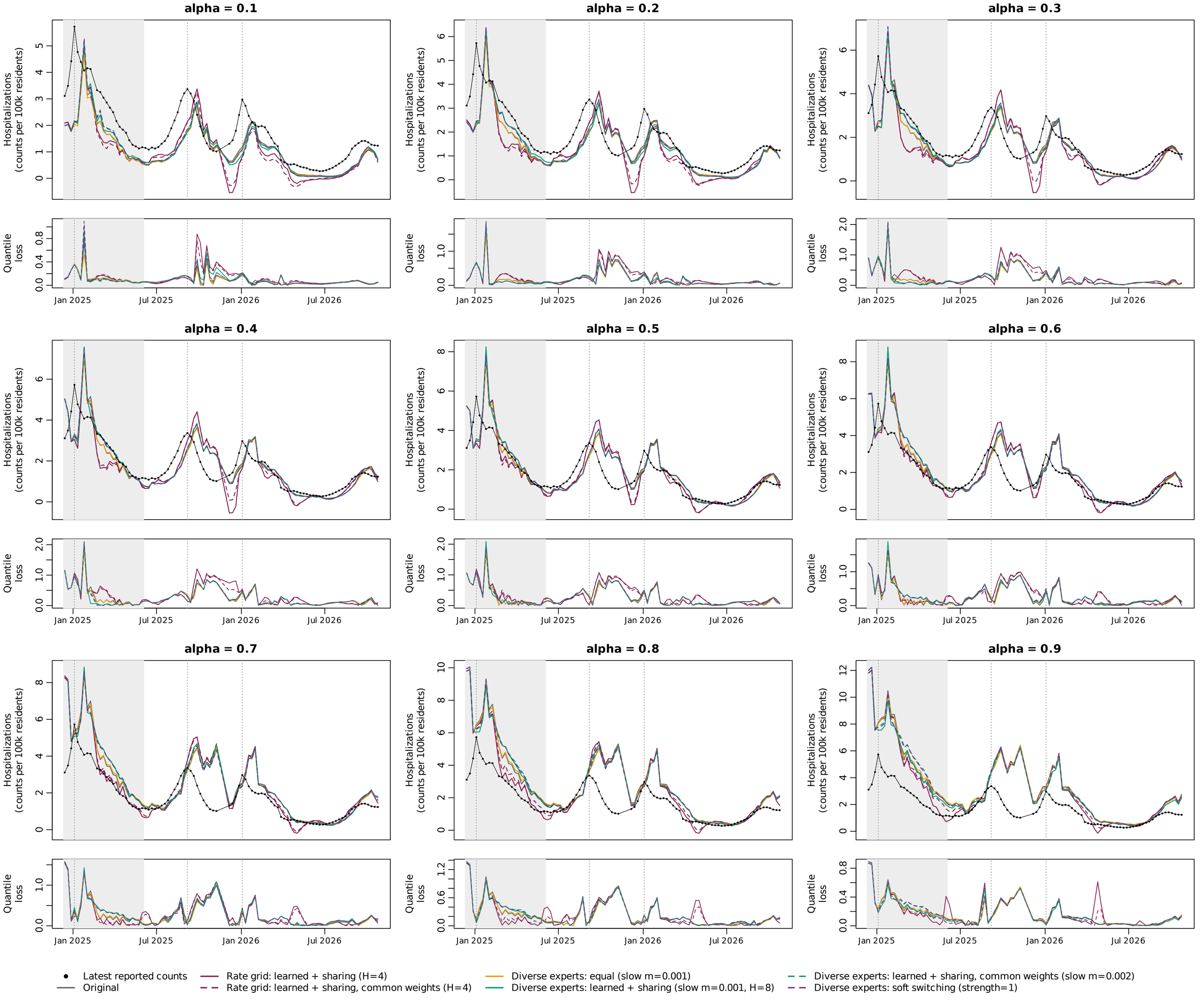
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0 | Rate-grid experts

Issued alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Dotted lines: peaks | Gray: tuning period (not scored)



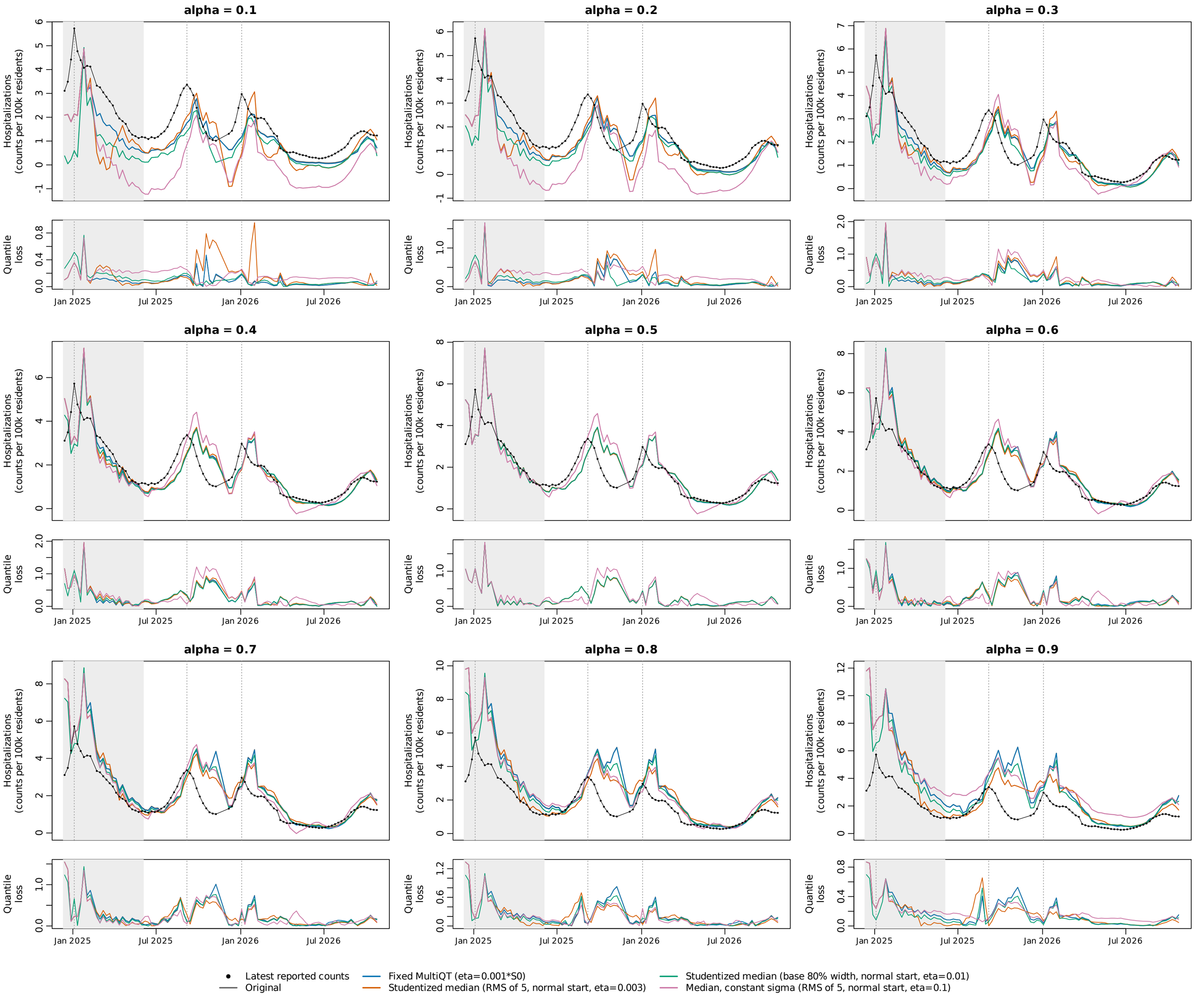
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0 | Diverse experts

Issued alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Dotted lines: peaks | Gray: tuning period (not scored)



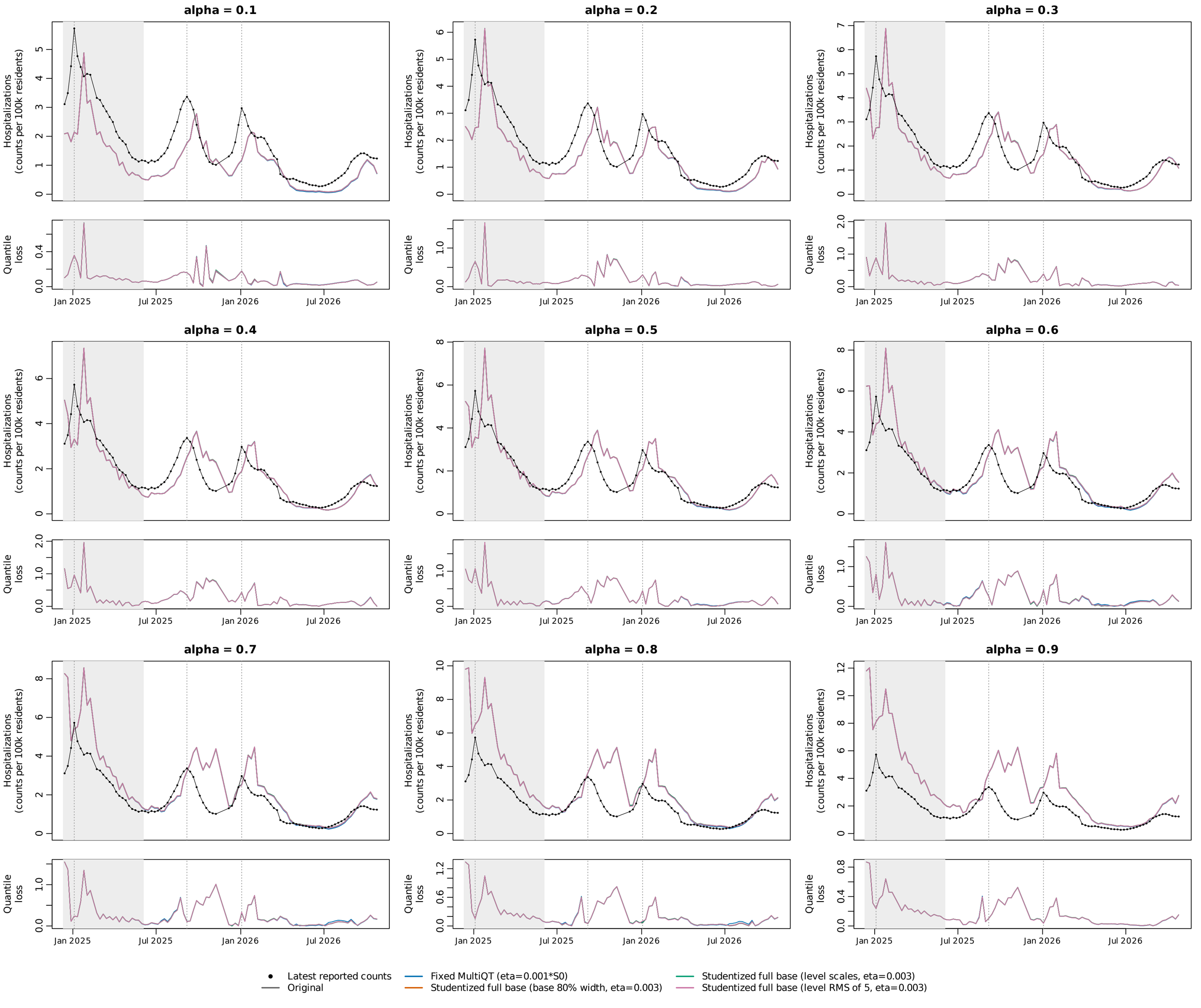
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0 | Studentized median

Issued alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Dotted lines: peaks | Gray: tuning period (not scored)



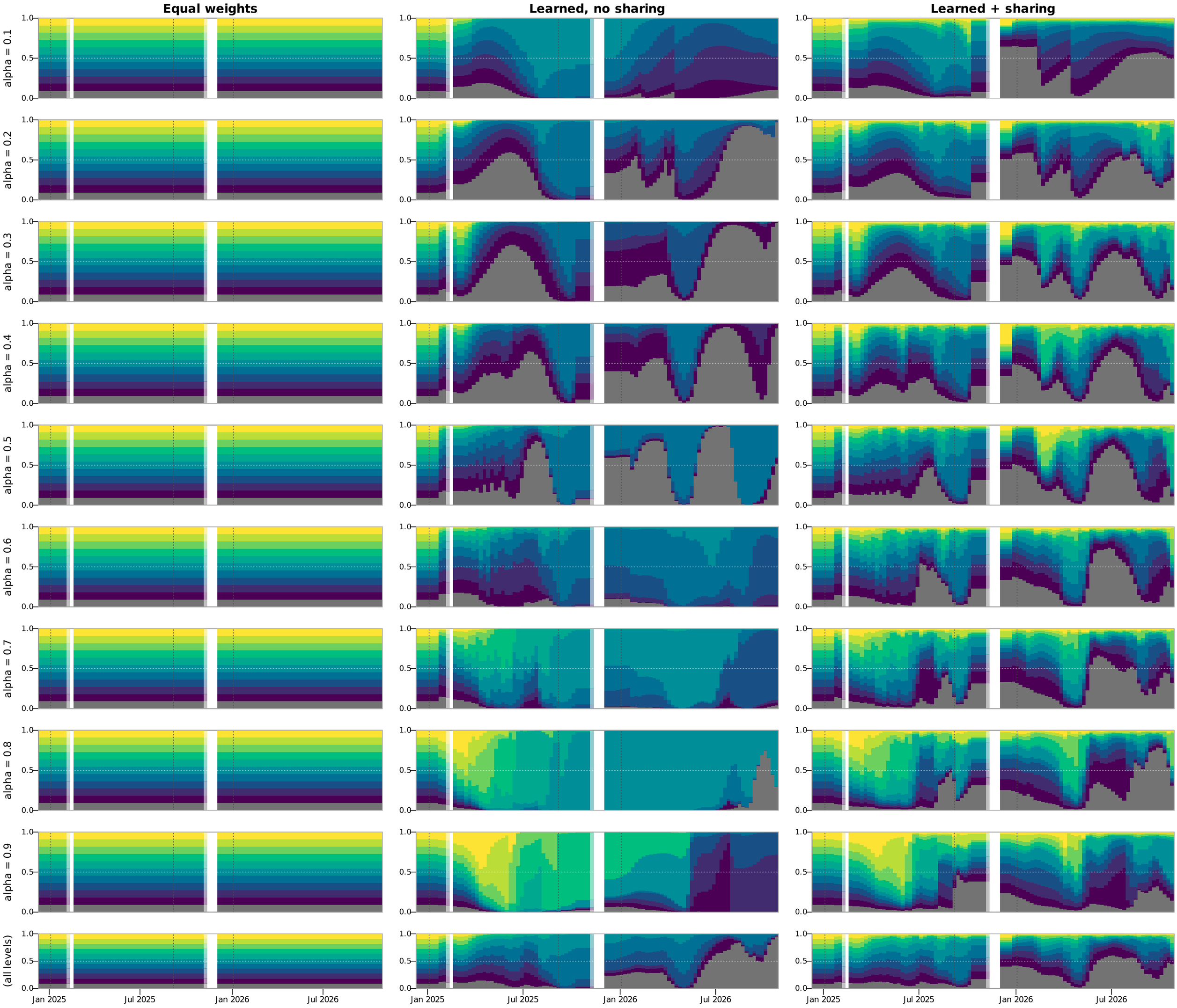
covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0 | Studentized full base

Issued alpha-quantile forecasts and their weekly quantile loss in counts per 100k residents against the latest reported counts | Dotted lines: peaks | Gray: tuning period (not scored)



Frozen rate-grid weights

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0



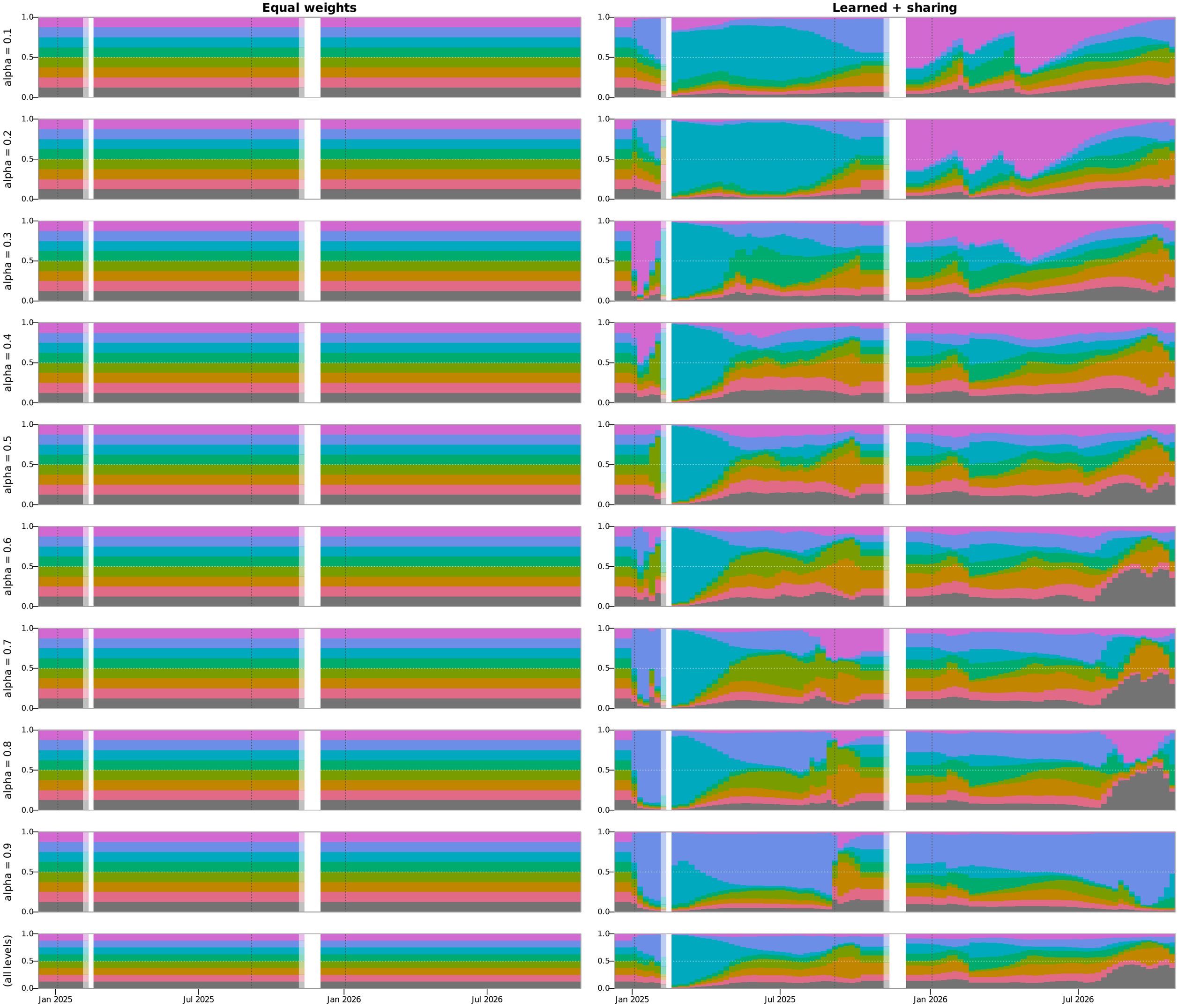
Target date | Weight at issuance for each target | Dotted lines: peaks

Rows alpha = 0.1-0.9: one weight vector per quantile level. Last row: one weight vector for all levels, learned from the mean quantile loss.

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

Diverse expert weights

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0



Target date | Weight at issuance for each target | Dotted lines: peaks

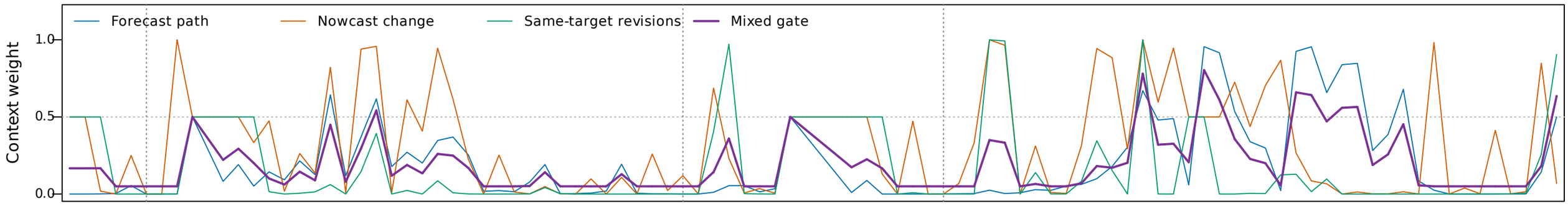
Rows alpha = 0.1-0.9: one weight vector per quantile level. Last row: one weight vector for all levels, learned from the mean quantile loss.

Lighter bars: rounds without a CMU forecast, filled with the latest earlier CMU forecast of the same target week.

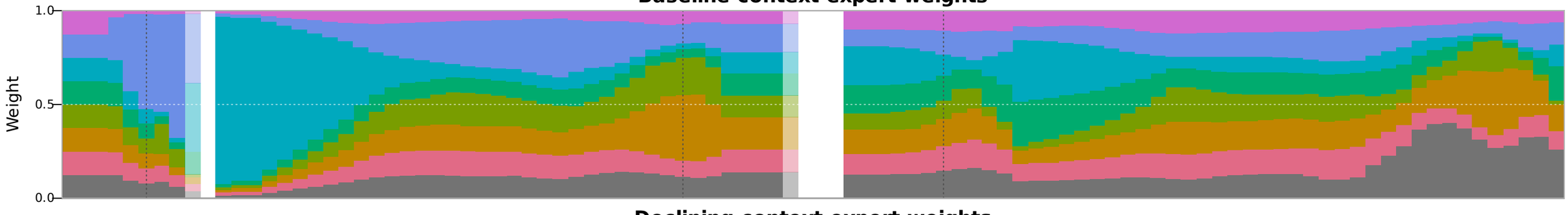
Diverse experts: soft switching (strength=1)

covid | horizon 2 | United States (US) | evaluation: Jun 2025 to the end of season 2025/26 | raw-count updates | feedback delay 0

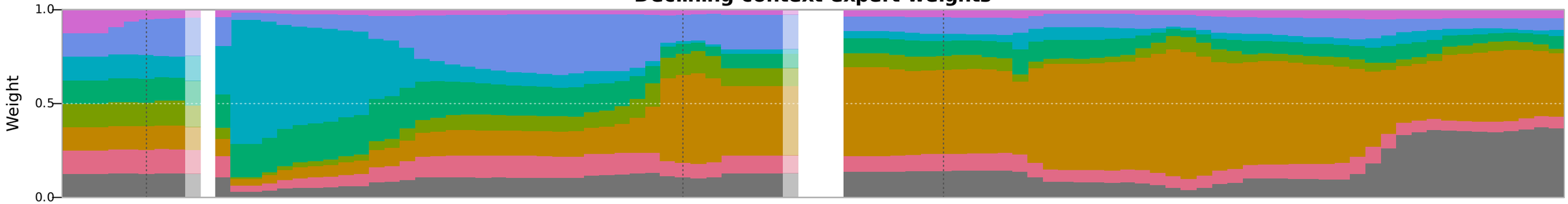
Declining context at issuance



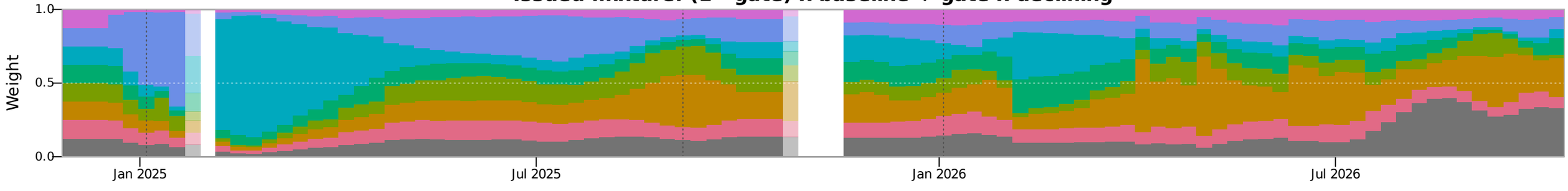
Baseline-context expert weights



Declining-context expert weights



Issued mixture: (1 - gate) x baseline + gate x declining



base
slow
leaky
damp=-0.5
damp=0.5
damp=1
width=0.75
width=1.5

Issue date | Common expert weights across quantile levels | Dotted lines: hindsight peaks

The gate is a context weight, not a peak probability; missing signals contribute a neutral score of 0.5.

Lighter bars: filled forecast rounds.